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|---------------------------|-----|------------------------|---------|-------------|---|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |   |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 1 | <b>of</b> | 98 |

# NSP/004/042/001 - Conductor Sag Charts For Wood Pole Lines up to and including 33kV

## 1. Purpose

The purpose of this document is to provide a series of sag charts for use with NSP/004/042

This document supersedes the following documents, all copies of which should be destroyed.

| Ref                 | Version         | Title                                                                                  |
|---------------------|-----------------|----------------------------------------------------------------------------------------|
| NSP/004/042/001     | 1.0             | Conductor Sag Charts For Wood Pole Lines up to and including 33kV                      |
| DSS/004/042         | 1.0             | Specification for Yorkshire Electricity Overhead Lines to ESI 43-40                    |
| O.309               | March 1996      | High Voltage Single Circuit Lines on Wood Poles for Voltages up to and Including 33kV  |
| TS 71-01 to TS 7113 | Initial Version | Erection sag and tension tables for lines designed formally to DSS/004/042 version 1.0 |

## 2. Scope

This document contains sag charts for the erection of 3-phase and single-phase overhead lines on wood poles up to and including 33kV. It has been designed to be in accordance with ENA TS 43-40: Issue 2

Additionally this document contains sag charts to be used for the renovation of existing HV lines using various historical conductor types

|                           |     |                        |         |             |         |
|---------------------------|-----|------------------------|---------|-------------|---------|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |         |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 2 of 98 |

## 2.1 Contents

|                                                                                         |    |
|-----------------------------------------------------------------------------------------|----|
| 1. Purpose .....                                                                        | 1  |
| 2. Scope.....                                                                           | 1  |
| 2.1 Contents .....                                                                      | 2  |
| 3. Technical specification .....                                                        | 4  |
| 3.1 Sag Charts.....                                                                     | 4  |
| 3.2 Pre-Stress Tensions for conductor erection .....                                    | 4  |
| 4. References .....                                                                     | 5  |
| 4.1 External Documentation .....                                                        | 5  |
| 4.2 Internal documentation .....                                                        | 5  |
| 4.3 Summary Of Changes .....                                                            | 5  |
| 5 Definitions.....                                                                      | 5  |
| 6. Authority for issue .....                                                            | 6  |
| Appendix 1 - Schedule of sag charts.....                                                | 7  |
| Section 1 - Sag charts for new construction.....                                        | 7  |
| Appendix 2 - 50mm AAAC (Hazel) - (90m Basic) Normal altitude .....                      | 8  |
| Appendix 3 - 50mm AAAC (Hazel) - (120m Basic) Normal altitude .....                     | 10 |
| Appendix 3a - 50mm AAAC (Hazel) - (120m Basic) Normal altitude .....                    | 11 |
| Appendix 4 - 100mm AAAC (Oak) - (110m Basic) Normal altitude.....                       | 12 |
| Appendix 4a - 100mm AAAC (Oak) - (110m Basic) Normal altitude.....                      | 13 |
| Appendix 5 - 175mm AAAC (Elm) - (110m Basic) Normal altitude .....                      | 14 |
| Appendix 5a - 175mm AAAC (Elm) - (110m Basic) Normal altitude .....                     | 15 |
| Appendix 6 - 200mm AAAC (Poplar) - (120m Basic) Normal altitude .....                   | 16 |
| Appendix 6a - 200mm AAAC (Poplar) - (120m Basic) Normal altitude.....                   | 17 |
| Appendix 7 - 50mm AAAC (Hazel) - (75m Basic) High altitude .....                        | 18 |
| Appendix 7a - 50mm AAAC (Hazel) - (75m Basic) - High altitude .....                     | 19 |
| Appendix 8 - 50mm AAAC (Hazel) - (80m Basic) High altitude .....                        | 20 |
| Appendix 8a - 50mm AAAC (Hazel) - (80m Basic) High altitude .....                       | 21 |
| Appendix 9 - 50mm AAAC (Hazel) - (100m Basic) High altitude .....                       | 22 |
| Appendix 9a - 50mm AAAC (Hazel) - (100m Basic) High altitude .....                      | 23 |
| Appendix 10 - 100mm AAAC (Oak) - (80m Basic) High altitude.....                         | 24 |
| Appendix 10a - 100mm AAAC (Oak) - (80m Basic) High altitude.....                        | 25 |
| Appendix 11 - 100mm AAAC (Oak) - (90m Basic) High altitude.....                         | 26 |
| Appendix 11a - 100mm AAAC - (Oak) - (90m Basic) High altitude.....                      | 27 |
| Appendix 12 - 175mm AAAC (Elm) - (90m Basic)- High altitude .....                       | 28 |
| Appendix 12a - 175mm AAAC (Elm) - (90m Basic) High altitude .....                       | 29 |
| Appendix 13 - 175mm AAAC (Elm)- (100m Basic) - High altitude .....                      | 30 |
| Appendix 13a - 175mm AAAC (Elm)- (100m Basic) Severe altitude .....                     | 31 |
| Appendix 14 - 200mm AAAC (Poplar) - (90m Basic)- High altitude.....                     | 32 |
| Appendix 14a - 200mm AAAC (Poplar) - (90m Basic)- High altitude.....                    | 33 |
| Appendix 15 - 200mm AAAC (Poplar) - (100m Basic)- High altitude.....                    | 34 |
| Appendix 15a - 200mm AAAC (Poplar) - (100m Basic)- High altitude.....                   | 35 |
| Appendix 16 - Section 2 - Sag charts for the renovation of existing overhead lines..... | 36 |
| Appendix 17 - .017" or 13mm <sup>2</sup> CACU - (All altitudes) - 80m Basic .....       | 37 |
| Appendix 17a - .017" or 13mm <sup>2</sup> CACU - (All altitudes) - 80m Basic .....      | 38 |
| Appendix 18 - .017" or 13mm <sup>2</sup> CACU - (All altitudes) - 90m Basic .....       | 39 |
| Appendix 18a - .017" or 13mm <sup>2</sup> CACU - (All altitudes) - 90m Basic .....      | 40 |
| Appendix 19 - .017" or 13mm <sup>2</sup> CACU - (All altitudes) - 100m Basic .....      | 41 |
| Appendix 19a - .017" or 13mm <sup>2</sup> CACU - (All altitudes) - 100m Basic .....     | 42 |
| Appendix 20 - .017" or 13mm <sup>2</sup> CACU - (All altitudes) - 120m Basic .....      | 43 |
| Appendix 20a - .017" or 13mm <sup>2</sup> CACU - (All altitudes) - 120m Basic .....     | 44 |
| Appendix 21 - .05" or 32mm <sup>2</sup> HDBC - (All altitudes) - 80m Basic .....        | 45 |
| Appendix 21a - .05" or 32mm <sup>2</sup> HDBC - (All altitudes) - 80m Basic .....       | 46 |
| Appendix 22 - .05" or 32mm <sup>2</sup> HDBC - (All altitudes) - 90m Basic .....        | 47 |
| Appendix 22a - .05" or 32mm <sup>2</sup> HDBC - (All altitudes) - 90m Basic .....       | 48 |

|                           |     |                        |         |             |         |
|---------------------------|-----|------------------------|---------|-------------|---------|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |         |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 3 of 98 |

|                                                                                     |    |
|-------------------------------------------------------------------------------------|----|
| Appendix 23 - .05" or 32mm <sup>2</sup> HDBC - (All altitudes) - 100m Basic .....   | 49 |
| Appendix 23a - .05" or 32mm <sup>2</sup> HDBC - (All altitudes) - 100m Basic .....  | 50 |
| Appendix 24 - .05" or 32mm <sup>2</sup> HDBC - (All altitudes) - 120m Basic .....   | 51 |
| Appendix 24a - .05" or 32mm <sup>2</sup> HDBC - (All altitudes) - 120m Basic .....  | 52 |
| Appendix 25 - .05" or 50mm <sup>2</sup> AAAC - (All altitudes) - 80m Basic .....    | 53 |
| Appendix 25a - .05" or 50mm <sup>2</sup> AAAC - (All altitudes) - 80m Basic .....   | 54 |
| Appendix 26 - .05" or 50mm <sup>2</sup> AAAC - (All altitudes) - 100m Basic .....   | 55 |
| Appendix 26a - .05" or 50mm <sup>2</sup> AAAC - (All altitudes) - 100m Basic .....  | 56 |
| Appendix 27 - .05" or 50mm <sup>2</sup> AAAC - (All altitudes) - 120m Basic .....   | 57 |
| Appendix 27a - .05" or 50mm <sup>2</sup> AAAC - (All altitudes) - 120m Basic .....  | 58 |
| Appendix 28 - .025" or 16mm <sup>2</sup> HDBC - (All altitudes) - 90m Basic .....   | 59 |
| Appendix 28a - .025" or 16mm <sup>2</sup> HDBC - (All altitudes) - 90m Basic .....  | 60 |
| Appendix 29 - .025" or 16mm <sup>2</sup> HDBC - (All altitudes) - 100m Basic .....  | 61 |
| Appendix 29a - .025" or 16mm <sup>2</sup> HDBC - (All altitudes) - 100m Basic ..... | 62 |
| Appendix 30 - 3 SWG HDBC - (All altitudes) - 90m Basic .....                        | 63 |
| Appendix 30a - 3 SWG HDBC - (All altitudes) - 90m Basic .....                       | 64 |
| Appendix 31 - 3 SWG HDBC - (All altitude) - 100m Basic .....                        | 65 |
| Appendix 31 - a3 SWG HDBC - (All altitudes) - 100m Basic .....                      | 66 |
| Appendix 32 - 5SWG HDBC - (All altitudes) - 90m Basic .....                         | 67 |
| Appendix 32a - 5SWG HDBC - (All altitudes) - 90m Basic .....                        | 68 |
| Appendix 33 - 5SWG HDBC - (All altitude) - 100m Basic .....                         | 69 |
| Appendix 33a - 5SWG HDBC - (All altitudes) - 100m Basic .....                       | 70 |
| Appendix 34 - 6SWG HDBC - (All altitudes) - 90m Basic .....                         | 71 |
| Appendix 34a - 6SWG HDBC - (All altitudes) - 90m Basic .....                        | 72 |
| Appendix 35 - 6SWG HDBC - (All altitudes) - 100m Basic .....                        | 73 |
| Appendix 35a - 6SWG HDBC - (All altitudes) - 100m Basic .....                       | 74 |
| Appendix 36 - 7SWG HDBC - (All altitudes) - 90m Basic .....                         | 75 |
| Appendix 36a - 7SWG HDBC - (All altitudes) - 90m Basic .....                        | 76 |
| Appendix 37 - 7SWG HDBC - (All altitude) - 100m Basic .....                         | 77 |
| Appendix 37a - 7SWG HDBC - (All altitude) - 100m Basic .....                        | 78 |
| Appendix 38 - .175" ACSR / 175mm ACSR (Lynx) - (All altitudes) - 76.2mm Basic.....  | 79 |
| Appendix 38a - .175" ACSR / 175mm ACSR (Lynx) - (All altitudes) - 76.2mm Basic..... | 80 |
| Appendix 39 - .1" / 100mm <sup>2</sup> ACSR (Normal altitude) - 90m Basic .....     | 81 |
| Appendix 39a - .1" / 100mm <sup>2</sup> ACSR (Normal altitude) - 90m Basic .....    | 82 |
| Appendix 40 - .1" / 100mm <sup>2</sup> ACSR (Normal altitude) - 100m Basic .....    | 83 |
| Appendix 40a - .1" / 100mm <sup>2</sup> ACSR (Normal altitude) - 100m Basic .....   | 84 |
| Appendix 41 - .1" / 70mm <sup>2</sup> HDBC (Normal altitude) - 90m Basic .....      | 85 |
| Appendix 41a - .1" / 70mm <sup>2</sup> HDBC (Normal altitude) - 90m Basic .....     | 86 |
| Appendix 42 - .15" / 100mm <sup>2</sup> HDBC (Normal altitude) - 90m Basic .....    | 87 |
| Appendix 42a - .15" / 100mm <sup>2</sup> HDBC (Normal altitude) - 90m Basic .....   | 88 |
| Appendix 43 - .2" / 125mm <sup>2</sup> HDBC (Normal altitude) - 100m Basic .....    | 89 |
| Appendix 43a - .2" / 125mm <sup>2</sup> HDBC (Normal altitude) - 100m Basic .....   | 90 |
| Appendix 44 - .1" / 100mm <sup>2</sup> ACSR (High altitude) - 90m Basic .....       | 91 |
| Appendix 44a - .1" / 100mm <sup>2</sup> ACSR (High altitude) - 90m Basic .....      | 92 |
| Appendix 45 - .1" / 70mm <sup>2</sup> HDBC (High altitude) - 90m Basic .....        | 93 |
| Appendix 45a - .1" / 70mm <sup>2</sup> HDBC (High altitude) - 90m Basic .....       | 94 |
| Appendix 46 - .15" / 100mm <sup>2</sup> HDBC (High altitude) - 90m Basic .....      | 95 |
| Appendix 46a - .15" / 100mm <sup>2</sup> HDBC (High altitude) - 90m Basic .....     | 96 |
| Appendix 47 - .2" / 125mm <sup>2</sup> HDBC (High altitude) - 100m Basic .....      | 97 |
| Appendix 47a - .2" / 125mm <sup>2</sup> HDBC (High altitude) - 100m Basic .....     | 98 |

|                           |     |                        |         |             |   |           |    |
|---------------------------|-----|------------------------|---------|-------------|---|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |   |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 4 | <b>of</b> | 98 |

## 3. Technical specification

### 3.1 Sag Charts

Each conductor type has two pages of information specific to a particular conductor and the altitude in which it is constructed.

#### Sheet 1

Describes the conductor technical data and applied weather loads applicable in the selected environment. This information is used to derive the applicable conductor MCT that is then used to create the sag charts.

Additionally this sheet contains the MCW, MCP and FPT loading values, which together with the MCT are used in the main specification to determine the structure, stay and crossarm requirements.

#### Sheet 2

The erection chart shall be used to determine the appropriate sags and tension for new conductors only, as it contains additional factors that increase the conductor tensions. These factors are designed to compensate for long term creep.

Design charts contain no additional factors, they shall be used for the re-sagging of existing conductors or to review the likely sags or tensions in existing overhead lines. This chart shall be used for any clearance enquiries.

Appendix 2 to 15 - New construction

Appendix 16 to 47 - Renovation or modification to existing lines

### 3.2 Pre-Stress Tensions for conductor erection

| Conductor Type          | Pre-stress Erection Tension |
|-------------------------|-----------------------------|
| 50mm <sup>2</sup> AAAC  | 5884N (600kgf)              |
| 100mm <sup>2</sup> AAAC | 12258N (1250kgf)            |
| 175mm <sup>2</sup> AAAC | 14,710N (1500kgf)           |
| 200mm <sup>2</sup> AAAC | 14,710N (1500kgf)           |

|                           |     |                        |         |             |   |           |    |
|---------------------------|-----|------------------------|---------|-------------|---|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |   |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 5 | <b>of</b> | 98 |

## 4. References

### 4.1 External Documentation

| Reference    | Title                                                    |
|--------------|----------------------------------------------------------|
| ENA TS 43-40 | High voltage single circuit overhead lines on wood poles |

### 4.2 Internal documentation

| Reference   | Title                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------------|
| NSP/004/042 | Specification for High Voltage Single Circuit Lines on Wood Poles for Voltages upto and including 33kV |

### 4.3 Summary Of Changes

| Clause           | Changes or amendments                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1              | Table of contents inserted                                                                                                                                         |
| 4.3              | Summary of changes table inserted                                                                                                                                  |
| Appendix 1       | Simplified list of allowable sag charts                                                                                                                            |
| Appendix 2 – 15  | Revised sag charts to incorporate higher tension limits for the 175mm AAAC and 200mm AAAC conductors and revisions to the % reduction used in all ordinate tables. |
| Appendix 16 – 47 | Revised Over tension and ordinate values applied to charts                                                                                                         |

## 5 Definitions

| Term                             | Definition                                                                                                                                                       |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Basic Span                       | The span length adopted for sag/tension calculations                                                                                                             |
| Everyday Tension (EDT)           | The design stress in an unloaded conductor at 5°C. Used in sag / tension calculations to limit harmful conductor vibrations                                      |
| Freezing Point Tension (FPT)     | The design tension of an unloaded conductor at 0°C in still air.                                                                                                 |
| Maximum Conductor Pressure (MCP) | The maximum transverse component of applied conductor load when subjected to a wind load.                                                                        |
| Maximum Conductor Weight (MCW)   | The maximum vertical component of applied conductor load, including the weight of accreted ice, if present.                                                      |
| MCT                              | The maximum conductor tension under design loading conditions                                                                                                    |
| Maximum Span                     | The maximum permitted length of any span                                                                                                                         |
| Maximum Working Tension (MWT)    | The absolute maximum conductor tension assessed at -5.6°C with wind and ice loading.                                                                             |
| Rated Breaking Strength (RBS)    | The rating assigned to a component (usually conductors) that defines the maximum mechanical load that the component will withstand without damage.               |
| Recommended Span                 | The average span length in any section to which the line shall be planned. Individual spans will normally be within +/- 20% of the chosen basic span.            |
| Sag                              | The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured at mid-span |
| Span                             | The horizontal distance between adjacent supports. Individual spans will normally be within 20% of the chosen basic span                                         |

|                           |     |                        |         |             |   |           |    |
|---------------------------|-----|------------------------|---------|-------------|---|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |   |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 6 | <b>of</b> | 98 |

## 6. Authority for issue

### 6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

|          |                                 | Sign | Date |
|----------|---------------------------------|------|------|
| G Hammel | Overhead Line Standards Manager |      |      |

### 6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

|             |                            | Sign | Date |
|-------------|----------------------------|------|------|
| A Bilclough | Overhead Programme Manager |      |      |

### 6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

|             |                            | Sign | Date |
|-------------|----------------------------|------|------|
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### 6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

|           |                   | Sign | Date |
|-----------|-------------------|------|------|
| S Johnson | DBD Administrator |      |      |

### 6.5 Approval

Approval is given for the content of this document

|              |                                  | Sign | Date |
|--------------|----------------------------------|------|------|
| C Holdsworth | Standards and Assessment Manager |      |      |

### 6.6 Authorisation

Authorisation is granted for publication of this document

|             |                         | Sign | Date |
|-------------|-------------------------|------|------|
| M Nicholson | Head of System Strategy |      |      |

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|---------------------------|-----|------------------------|---------|-------------|---|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |   |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 7 | <b>of</b> | 98 |

## Appendix 1 - Schedule of sag charts

### Section 1 - Sag charts for new construction

| Conductor Type          | Basic Span | Altitude or Loading | Appendix No. |
|-------------------------|------------|---------------------|--------------|
| 50mm <sup>2</sup> AAAC  | 90m        | Normal              | 2            |
|                         | 120m       |                     | 3            |
| 100mm <sup>2</sup> AAAC | 110m       |                     | 4            |
| 175mm <sup>2</sup> AAAC | 110m       |                     | 5            |
| 200mm <sup>2</sup> AAAC | 120m       |                     | 6            |
| 50mm <sup>2</sup> AAAC  | 75m        | High                | 7            |
|                         | 80m        |                     | 8            |
|                         | 100m       |                     | 9            |
| 100mm <sup>2</sup> AAAC | 80m        |                     | 10           |
|                         | 90m        |                     | 11           |
| 175mm <sup>2</sup> AAAC | 90m        |                     | 12           |
|                         | 100m       |                     | 13           |
| 200mm <sup>2</sup> AAAC | 90m        |                     | 14           |
|                         | 100m       |                     | 15           |

**Note:**

See Appendix 16 for details of historical renovation conductors

|                           |                 |                        |         |             |   |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|---|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |   |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 8 | <b>of</b> | 98 |

## Appendix 2 - 50mm AAAC (Hazel) - (90m Basic) Normal altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | HAZEL   |
| Greased Conductor Weight (kg/m)                    | 0.1648  |
| Cross Sectional Area of Conductor (mm²)            | 59.9    |
| Conductor Overall Diameter (mm)                    | 9.9     |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)         | 1800.8  |
| Basic / Recommended Span (m)                       | 90      |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 5       |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 720.3   |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 360.2   |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 641.0 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.378 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.771 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 397.1 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |       |                               |   |
|------------|-------|-------------------------------|---|
| Conductor: | HAZEL | Temperature Shift (Deg. C)    | 0 |
|            |       | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.07                              | 0.25                                        | 0.27  | 0.29  | 0.31  | 0.33  | 0.34  |
| 80             | 0.30                              | 1.00                                        | 1.08  | 1.16  | 1.23  | 1.31  | 1.38  |
| 120            | 0.67                              | 2.24                                        | 2.43  | 2.60  | 2.77  | 2.94  | 3.10  |
| 160            | 1.20                              | 3.99                                        | 4.31  | 4.62  | 4.93  | 5.22  | 5.51  |
| 200            | 1.87                              | 6.24                                        | 6.74  | 7.23  | 7.70  | 8.16  | 8.60  |
| 240            | 2.70                              | 8.98                                        | 9.70  | 10.40 | 11.09 | 11.75 | 12.39 |
| 280            | 3.67                              | 12.22                                       | 13.20 | 14.16 | 15.09 | 15.99 | 16.86 |
| 320            | 4.80                              | 15.96                                       | 17.25 | 18.50 | 19.71 | 20.89 | 22.02 |
| 360            | 6.07                              | 20.20                                       | 21.83 | 23.41 | 24.95 | 26.43 | 27.87 |
| 400            | 7.50                              | 24.94                                       | 26.95 | 28.90 | 30.80 | 32.63 | 34.41 |
| 440            | 9.07                              | 30.18                                       | 32.61 | 34.97 | 37.27 | 39.49 | 41.64 |
| 480            | 10.80                             | 35.92                                       | 38.80 | 41.62 | 44.35 | 46.99 | 49.55 |
| 520            | 12.67                             | 42.16                                       | 45.54 | 48.84 | 52.05 | 55.15 | 58.15 |

|                           |     |                        |         |             |   |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|---|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |   |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 9 | <b>of</b> | 98 |  |

## Appendix 2a - 50mm AAAC (Hazel) - (90m Basic) Normal altitude

### Common Design & Erection table

#### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | HAZEL                                                             |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 439.6            | 0.17                        | 0.23 | 0.30 | 0.38 | 0.47 | 0.57 | 0.67 | 0.79 | 1.52 |
| 0                 | 397.1            | 0.19                        | 0.25 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.88 | 1.68 |
| 5                 | 360.2            | 0.21                        | 0.28 | 0.37 | 0.46 | 0.57 | 0.69 | 0.82 | 0.97 | 1.85 |
| 10                | 324.6            | 0.23                        | 0.31 | 0.41 | 0.51 | 0.63 | 0.77 | 0.91 | 1.07 | 2.06 |
| 15                | 290.9            | 0.25                        | 0.35 | 0.45 | 0.57 | 0.71 | 0.86 | 1.02 | 1.20 | 2.29 |
| 20                | 259.4            | 0.29                        | 0.39 | 0.51 | 0.64 | 0.79 | 0.96 | 1.14 | 1.34 | 2.57 |
| 25                | 230.9            | 0.32                        | 0.44 | 0.57 | 0.72 | 0.89 | 1.08 | 1.28 | 1.51 | 2.89 |
| 30                | 205.6            | 0.36                        | 0.49 | 0.64 | 0.81 | 1.00 | 1.21 | 1.44 | 1.69 | 3.25 |
| 35                | 183.8            | 0.40                        | 0.55 | 0.72 | 0.91 | 1.12 | 1.36 | 1.61 | 1.89 | 3.63 |
| 40                | 165.4            | 0.45                        | 0.61 | 0.80 | 1.01 | 1.25 | 1.51 | 1.79 | 2.11 | 4.04 |
| 45                | 150.0            | 0.49                        | 0.67 | 0.88 | 1.11 | 1.37 | 1.66 | 1.98 | 2.32 | 4.45 |
| 50                | 137.2            | 0.54                        | 0.74 | 0.96 | 1.22 | 1.50 | 1.82 | 2.16 | 2.54 | 4.86 |
| 55                | 126.5            | 0.59                        | 0.80 | 1.04 | 1.32 | 1.63 | 1.97 | 2.34 | 2.75 | 5.27 |
| 60                | 117.6            | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 5.67 |
| 65                | 110.1            | 0.67                        | 0.92 | 1.20 | 1.52 | 1.87 | 2.26 | 2.70 | 3.16 | 6.06 |
| 70                | 103.6            | 0.72                        | 0.97 | 1.27 | 1.61 | 1.99 | 2.41 | 2.86 | 3.36 | 6.44 |
| 75                | 98.1             | 0.76                        | 1.03 | 1.34 | 1.70 | 2.10 | 2.54 | 3.03 | 3.55 | 6.81 |
| 80                | 93.2             | 0.80                        | 1.08 | 1.41 | 1.79 | 2.21 | 2.67 | 3.18 | 3.73 | 7.16 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 10 | <b>of</b> | 98 |

## Appendix 3 - 50mm AAAC (Hazel) - (120m Basic) Normal altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | HAZEL   |
| Greased Conductor Weight (kg/m)                    | 0.1648  |
| Cross Sectional Area of Conductor (mm²)            | 59.9    |
| Conductor Overall Diameter (mm)                    | 9.9     |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)         | 1800.8  |
| Basic / Recommended Span (m)                       | 120     |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 5       |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 720.3   |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 360.2   |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1288.7 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.968  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.279  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 786.7  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |       |                               |   |
|------------|-------|-------------------------------|---|
| Conductor: | HAZEL | Temperature Shift (Deg. C)    | 0 |
|            |       | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.21                                        | 0.22  | 0.24  | 0.25  | 0.26  | 0.27  |
| 80             | 0.30                              | 0.84                                        | 0.89  | 0.95  | 1.00  | 1.05  | 1.10  |
| 120            | 0.68                              | 1.89                                        | 2.01  | 2.13  | 2.25  | 2.36  | 2.47  |
| 160            | 1.22                              | 3.35                                        | 3.57  | 3.78  | 3.99  | 4.20  | 4.39  |
| 200            | 1.90                              | 5.24                                        | 5.58  | 5.91  | 6.24  | 6.55  | 6.86  |
| 240            | 2.73                              | 7.54                                        | 8.03  | 8.51  | 8.98  | 9.44  | 9.88  |
| 280            | 3.72                              | 10.27                                       | 10.94 | 11.59 | 12.23 | 12.85 | 13.45 |
| 320            | 4.86                              | 13.41                                       | 14.28 | 15.14 | 15.97 | 16.78 | 17.57 |
| 360            | 6.15                              | 16.98                                       | 18.08 | 19.16 | 20.21 | 21.24 | 22.24 |
| 400            | 7.60                              | 20.96                                       | 22.32 | 23.65 | 24.95 | 26.22 | 27.45 |
| 440            | 9.19                              | 25.36                                       | 27.01 | 28.62 | 30.19 | 31.73 | 33.22 |
| 480            | 10.94                             | 30.18                                       | 32.14 | 34.06 | 35.93 | 37.76 | 39.53 |
| 520            | 12.84                             | 35.42                                       | 37.72 | 39.97 | 42.17 | 44.31 | 46.40 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 11 | <b>of</b> | 98 |  |

## Appendix 3a - 50mm AAAC (Hazel) - (120m Basic) Normal altitude

### Common Design and erection Tables DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |       |                                |
|-----------------------------------------------|-------|--------------------------------|
| Conductor Code Name                           | HAZEL |                                |
| Basic / Recommended Span (m)                  | 120   |                                |
| Temperature Shift for Creep (Deg. C)          | 0     | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 0.0   | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 0     | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 433.9            | 0.17                        | 0.23 | 0.30 | 0.38 | 0.47 | 0.57 | 0.68 | 0.80 | 0.93 |
| 0                 | 394.1            | 0.19                        | 0.26 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.88 | 1.02 |
| 5                 | 360.2            | 0.21                        | 0.28 | 0.37 | 0.46 | 0.57 | 0.69 | 0.82 | 0.97 | 1.12 |
| 10                | 328.0            | 0.23                        | 0.31 | 0.40 | 0.51 | 0.63 | 0.76 | 0.90 | 1.06 | 1.23 |
| 15                | 298.1            | 0.25                        | 0.34 | 0.44 | 0.56 | 0.69 | 0.84 | 1.00 | 1.17 | 1.35 |
| 20                | 270.7            | 0.27                        | 0.37 | 0.49 | 0.62 | 0.76 | 0.92 | 1.10 | 1.29 | 1.49 |
| 25                | 246.1            | 0.30                        | 0.41 | 0.54 | 0.68 | 0.84 | 1.01 | 1.21 | 1.41 | 1.64 |
| 30                | 224.3            | 0.33                        | 0.45 | 0.59 | 0.74 | 0.92 | 1.11 | 1.32 | 1.55 | 1.80 |
| 35                | 205.4            | 0.36                        | 0.49 | 0.64 | 0.81 | 1.00 | 1.21 | 1.44 | 1.69 | 1.97 |
| 40                | 189.1            | 0.39                        | 0.53 | 0.70 | 0.88 | 1.09 | 1.32 | 1.57 | 1.84 | 2.14 |
| 45                | 175.0            | 0.42                        | 0.58 | 0.75 | 0.95 | 1.18 | 1.42 | 1.69 | 1.99 | 2.31 |
| 50                | 163.0            | 0.46                        | 0.62 | 0.81 | 1.02 | 1.26 | 1.53 | 1.82 | 2.14 | 2.48 |
| 55                | 152.6            | 0.49                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.28 | 2.65 |
| 60                | 143.6            | 0.52                        | 0.70 | 0.92 | 1.16 | 1.43 | 1.74 | 2.07 | 2.42 | 2.81 |
| 65                | 135.8            | 0.55                        | 0.74 | 0.97 | 1.23 | 1.52 | 1.83 | 2.18 | 2.56 | 2.97 |
| 70                | 129.0            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.60 | 1.93 | 2.30 | 2.70 | 3.13 |
| 75                | 123.0            | 0.60                        | 0.82 | 1.07 | 1.36 | 1.68 | 2.03 | 2.41 | 2.83 | 3.28 |
| 80                | 117.6            | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 3.43 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 12 | <b>of</b> | 98 |

## Appendix 4 - 100mm AAAC (Oak) - (110m Basic) Normal altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | OAK     |
| Greased Conductor Weight (kg/m)                    | 0.3273  |
| Cross Sectional Area of Conductor (mm²)            | 118.9   |
| Conductor Overall Diameter (mm)                    | 14      |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)         | 3576.1  |
| Basic / Recommended Span (m)                       | 110     |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 9.5     |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 1788.1  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 715.2   |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 714.9 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.378 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.771 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 394.1 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | OAK | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.23                                        | 0.24  | 0.26  | 0.27  | 0.29  | 0.30  |
| 80             | 0.30                              | 0.92                                        | 0.98  | 1.03  | 1.09  | 1.15  | 1.20  |
| 120            | 0.68                              | 2.06                                        | 2.20  | 2.33  | 2.46  | 2.58  | 2.71  |
| 160            | 1.21                              | 3.66                                        | 3.90  | 4.14  | 4.37  | 4.59  | 4.81  |
| 200            | 1.89                              | 5.72                                        | 6.10  | 6.47  | 6.83  | 7.18  | 7.52  |
| 240            | 2.72                              | 8.24                                        | 8.78  | 9.31  | 9.83  | 10.33 | 10.82 |
| 280            | 3.71                              | 11.21                                       | 11.95 | 12.67 | 13.38 | 14.06 | 14.73 |
| 320            | 4.84                              | 14.64                                       | 15.61 | 16.55 | 17.48 | 18.37 | 19.24 |
| 360            | 6.13                              | 18.53                                       | 19.76 | 20.95 | 22.12 | 23.25 | 24.35 |
| 400            | 7.56                              | 22.88                                       | 24.39 | 25.87 | 27.30 | 28.70 | 30.06 |
| 440            | 9.15                              | 27.68                                       | 29.51 | 31.30 | 33.04 | 34.73 | 36.37 |
| 480            | 10.89                             | 32.94                                       | 35.12 | 37.25 | 39.32 | 41.33 | 43.29 |
| 520            | 12.78                             | 38.66                                       | 41.22 | 43.71 | 46.15 | 48.51 | 50.80 |

|                           |                 |                        |         |             |    |           |    |  |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 13 | <b>of</b> | 98 |  |

## Appendix 4a - 100mm AAAC (Oak) - (110m Basic) Normal altitude

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | OAK                                                               |
| Basic / Recommended Span (m)                  | 110                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 865.7            | 0.17                        | 0.23 | 0.30 | 0.38 | 0.47 | 0.57 | 0.68 | 0.80 | 0.93 |
| 0                 | 784.7            | 0.19                        | 0.26 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.88 | 1.02 |
| 5                 | 715.2            | 0.21                        | 0.28 | 0.37 | 0.46 | 0.57 | 0.69 | 0.82 | 0.97 | 1.12 |
| 10                | 649.1            | 0.23                        | 0.31 | 0.40 | 0.51 | 0.63 | 0.76 | 0.91 | 1.07 | 1.24 |
| 15                | 587.2            | 0.25                        | 0.34 | 0.45 | 0.56 | 0.70 | 0.84 | 1.00 | 1.18 | 1.37 |
| 20                | 530.3            | 0.28                        | 0.38 | 0.49 | 0.62 | 0.77 | 0.93 | 1.11 | 1.30 | 1.51 |
| 25                | 479.0            | 0.31                        | 0.42 | 0.55 | 0.69 | 0.85 | 1.03 | 1.23 | 1.44 | 1.67 |
| 30                | 433.7            | 0.34                        | 0.46 | 0.60 | 0.76 | 0.94 | 1.14 | 1.36 | 1.59 | 1.85 |
| 35                | 394.4            | 0.37                        | 0.51 | 0.66 | 0.84 | 1.04 | 1.26 | 1.49 | 1.75 | 2.03 |
| 40                | 360.7            | 0.41                        | 0.56 | 0.73 | 0.92 | 1.13 | 1.37 | 1.63 | 1.92 | 2.22 |
| 45                | 332.0            | 0.44                        | 0.60 | 0.79 | 1.00 | 1.23 | 1.49 | 1.77 | 2.08 | 2.42 |
| 50                | 307.6            | 0.48                        | 0.65 | 0.85 | 1.08 | 1.33 | 1.61 | 1.92 | 2.25 | 2.61 |
| 55                | 286.8            | 0.51                        | 0.70 | 0.91 | 1.16 | 1.43 | 1.73 | 2.05 | 2.41 | 2.80 |
| 60                | 268.9            | 0.55                        | 0.75 | 0.97 | 1.23 | 1.52 | 1.84 | 2.19 | 2.57 | 2.98 |
| 65                | 253.5            | 0.58                        | 0.79 | 1.03 | 1.31 | 1.61 | 1.95 | 2.32 | 2.73 | 3.16 |
| 70                | 240.2            | 0.61                        | 0.83 | 1.09 | 1.38 | 1.70 | 2.06 | 2.45 | 2.88 | 3.34 |
| 75                | 228.4            | 0.64                        | 0.88 | 1.15 | 1.45 | 1.79 | 2.17 | 2.58 | 3.03 | 3.51 |
| 80                | 218.1            | 0.68                        | 0.92 | 1.20 | 1.52 | 1.88 | 2.27 | 2.70 | 3.17 | 3.68 |

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | OAK                                                               |
| Basic / Recommended Span (m)                  | 110                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 936.3            | 0.16                        | 0.21 | 0.28 | 0.35 | 0.44 | 0.53 | 0.63 | 0.74 | 0.86 |
| 0                 | 853.2            | 0.17                        | 0.23 | 0.31 | 0.39 | 0.48 | 0.58 | 0.69 | 0.81 | 0.94 |
| 5                 | 781.2            | 0.19                        | 0.26 | 0.34 | 0.42 | 0.52 | 0.63 | 0.75 | 0.89 | 1.03 |
| 10                | 711.9            | 0.21                        | 0.28 | 0.37 | 0.47 | 0.57 | 0.70 | 0.83 | 0.97 | 1.13 |
| 15                | 645.9            | 0.23                        | 0.31 | 0.41 | 0.51 | 0.63 | 0.77 | 0.91 | 1.07 | 1.24 |
| 20                | 584.3            | 0.25                        | 0.34 | 0.45 | 0.57 | 0.70 | 0.85 | 1.01 | 1.18 | 1.37 |
| 25                | 527.6            | 0.28                        | 0.38 | 0.50 | 0.63 | 0.78 | 0.94 | 1.12 | 1.31 | 1.52 |
| 30                | 476.6            | 0.31                        | 0.42 | 0.55 | 0.70 | 0.86 | 1.04 | 1.24 | 1.45 | 1.68 |
| 35                | 431.6            | 0.34                        | 0.46 | 0.61 | 0.77 | 0.95 | 1.15 | 1.36 | 1.60 | 1.86 |
| 40                | 392.6            | 0.38                        | 0.51 | 0.67 | 0.84 | 1.04 | 1.26 | 1.50 | 1.76 | 2.04 |
| 45                | 359.2            | 0.41                        | 0.56 | 0.73 | 0.92 | 1.14 | 1.38 | 1.64 | 1.92 | 2.23 |
| 50                | 330.7            | 0.45                        | 0.61 | 0.79 | 1.00 | 1.24 | 1.50 | 1.78 | 2.09 | 2.42 |
| 55                | 306.5            | 0.48                        | 0.65 | 0.85 | 1.08 | 1.33 | 1.62 | 1.92 | 2.26 | 2.62 |
| 60                | 285.8            | 0.52                        | 0.70 | 0.92 | 1.16 | 1.43 | 1.73 | 2.06 | 2.42 | 2.81 |
| 65                | 268.1            | 0.55                        | 0.75 | 0.98 | 1.24 | 1.53 | 1.85 | 2.20 | 2.58 | 2.99 |
| 70                | 252.8            | 0.58                        | 0.79 | 1.04 | 1.31 | 1.62 | 1.96 | 2.33 | 2.73 | 3.17 |
| 75                | 239.5            | 0.61                        | 0.84 | 1.09 | 1.38 | 1.71 | 2.07 | 2.46 | 2.89 | 3.35 |
| 80                | 227.9            | 0.65                        | 0.88 | 1.15 | 1.45 | 1.80 | 2.17 | 2.59 | 3.03 | 3.52 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 14 | <b>of</b> | 98 |

## Appendix 5 - 175mm AAAC (Elm) - (110m Basic) Normal altitude

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | ELM     |
| Greased Conductor Weight (kg/m)                    | 0.5922  |
| Cross Sectional Area of Conductor (mm²)            | 211     |
| Conductor Overall Diameter (mm)                    | 18.8    |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 5710.41 |
| Rated Breaking Strength of Conductor (kgf)         | 6346.7  |
| Basic / Recommended Span (m)                       | 110     |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 9.5     |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 2379.3  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 1269.3  |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 2024.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.363  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.465  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 1386.5 |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | ELM | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.23                                        | 0.24  | 0.26  | 0.27  | 0.28  | 0.30  |
| 80             | 0.31                              | 0.91                                        | 0.97  | 1.03  | 1.08  | 1.14  | 1.19  |
| 120            | 0.70                              | 2.04                                        | 2.18  | 2.31  | 2.44  | 2.56  | 2.68  |
| 160            | 1.24                              | 3.63                                        | 3.87  | 4.10  | 4.33  | 4.55  | 4.77  |
| 200            | 1.94                              | 5.67                                        | 6.05  | 6.41  | 6.77  | 7.12  | 7.45  |
| 240            | 2.80                              | 8.17                                        | 8.71  | 9.23  | 9.75  | 10.25 | 10.73 |
| 280            | 3.81                              | 11.12                                       | 11.85 | 12.57 | 13.27 | 13.95 | 14.61 |
| 320            | 4.98                              | 14.52                                       | 15.48 | 16.42 | 17.33 | 18.22 | 19.08 |
| 360            | 6.30                              | 18.38                                       | 19.59 | 20.78 | 21.93 | 23.06 | 24.15 |
| 400            | 7.78                              | 22.69                                       | 24.19 | 25.65 | 27.08 | 28.47 | 29.82 |
| 440            | 9.41                              | 27.46                                       | 29.27 | 31.04 | 32.76 | 34.44 | 36.08 |
| 480            | 11.20                             | 32.68                                       | 34.83 | 36.94 | 38.99 | 40.99 | 42.94 |
| 520            | 13.14                             | 38.35                                       | 40.88 | 43.35 | 45.76 | 48.11 | 50.39 |

|                           |                 |                        |         |             |    |           |    |  |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 15 | <b>of</b> | 98 |  |

## Appendix 5a - 175mm AAAC (Elm) - (110m Basic) Normal altitude

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | ELM                                                               |
| Basic / Recommended Span (m)                  | 110                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 1522.8           | 0.18                        | 0.24 | 0.31 | 0.39 | 0.49 | 0.59 | 0.70 | 0.82 | 0.95 |
| 0                 | 1386.5           | 0.19                        | 0.26 | 0.34 | 0.43 | 0.53 | 0.65 | 0.77 | 0.90 | 1.05 |
| 5                 | 1269.3           | 0.21                        | 0.29 | 0.37 | 0.47 | 0.58 | 0.71 | 0.84 | 0.99 | 1.14 |
| 10                | 1157.6           | 0.23                        | 0.31 | 0.41 | 0.52 | 0.64 | 0.77 | 0.92 | 1.08 | 1.25 |
| 15                | 1052.4           | 0.25                        | 0.34 | 0.45 | 0.57 | 0.70 | 0.85 | 1.01 | 1.19 | 1.38 |
| 20                | 955.0            | 0.28                        | 0.38 | 0.50 | 0.63 | 0.78 | 0.94 | 1.12 | 1.31 | 1.52 |
| 25                | 866.6            | 0.31                        | 0.42 | 0.55 | 0.69 | 0.85 | 1.03 | 1.23 | 1.44 | 1.67 |
| 30                | 787.7            | 0.34                        | 0.46 | 0.60 | 0.76 | 0.94 | 1.14 | 1.35 | 1.59 | 1.84 |
| 35                | 718.5            | 0.37                        | 0.50 | 0.66 | 0.83 | 1.03 | 1.25 | 1.48 | 1.74 | 2.02 |
| 40                | 658.5            | 0.40                        | 0.55 | 0.72 | 0.91 | 1.12 | 1.36 | 1.62 | 1.90 | 2.20 |
| 45                | 606.9            | 0.44                        | 0.60 | 0.78 | 0.99 | 1.22 | 1.48 | 1.76 | 2.06 | 2.39 |
| 50                | 562.8            | 0.47                        | 0.64 | 0.84 | 1.07 | 1.32 | 1.59 | 1.89 | 2.22 | 2.58 |
| 55                | 524.9            | 0.51                        | 0.69 | 0.90 | 1.14 | 1.41 | 1.71 | 2.03 | 2.38 | 2.76 |
| 60                | 492.2            | 0.54                        | 0.74 | 0.96 | 1.22 | 1.50 | 1.82 | 2.17 | 2.54 | 2.95 |
| 65                | 464.0            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.60 | 1.93 | 2.30 | 2.70 | 3.13 |
| 70                | 439.4            | 0.61                        | 0.83 | 1.08 | 1.36 | 1.68 | 2.04 | 2.43 | 2.85 | 3.30 |
| 75                | 417.8            | 0.64                        | 0.87 | 1.13 | 1.44 | 1.77 | 2.14 | 2.55 | 2.99 | 3.47 |
| 80                | 398.7            | 0.67                        | 0.91 | 1.19 | 1.50 | 1.86 | 2.25 | 2.67 | 3.14 | 3.64 |

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | ELM                                                               |
| Basic / Recommended Span (m)                  | 110                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 1647.9           | 0.16                        | 0.22 | 0.29 | 0.36 | 0.45 | 0.54 | 0.65 | 0.76 | 0.88 |
| 0                 | 1508.0           | 0.18                        | 0.24 | 0.31 | 0.40 | 0.49 | 0.59 | 0.71 | 0.83 | 0.96 |
| 5                 | 1386.6           | 0.19                        | 0.26 | 0.34 | 0.43 | 0.53 | 0.65 | 0.77 | 0.90 | 1.05 |
| 10                | 1269.4           | 0.21                        | 0.29 | 0.37 | 0.47 | 0.58 | 0.71 | 0.84 | 0.99 | 1.14 |
| 15                | 1157.6           | 0.23                        | 0.31 | 0.41 | 0.52 | 0.64 | 0.77 | 0.92 | 1.08 | 1.25 |
| 20                | 1052.4           | 0.25                        | 0.34 | 0.45 | 0.57 | 0.70 | 0.85 | 1.01 | 1.19 | 1.38 |
| 25                | 955.1            | 0.28                        | 0.38 | 0.50 | 0.63 | 0.78 | 0.94 | 1.12 | 1.31 | 1.52 |
| 30                | 866.6            | 0.31                        | 0.42 | 0.55 | 0.69 | 0.85 | 1.03 | 1.23 | 1.44 | 1.67 |
| 35                | 787.7            | 0.34                        | 0.46 | 0.60 | 0.76 | 0.94 | 1.14 | 1.35 | 1.59 | 1.84 |
| 40                | 718.5            | 0.37                        | 0.50 | 0.66 | 0.83 | 1.03 | 1.25 | 1.48 | 1.74 | 2.02 |
| 45                | 658.5            | 0.40                        | 0.55 | 0.72 | 0.91 | 1.12 | 1.36 | 1.62 | 1.90 | 2.20 |
| 50                | 607.0            | 0.44                        | 0.60 | 0.78 | 0.99 | 1.22 | 1.48 | 1.76 | 2.06 | 2.39 |
| 55                | 562.8            | 0.47                        | 0.64 | 0.84 | 1.07 | 1.32 | 1.59 | 1.89 | 2.22 | 2.58 |
| 60                | 524.9            | 0.51                        | 0.69 | 0.90 | 1.14 | 1.41 | 1.71 | 2.03 | 2.38 | 2.76 |
| 65                | 492.2            | 0.54                        | 0.74 | 0.96 | 1.22 | 1.50 | 1.82 | 2.17 | 2.54 | 2.95 |
| 70                | 464.0            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.60 | 1.93 | 2.30 | 2.70 | 3.13 |
| 75                | 439.4            | 0.61                        | 0.83 | 1.08 | 1.36 | 1.68 | 2.04 | 2.43 | 2.85 | 3.30 |
| 80                | 417.8            | 0.64                        | 0.87 | 1.13 | 1.44 | 1.77 | 2.14 | 2.55 | 2.99 | 3.47 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 16 | <b>of</b> | 98 |

## Appendix 6 - 200mm AAAC (Poplar) - (120m Basic) Normal altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | POPLAR  |
| Greased Conductor Weight (kg/m)                    | 0.68992 |
| Cross Sectional Area of Conductor (mm²)            | 239.4   |
| Conductor Overall Diameter (mm)                    | 20.1    |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 5700    |
| Rated Breaking Strength of Conductor (kgf)         | 7200.2  |
| Basic / Recommended Span (m)                       | 120     |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 9.5     |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 2379.3  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 1440.04 |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 2291.7 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.496  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.515  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 1567.6 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |        |                               |    |
|------------|--------|-------------------------------|----|
| Conductor: | POPLAR | Temperature Shift (Deg. C)    | 0  |
|            |        | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.22                                        | 0.23  | 0.24  | 0.26  | 0.27  | 0.28  |
| 80             | 0.32                              | 0.87                                        | 0.93  | 0.98  | 1.03  | 1.08  | 1.12  |
| 120            | 0.72                              | 1.96                                        | 2.08  | 2.20  | 2.31  | 2.42  | 2.53  |
| 160            | 1.29                              | 3.49                                        | 3.70  | 3.91  | 4.11  | 4.31  | 4.50  |
| 200            | 2.01                              | 5.46                                        | 5.79  | 6.11  | 6.42  | 6.73  | 7.03  |
| 240            | 2.89                              | 7.86                                        | 8.33  | 8.80  | 9.25  | 9.69  | 10.12 |
| 280            | 3.94                              | 10.70                                       | 11.34 | 11.97 | 12.59 | 13.19 | 13.78 |
| 320            | 5.14                              | 13.97                                       | 14.81 | 15.64 | 16.45 | 17.23 | 18.00 |
| 360            | 6.51                              | 17.68                                       | 18.75 | 19.79 | 20.81 | 21.81 | 22.78 |
| 400            | 8.04                              | 21.83                                       | 23.15 | 24.44 | 25.70 | 26.92 | 28.12 |
| 440            | 9.72                              | 26.42                                       | 28.01 | 29.57 | 31.09 | 32.58 | 34.03 |
| 480            | 11.57                             | 31.44                                       | 33.33 | 35.19 | 37.00 | 38.77 | 40.50 |
| 520            | 13.58                             | 36.90                                       | 39.12 | 41.30 | 43.43 | 45.50 | 47.53 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 17 | <b>of</b> | 98 |  |

## Appendix 6a - 200mm AAAC (Poplar) - (120m Basic) Normal altitude

Conductor Code Name **POPLAR**  
 Basic / Recommended Span (m) **120**  
 Temperature Shift for Creep (Deg. C) **0** Insert minus sign as necessary  
 Equivalent Percentage Increase in Tension (%) **0.0** at 15°C  
 Required Percentage Increase in Tension (%) **0** at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 1717.0           | 0.18                        | 0.25 | 0.32 | 0.41 | 0.50 | 0.61 | 0.72 | 0.85 | 0.98 |
| 0                 | 1567.6           | 0.20                        | 0.27 | 0.35 | 0.45 | 0.55 | 0.67 | 0.79 | 0.93 | 1.08 |
| 5                 | 1440.0           | 0.22                        | 0.29 | 0.38 | 0.49 | 0.60 | 0.72 | 0.86 | 1.01 | 1.17 |
| 10                | 1319.1           | 0.24                        | 0.32 | 0.42 | 0.53 | 0.65 | 0.79 | 0.94 | 1.10 | 1.28 |
| 15                | 1206.1           | 0.26                        | 0.35 | 0.46 | 0.58 | 0.72 | 0.87 | 1.03 | 1.21 | 1.40 |
| 20                | 1102.2           | 0.28                        | 0.38 | 0.50 | 0.63 | 0.78 | 0.95 | 1.13 | 1.32 | 1.53 |
| 25                | 1008.0           | 0.31                        | 0.42 | 0.55 | 0.69 | 0.86 | 1.04 | 1.23 | 1.45 | 1.68 |
| 30                | 924.0            | 0.34                        | 0.46 | 0.60 | 0.76 | 0.93 | 1.13 | 1.34 | 1.58 | 1.83 |
| 35                | 850.0            | 0.37                        | 0.50 | 0.65 | 0.82 | 1.01 | 1.23 | 1.46 | 1.71 | 1.99 |
| 40                | 785.4            | 0.40                        | 0.54 | 0.70 | 0.89 | 1.10 | 1.33 | 1.58 | 1.86 | 2.15 |
| 45                | 729.3            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.18 | 1.43 | 1.70 | 2.00 | 2.32 |
| 50                | 680.7            | 0.46                        | 0.62 | 0.81 | 1.03 | 1.27 | 1.53 | 1.82 | 2.14 | 2.48 |
| 55                | 638.6            | 0.49                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.28 | 2.65 |
| 60                | 601.8            | 0.52                        | 0.70 | 0.92 | 1.16 | 1.43 | 1.73 | 2.06 | 2.42 | 2.81 |
| 65                | 569.7            | 0.54                        | 0.74 | 0.97 | 1.23 | 1.51 | 1.83 | 2.18 | 2.56 | 2.97 |
| 70                | 541.4            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.59 | 1.93 | 2.29 | 2.69 | 3.12 |
| 75                | 516.4            | 0.60                        | 0.82 | 1.07 | 1.35 | 1.67 | 2.02 | 2.40 | 2.82 | 3.27 |
| 80                | 494.2            | 0.63                        | 0.86 | 1.12 | 1.41 | 1.75 | 2.11 | 2.51 | 2.95 | 3.42 |

Conductor Code Name **POPLAR**  
 Basic / Recommended Span (m) **120**  
 Temperature Shift for Creep (Deg. C) **0** Insert minus sign as necessary  
 Equivalent Percentage Increase in Tension (%) **10.0** at 15°C  
 Required Percentage Increase in Tension (%) **10** at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 1864.1           | 0.17                        | 0.23 | 0.30 | 0.37 | 0.46 | 0.56 | 0.67 | 0.78 | 0.91 |
| 0                 | 1709.5           | 0.18                        | 0.25 | 0.32 | 0.41 | 0.50 | 0.61 | 0.73 | 0.85 | 0.99 |
| 5                 | 1576.1           | 0.20                        | 0.27 | 0.35 | 0.44 | 0.55 | 0.66 | 0.79 | 0.92 | 1.07 |
| 10                | 1448.1           | 0.21                        | 0.29 | 0.38 | 0.48 | 0.60 | 0.72 | 0.86 | 1.01 | 1.17 |
| 15                | 1326.8           | 0.23                        | 0.32 | 0.42 | 0.53 | 0.65 | 0.79 | 0.94 | 1.10 | 1.27 |
| 20                | 1213.2           | 0.26                        | 0.35 | 0.45 | 0.58 | 0.71 | 0.86 | 1.02 | 1.20 | 1.39 |
| 25                | 1108.6           | 0.28                        | 0.38 | 0.50 | 0.63 | 0.78 | 0.94 | 1.12 | 1.31 | 1.52 |
| 30                | 1013.8           | 0.31                        | 0.42 | 0.54 | 0.69 | 0.85 | 1.03 | 1.22 | 1.44 | 1.67 |
| 35                | 929.1            | 0.33                        | 0.45 | 0.59 | 0.75 | 0.93 | 1.12 | 1.34 | 1.57 | 1.82 |
| 40                | 854.5            | 0.36                        | 0.49 | 0.65 | 0.82 | 1.01 | 1.22 | 1.45 | 1.71 | 1.98 |
| 45                | 789.3            | 0.39                        | 0.54 | 0.70 | 0.88 | 1.09 | 1.32 | 1.57 | 1.85 | 2.14 |
| 50                | 732.7            | 0.42                        | 0.58 | 0.75 | 0.95 | 1.18 | 1.42 | 1.69 | 1.99 | 2.31 |
| 55                | 683.7            | 0.45                        | 0.62 | 0.81 | 1.02 | 1.26 | 1.53 | 1.82 | 2.13 | 2.47 |
| 60                | 641.1            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.27 | 2.64 |
| 65                | 604.1            | 0.51                        | 0.70 | 0.91 | 1.16 | 1.43 | 1.73 | 2.06 | 2.41 | 2.80 |
| 70                | 571.7            | 0.54                        | 0.74 | 0.97 | 1.22 | 1.51 | 1.83 | 2.17 | 2.55 | 2.96 |
| 75                | 543.2            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.59 | 1.92 | 2.29 | 2.68 | 3.11 |
| 80                | 518.0            | 0.60                        | 0.82 | 1.07 | 1.35 | 1.67 | 2.01 | 2.40 | 2.81 | 3.26 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 18 | <b>of</b> | 98 |

## Appendix 7 - 50mm AAAC (Hazel) - (75m Basic) High altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |         |
|------------------------------------------------------|---------|
| Conductor Code Name (if any)                         | HAZEL   |
| Greased Conductor Weight (kg/m)                      | 0.1648  |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 59.9    |
| Conductor Overall Diameter (mm)                      | 9.9     |
| Coefficient of Linear Expansion (/Degree C)          | 2.3E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)           | 1800.8  |
| Basic / Recommended Span (m)                         | 75      |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 380     |
| Radial Ice Thickness (mm)                            | 5       |
| Ice Density (kg/m <sup>3</sup> )                     | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 720.3   |
| Temperature at MWT Limit (Degrees C)                 | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 360.2   |
| Temperature at EDT Limit (Degrees C)                 | 5       |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 602.2 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.378 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.771 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 398.4 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |       |                               |   |
|------------|-------|-------------------------------|---|
| Conductor: | HAZEL | Temperature Shift (Deg. C)    | 0 |
|            |       | Tension Reduction at 15°C (%) | 0 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.07                              | 0.27                                        | 0.29  | 0.32  | 0.34  | 0.37  | 0.39  |
| 80             | 0.30                              | 1.08                                        | 1.18  | 1.28  | 1.38  | 1.47  | 1.56  |
| 120            | 0.67                              | 2.42                                        | 2.65  | 2.88  | 3.10  | 3.31  | 3.52  |
| 160            | 1.19                              | 4.31                                        | 4.72  | 5.12  | 5.51  | 5.89  | 6.25  |
| 200            | 1.86                              | 6.73                                        | 7.37  | 8.00  | 8.61  | 9.20  | 9.76  |
| 240            | 2.68                              | 9.69                                        | 10.62 | 11.52 | 12.40 | 13.25 | 14.06 |
| 280            | 3.65                              | 13.19                                       | 14.45 | 15.69 | 16.88 | 18.03 | 19.14 |
| 320            | 4.77                              | 17.23                                       | 18.88 | 20.49 | 22.04 | 23.55 | 25.00 |
| 360            | 6.04                              | 21.80                                       | 23.89 | 25.93 | 27.90 | 29.80 | 31.64 |
| 400            | 7.46                              | 26.92                                       | 29.50 | 32.01 | 34.44 | 36.79 | 39.06 |
| 440            | 9.02                              | 32.57                                       | 35.69 | 38.73 | 41.68 | 44.52 | 47.26 |
| 480            | 10.74                             | 38.76                                       | 42.47 | 46.09 | 49.60 | 52.98 | 56.24 |
| 520            | 12.60                             | 45.49                                       | 49.85 | 54.10 | 58.21 | 62.18 | 66.01 |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 19 | of | 98 |  |

## Appendix 7a - 50mm AAAC (Hazel) - (75m Basic) - High altitude

### Common Design & Erection Table

#### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |       |                                |
|-----------------------------------------------|-------|--------------------------------|
| Conductor Code Name                           | HAZEL |                                |
| Basic / Recommended Span (m)                  | 75    |                                |
| Temperature Shift for Creep (Deg. C)          | 0     | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 5.0   | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 5     | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 458.2            | 0.16                        | 0.22 | 0.29 | 0.36 | 0.45 | 0.54 | 0.65 | 0.76 | 0.88 |
| 0                 | 414.2            | 0.18                        | 0.24 | 0.32 | 0.40 | 0.50 | 0.60 | 0.72 | 0.84 | 0.97 |
| 5                 | 375.7            | 0.20                        | 0.27 | 0.35 | 0.44 | 0.55 | 0.66 | 0.79 | 0.93 | 1.07 |
| 10                | 338.1            | 0.22                        | 0.30 | 0.39 | 0.49 | 0.61 | 0.74 | 0.88 | 1.03 | 1.19 |
| 15                | 301.7            | 0.25                        | 0.33 | 0.44 | 0.55 | 0.68 | 0.83 | 0.98 | 1.15 | 1.34 |
| 20                | 267.2            | 0.28                        | 0.38 | 0.49 | 0.62 | 0.77 | 0.93 | 1.11 | 1.30 | 1.51 |
| 25                | 235.1            | 0.32                        | 0.43 | 0.56 | 0.71 | 0.88 | 1.06 | 1.26 | 1.48 | 1.72 |
| 30                | 206.2            | 0.36                        | 0.49 | 0.64 | 0.81 | 1.00 | 1.21 | 1.44 | 1.69 | 1.96 |
| 35                | 180.9            | 0.41                        | 0.56 | 0.73 | 0.92 | 1.14 | 1.38 | 1.64 | 1.92 | 2.23 |
| 40                | 159.5            | 0.46                        | 0.63 | 0.83 | 1.05 | 1.29 | 1.56 | 1.86 | 2.18 | 2.53 |
| 45                | 141.9            | 0.52                        | 0.71 | 0.93 | 1.18 | 1.45 | 1.76 | 2.09 | 2.45 | 2.85 |
| 50                | 127.5            | 0.58                        | 0.79 | 1.03 | 1.31 | 1.62 | 1.95 | 2.33 | 2.73 | 3.17 |
| 55                | 115.9            | 0.64                        | 0.87 | 1.14 | 1.44 | 1.78 | 2.15 | 2.56 | 3.00 | 3.48 |
| 60                | 106.4            | 0.70                        | 0.95 | 1.24 | 1.57 | 1.94 | 2.34 | 2.79 | 3.27 | 3.80 |
| 65                | 98.5             | 0.75                        | 1.02 | 1.34 | 1.69 | 2.09 | 2.53 | 3.01 | 3.53 | 4.10 |
| 70                | 92.0             | 0.81                        | 1.10 | 1.43 | 1.81 | 2.24 | 2.71 | 3.23 | 3.79 | 4.39 |
| 75                | 86.4             | 0.86                        | 1.17 | 1.53 | 1.93 | 2.38 | 2.88 | 3.43 | 4.03 | 4.67 |
| 80                | 81.7             | 0.91                        | 1.24 | 1.61 | 2.04 | 2.52 | 3.05 | 3.63 | 4.26 | 4.94 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 20 | <b>of</b> | 98 |

## Appendix 8 - 50mm AAAC (Hazel) - (80m Basic) High altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | HAZEL   |
| Greased Conductor Weight (kg/m)                    | 0.1648  |
| Cross Sectional Area of Conductor (mm²)            | 59.9    |
| Conductor Overall Diameter (mm)                    | 9.9     |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)         | 1800.8  |
| Basic / Recommended Span (m)                       | 80      |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 5       |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 720.3   |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 360.2   |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 615.2 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.378 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.771 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 398.0 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |       |                               |   |
|------------|-------|-------------------------------|---|
| Conductor: | HAZEL | Temperature Shift (Deg. C)    | 0 |
|            |       | Tension Reduction at 15°C (%) | 0 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.07                              | 0.26                                        | 0.28  | 0.31  | 0.33  | 0.35  | 0.37  |
| 80             | 0.30                              | 1.03                                        | 1.13  | 1.22  | 1.31  | 1.40  | 1.48  |
| 120            | 0.67                              | 2.33                                        | 2.54  | 2.75  | 2.95  | 3.14  | 3.33  |
| 160            | 1.20                              | 4.14                                        | 4.51  | 4.88  | 5.24  | 5.59  | 5.92  |
| 200            | 1.87                              | 6.46                                        | 7.05  | 7.63  | 8.19  | 8.73  | 9.25  |
| 240            | 2.69                              | 9.30                                        | 10.16 | 10.99 | 11.79 | 12.57 | 13.33 |
| 280            | 3.66                              | 12.66                                       | 13.82 | 14.96 | 16.05 | 17.11 | 18.14 |
| 320            | 4.78                              | 16.54                                       | 18.06 | 19.53 | 20.97 | 22.35 | 23.69 |
| 360            | 6.05                              | 20.93                                       | 22.85 | 24.72 | 26.54 | 28.29 | 29.98 |
| 400            | 7.47                              | 25.84                                       | 28.21 | 30.52 | 32.76 | 34.93 | 37.02 |
| 440            | 9.04                              | 31.27                                       | 34.14 | 36.93 | 39.64 | 42.26 | 44.79 |
| 480            | 10.76                             | 37.22                                       | 40.63 | 43.95 | 47.18 | 50.30 | 53.30 |
| 520            | 12.62                             | 43.68                                       | 47.68 | 51.58 | 55.37 | 59.03 | 62.56 |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 21 | of | 98 |  |

## Appendix 8a - 50mm AAAC (Hazel) - (80m Basic) High altitude

### Common Design & Erection Table

#### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |       |                                |
|-----------------------------------------------|-------|--------------------------------|
| Conductor Code Name                           | HAZEL |                                |
| Basic / Recommended Span (m)                  | 80    |                                |
| Temperature Shift for Creep (Deg. C)          | 0     | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 5.0   | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 5     | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 457.7            | 0.16                        | 0.22 | 0.29 | 0.36 | 0.45 | 0.54 | 0.65 | 0.76 | 0.88 |
| 0                 | 414.1            | 0.18                        | 0.24 | 0.32 | 0.40 | 0.50 | 0.60 | 0.72 | 0.84 | 0.98 |
| 5                 | 375.9            | 0.20                        | 0.27 | 0.35 | 0.44 | 0.55 | 0.66 | 0.79 | 0.93 | 1.07 |
| 10                | 338.7            | 0.22                        | 0.30 | 0.39 | 0.49 | 0.61 | 0.74 | 0.88 | 1.03 | 1.19 |
| 15                | 302.9            | 0.24                        | 0.33 | 0.44 | 0.55 | 0.68 | 0.82 | 0.98 | 1.15 | 1.33 |
| 20                | 269.1            | 0.28                        | 0.38 | 0.49 | 0.62 | 0.77 | 0.93 | 1.10 | 1.29 | 1.50 |
| 25                | 237.8            | 0.31                        | 0.42 | 0.55 | 0.70 | 0.87 | 1.05 | 1.25 | 1.46 | 1.70 |
| 30                | 209.6            | 0.35                        | 0.48 | 0.63 | 0.80 | 0.98 | 1.19 | 1.42 | 1.66 | 1.93 |
| 35                | 185.0            | 0.40                        | 0.55 | 0.71 | 0.90 | 1.11 | 1.35 | 1.60 | 1.88 | 2.18 |
| 40                | 164.1            | 0.45                        | 0.61 | 0.80 | 1.02 | 1.25 | 1.52 | 1.81 | 2.12 | 2.46 |
| 45                | 146.9            | 0.50                        | 0.69 | 0.90 | 1.14 | 1.40 | 1.70 | 2.02 | 2.37 | 2.75 |
| 50                | 132.7            | 0.56                        | 0.76 | 0.99 | 1.26 | 1.55 | 1.88 | 2.24 | 2.62 | 3.04 |
| 55                | 121.1            | 0.61                        | 0.83 | 1.09 | 1.38 | 1.70 | 2.06 | 2.45 | 2.88 | 3.34 |
| 60                | 111.5            | 0.67                        | 0.91 | 1.18 | 1.50 | 1.85 | 2.24 | 2.66 | 3.12 | 3.62 |
| 65                | 103.5            | 0.72                        | 0.97 | 1.27 | 1.61 | 1.99 | 2.41 | 2.87 | 3.36 | 3.90 |
| 70                | 96.9             | 0.77                        | 1.04 | 1.36 | 1.72 | 2.13 | 2.57 | 3.06 | 3.59 | 4.17 |
| 75                | 91.2             | 0.81                        | 1.11 | 1.45 | 1.83 | 2.26 | 2.73 | 3.25 | 3.82 | 4.43 |
| 80                | 86.3             | 0.86                        | 1.17 | 1.53 | 1.93 | 2.39 | 2.89 | 3.44 | 4.03 | 4.68 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 22 | <b>of</b> | 98 |

## Appendix 9 - 50mm AAAC (Hazel) - (100m Basic) High altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | HAZEL   |
| Greased Conductor Weight (kg/m)                    | 0.1648  |
| Cross Sectional Area of Conductor (mm²)            | 59.9    |
| Conductor Overall Diameter (mm)                    | 9.9     |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)         | 1800.8  |
| Basic / Recommended Span (m)                       | 100     |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 5       |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 720.3   |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 360.2   |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 666.2 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.378 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.771 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 396.2 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |       |                               |   |
|------------|-------|-------------------------------|---|
| Conductor: | HAZEL | Temperature Shift (Deg. C)    | 0 |
|            |       | Tension Reduction at 15°C (%) | 0 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.23                                        | 0.24  | 0.26  | 0.28  | 0.29  | 0.31  |
| 80             | 0.30                              | 0.90                                        | 0.97  | 1.04  | 1.11  | 1.17  | 1.24  |
| 120            | 0.68                              | 2.03                                        | 2.19  | 2.34  | 2.49  | 2.64  | 2.78  |
| 160            | 1.20                              | 3.60                                        | 3.89  | 4.16  | 4.43  | 4.69  | 4.94  |
| 200            | 1.88                              | 5.63                                        | 6.07  | 6.50  | 6.92  | 7.33  | 7.72  |
| 240            | 2.71                              | 8.11                                        | 8.74  | 9.36  | 9.97  | 10.55 | 11.12 |
| 280            | 3.69                              | 11.04                                       | 11.90 | 12.74 | 13.56 | 14.36 | 15.14 |
| 320            | 4.82                              | 14.42                                       | 15.54 | 16.64 | 17.72 | 18.76 | 19.77 |
| 360            | 6.10                              | 18.25                                       | 19.67 | 21.07 | 22.42 | 23.74 | 25.02 |
| 400            | 7.53                              | 22.53                                       | 24.29 | 26.01 | 27.68 | 29.31 | 30.89 |
| 440            | 9.11                              | 27.26                                       | 29.39 | 31.47 | 33.50 | 35.47 | 37.38 |
| 480            | 10.84                             | 32.44                                       | 34.97 | 37.45 | 39.86 | 42.21 | 44.48 |
| 520            | 12.72                             | 38.08                                       | 41.04 | 43.95 | 46.78 | 49.53 | 52.20 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 23 | <b>of</b> | 98 |  |

## Appendix 9a - 50mm AAAC (Hazel) - (100m Basic) High altitude

### Common Design & Erection Table

#### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |       |                                |
|-----------------------------------------------|-------|--------------------------------|
| Conductor Code Name                           | HAZEL |                                |
| Basic / Recommended Span (m)                  | 100   |                                |
| Temperature Shift for Creep (Deg. C)          | 0     | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 5.0   | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 5     | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 455.3            | 0.16                        | 0.22 | 0.29 | 0.37 | 0.45 | 0.55 | 0.65 | 0.76 | 0.89 |
| 0                 | 413.1            | 0.18                        | 0.24 | 0.32 | 0.40 | 0.50 | 0.60 | 0.72 | 0.84 | 0.98 |
| 5                 | 376.6            | 0.20                        | 0.27 | 0.35 | 0.44 | 0.55 | 0.66 | 0.79 | 0.92 | 1.07 |
| 10                | 341.4            | 0.22                        | 0.30 | 0.39 | 0.49 | 0.60 | 0.73 | 0.87 | 1.02 | 1.18 |
| 15                | 307.9            | 0.24                        | 0.33 | 0.43 | 0.54 | 0.67 | 0.81 | 0.96 | 1.13 | 1.31 |
| 20                | 276.7            | 0.27                        | 0.36 | 0.48 | 0.60 | 0.74 | 0.90 | 1.07 | 1.26 | 1.46 |
| 25                | 248.2            | 0.30                        | 0.41 | 0.53 | 0.67 | 0.83 | 1.00 | 1.20 | 1.40 | 1.63 |
| 30                | 222.8            | 0.33                        | 0.45 | 0.59 | 0.75 | 0.92 | 1.12 | 1.33 | 1.56 | 1.81 |
| 35                | 200.5            | 0.37                        | 0.50 | 0.66 | 0.83 | 1.03 | 1.24 | 1.48 | 1.74 | 2.01 |
| 40                | 181.5            | 0.41                        | 0.56 | 0.73 | 0.92 | 1.14 | 1.37 | 1.63 | 1.92 | 2.22 |
| 45                | 165.4            | 0.45                        | 0.61 | 0.80 | 1.01 | 1.25 | 1.51 | 1.79 | 2.11 | 2.44 |
| 50                | 151.8            | 0.49                        | 0.66 | 0.87 | 1.10 | 1.36 | 1.64 | 1.95 | 2.29 | 2.66 |
| 55                | 140.4            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.78 | 2.11 | 2.48 | 2.88 |
| 60                | 130.7            | 0.57                        | 0.77 | 1.01 | 1.28 | 1.58 | 1.91 | 2.27 | 2.66 | 3.09 |
| 65                | 122.5            | 0.61                        | 0.82 | 1.08 | 1.36 | 1.68 | 2.04 | 2.42 | 2.84 | 3.30 |
| 70                | 115.4            | 0.64                        | 0.87 | 1.14 | 1.45 | 1.79 | 2.16 | 2.57 | 3.02 | 3.50 |
| 75                | 109.3            | 0.68                        | 0.92 | 1.21 | 1.53 | 1.89 | 2.28 | 2.72 | 3.19 | 3.70 |
| 80                | 103.9            | 0.71                        | 0.97 | 1.27 | 1.61 | 1.98 | 2.40 | 2.85 | 3.35 | 3.89 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 24 | <b>of</b> | 98 |

## Appendix 10 - 100mm AAAC (Oak) -(80m Basic) High altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | OAK     |
| Greased Conductor Weight (kg/m)                    | 0.3273  |
| Cross Sectional Area of Conductor (mm²)            | 118.9   |
| Conductor Overall Diameter (mm)                    | 14      |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)         | 3576.1  |
| Basic / Recommended Span (m)                       | 80      |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 9.5     |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 1788.1  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 715.2   |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1193.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.968  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.279  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 790.3  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | OAK | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.07                              | 0.28                                        | 0.30  | 0.32  | 0.35  | 0.37  | 0.39  |
| 80             | 0.30                              | 1.12                                        | 1.21  | 1.30  | 1.39  | 1.47  | 1.55  |
| 120            | 0.67                              | 2.51                                        | 2.72  | 2.92  | 3.12  | 3.31  | 3.49  |
| 160            | 1.20                              | 4.47                                        | 4.84  | 5.20  | 5.54  | 5.88  | 6.20  |
| 200            | 1.87                              | 6.98                                        | 7.56  | 8.12  | 8.66  | 9.19  | 9.69  |
| 240            | 2.69                              | 10.05                                       | 10.88 | 11.69 | 12.48 | 13.23 | 13.96 |
| 280            | 3.66                              | 13.68                                       | 14.81 | 15.91 | 16.98 | 18.01 | 19.00 |
| 320            | 4.78                              | 17.86                                       | 19.35 | 20.79 | 22.18 | 23.52 | 24.82 |
| 360            | 6.05                              | 22.61                                       | 24.49 | 26.31 | 28.07 | 29.77 | 31.41 |
| 400            | 7.47                              | 27.91                                       | 30.23 | 32.48 | 34.65 | 36.75 | 38.78 |
| 440            | 9.04                              | 33.77                                       | 36.58 | 39.30 | 41.93 | 44.47 | 46.92 |
| 480            | 10.76                             | 40.19                                       | 43.53 | 46.77 | 49.90 | 52.92 | 55.84 |
| 520            | 12.62                             | 47.17                                       | 51.09 | 54.89 | 58.56 | 62.11 | 65.53 |

|                           |                 |                        |         |             |    |           |    |  |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 25 | <b>of</b> | 98 |  |

## Appendix 10a - 100mm AAAC (Oak) - (80m Basic) High altitude

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | OAK                                                               |
| Basic / Recommended Span (m)                  | 80                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 876.3            | 0.17                        | 0.23 | 0.30 | 0.38 | 0.47 | 0.56 | 0.67 | 0.79 | 0.92 |
| 0                 | 790.3            | 0.19                        | 0.25 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.87 | 1.01 |
| 5                 | 715.2            | 0.21                        | 0.28 | 0.37 | 0.46 | 0.57 | 0.69 | 0.82 | 0.97 | 1.12 |
| 10                | 642.5            | 0.23                        | 0.31 | 0.41 | 0.52 | 0.64 | 0.77 | 0.92 | 1.08 | 1.25 |
| 15                | 573.0            | 0.26                        | 0.35 | 0.46 | 0.58 | 0.71 | 0.86 | 1.03 | 1.21 | 1.40 |
| 20                | 507.8            | 0.29                        | 0.39 | 0.52 | 0.65 | 0.81 | 0.97 | 1.16 | 1.36 | 1.58 |
| 25                | 448.0            | 0.33                        | 0.45 | 0.58 | 0.74 | 0.91 | 1.10 | 1.31 | 1.54 | 1.79 |
| 30                | 395.0            | 0.37                        | 0.51 | 0.66 | 0.84 | 1.04 | 1.25 | 1.49 | 1.75 | 2.03 |
| 35                | 349.2            | 0.42                        | 0.57 | 0.75 | 0.95 | 1.17 | 1.42 | 1.69 | 1.98 | 2.30 |
| 40                | 310.9            | 0.47                        | 0.64 | 0.84 | 1.07 | 1.32 | 1.59 | 1.90 | 2.22 | 2.58 |
| 45                | 279.2            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.77 | 2.11 | 2.48 | 2.87 |
| 50                | 253.3            | 0.58                        | 0.79 | 1.03 | 1.31 | 1.61 | 1.95 | 2.33 | 2.73 | 3.17 |
| 55                | 232.1            | 0.63                        | 0.86 | 1.13 | 1.43 | 1.76 | 2.13 | 2.54 | 2.98 | 3.46 |
| 60                | 214.5            | 0.69                        | 0.93 | 1.22 | 1.54 | 1.91 | 2.31 | 2.75 | 3.22 | 3.74 |
| 65                | 199.8            | 0.74                        | 1.00 | 1.31 | 1.66 | 2.05 | 2.48 | 2.95 | 3.46 | 4.01 |
| 70                | 187.4            | 0.79                        | 1.07 | 1.40 | 1.77 | 2.18 | 2.64 | 3.14 | 3.69 | 4.28 |
| 75                | 176.9            | 0.83                        | 1.13 | 1.48 | 1.87 | 2.31 | 2.80 | 3.33 | 3.91 | 4.53 |
| 80                | 167.7            | 0.88                        | 1.20 | 1.56 | 1.98 | 2.44 | 2.95 | 3.51 | 4.12 | 4.78 |

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | OAK                                                               |
| Basic / Recommended Span (m)                  | 80                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 940.9            | 0.16                        | 0.21 | 0.28 | 0.35 | 0.43 | 0.53 | 0.63 | 0.73 | 0.85 |
| 0                 | 853.7            | 0.17                        | 0.23 | 0.31 | 0.39 | 0.48 | 0.58 | 0.69 | 0.81 | 0.94 |
| 5                 | 777.2            | 0.19                        | 0.26 | 0.34 | 0.43 | 0.53 | 0.64 | 0.76 | 0.89 | 1.03 |
| 10                | 702.5            | 0.21                        | 0.29 | 0.37 | 0.47 | 0.58 | 0.70 | 0.84 | 0.98 | 1.14 |
| 15                | 630.3            | 0.23                        | 0.32 | 0.42 | 0.53 | 0.65 | 0.79 | 0.93 | 1.10 | 1.27 |
| 20                | 561.4            | 0.26                        | 0.36 | 0.47 | 0.59 | 0.73 | 0.88 | 1.05 | 1.23 | 1.43 |
| 25                | 497.0            | 0.30                        | 0.40 | 0.53 | 0.67 | 0.82 | 1.00 | 1.19 | 1.39 | 1.61 |
| 30                | 438.4            | 0.34                        | 0.46 | 0.60 | 0.76 | 0.93 | 1.13 | 1.34 | 1.58 | 1.83 |
| 35                | 386.6            | 0.38                        | 0.52 | 0.68 | 0.86 | 1.06 | 1.28 | 1.52 | 1.79 | 2.07 |
| 40                | 342.1            | 0.43                        | 0.59 | 0.77 | 0.97 | 1.20 | 1.45 | 1.72 | 2.02 | 2.34 |
| 45                | 305.0            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.34 | 1.62 | 1.93 | 2.27 | 2.63 |
| 50                | 274.4            | 0.54                        | 0.73 | 0.95 | 1.21 | 1.49 | 1.80 | 2.15 | 2.52 | 2.92 |
| 55                | 249.4            | 0.59                        | 0.80 | 1.05 | 1.33 | 1.64 | 1.99 | 2.36 | 2.77 | 3.22 |
| 60                | 228.8            | 0.64                        | 0.88 | 1.14 | 1.45 | 1.79 | 2.16 | 2.57 | 3.02 | 3.50 |
| 65                | 211.8            | 0.70                        | 0.95 | 1.24 | 1.56 | 1.93 | 2.34 | 2.78 | 3.26 | 3.79 |
| 70                | 197.6            | 0.75                        | 1.01 | 1.33 | 1.68 | 2.07 | 2.51 | 2.98 | 3.50 | 4.06 |
| 75                | 185.5            | 0.79                        | 1.08 | 1.41 | 1.79 | 2.21 | 2.67 | 3.18 | 3.73 | 4.32 |
| 80                | 175.2            | 0.84                        | 1.14 | 1.49 | 1.89 | 2.34 | 2.83 | 3.36 | 3.95 | 4.58 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 26 | <b>of</b> | 98 |

## Appendix 11 - 100mm AAAC (Oak) -(90m Basic) High altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | OAK     |
| Greased Conductor Weight (kg/m)                    | 0.3273  |
| Cross Sectional Area of Conductor (mm²)            | 118.9   |
| Conductor Overall Diameter (mm)                    | 14      |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)         | 3576.1  |
| Basic / Recommended Span (m)                       | 90      |
| Wind Pressure on Conductor (N/m²)                  | 570     |
| Radial Ice Thickness (mm)                          | 12.5    |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 1788.1  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 715.2   |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1533.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.277  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 2.267  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 788.5  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | OAK | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.07                              | 0.26                                        | 0.28  | 0.30  | 0.32  | 0.33  | 0.35  |
| 80             | 0.30                              | 1.04                                        | 1.11  | 1.19  | 1.27  | 1.34  | 1.41  |
| 120            | 0.67                              | 2.33                                        | 2.51  | 2.68  | 2.85  | 3.01  | 3.17  |
| 160            | 1.20                              | 4.14                                        | 4.46  | 4.77  | 5.07  | 5.36  | 5.64  |
| 200            | 1.87                              | 6.47                                        | 6.97  | 7.45  | 7.92  | 8.37  | 8.81  |
| 240            | 2.70                              | 9.32                                        | 10.03 | 10.73 | 11.40 | 12.05 | 12.68 |
| 280            | 3.67                              | 12.69                                       | 13.66 | 14.60 | 15.52 | 16.40 | 17.26 |
| 320            | 4.80                              | 16.57                                       | 17.84 | 19.07 | 20.27 | 21.42 | 22.54 |
| 360            | 6.07                              | 20.97                                       | 22.57 | 24.14 | 25.65 | 27.11 | 28.53 |
| 400            | 7.50                              | 25.89                                       | 27.87 | 29.80 | 31.66 | 33.47 | 35.22 |
| 440            | 9.07                              | 31.32                                       | 33.72 | 36.05 | 38.31 | 40.50 | 42.62 |
| 480            | 10.80                             | 37.28                                       | 40.13 | 42.91 | 45.60 | 48.20 | 50.72 |
| 520            | 12.67                             | 43.75                                       | 47.10 | 50.36 | 53.51 | 56.57 | 59.53 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 27 | <b>of</b> | 98 |

## Appendix 11a - 100mm AAAC - (Oak) - (90m Basic) High altitude

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | OAK                                                               |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 873.0            | 0.17                        | 0.23 | 0.30 | 0.38 | 0.47 | 0.57 | 0.67 | 0.79 | 1.52 |
| 0                 | 788.5            | 0.19                        | 0.25 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.88 | 1.68 |
| 5                 | 715.2            | 0.21                        | 0.28 | 0.37 | 0.46 | 0.57 | 0.69 | 0.82 | 0.97 | 1.85 |
| 10                | 644.6            | 0.23                        | 0.31 | 0.41 | 0.51 | 0.63 | 0.77 | 0.91 | 1.07 | 2.06 |
| 15                | 577.7            | 0.25                        | 0.35 | 0.45 | 0.57 | 0.71 | 0.86 | 1.02 | 1.20 | 2.29 |
| 20                | 515.3            | 0.29                        | 0.39 | 0.51 | 0.64 | 0.79 | 0.96 | 1.14 | 1.34 | 2.57 |
| 25                | 458.6            | 0.32                        | 0.44 | 0.57 | 0.72 | 0.89 | 1.08 | 1.28 | 1.51 | 2.89 |
| 30                | 408.4            | 0.36                        | 0.49 | 0.64 | 0.81 | 1.00 | 1.21 | 1.44 | 1.69 | 3.25 |
| 35                | 365.1            | 0.40                        | 0.55 | 0.72 | 0.91 | 1.12 | 1.36 | 1.61 | 1.89 | 3.63 |
| 40                | 328.5            | 0.45                        | 0.61 | 0.80 | 1.01 | 1.25 | 1.51 | 1.79 | 2.10 | 4.04 |
| 45                | 297.9            | 0.49                        | 0.67 | 0.88 | 1.11 | 1.37 | 1.66 | 1.98 | 2.32 | 4.45 |
| 50                | 272.5            | 0.54                        | 0.74 | 0.96 | 1.22 | 1.50 | 1.82 | 2.16 | 2.54 | 4.86 |
| 55                | 251.4            | 0.59                        | 0.80 | 1.04 | 1.32 | 1.63 | 1.97 | 2.34 | 2.75 | 5.27 |
| 60                | 233.6            | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 5.67 |
| 65                | 218.6            | 0.67                        | 0.92 | 1.20 | 1.52 | 1.87 | 2.26 | 2.69 | 3.16 | 6.06 |
| 70                | 205.8            | 0.72                        | 0.97 | 1.27 | 1.61 | 1.99 | 2.41 | 2.86 | 3.36 | 6.44 |
| 75                | 194.8            | 0.76                        | 1.03 | 1.34 | 1.70 | 2.10 | 2.54 | 3.02 | 3.55 | 6.81 |
| 80                | 185.2            | 0.80                        | 1.08 | 1.41 | 1.79 | 2.21 | 2.67 | 3.18 | 3.73 | 7.16 |

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | OAK                                                               |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 939.6            | 0.16                        | 0.21 | 0.28 | 0.35 | 0.44 | 0.53 | 0.63 | 0.74 | 1.41 |
| 0                 | 853.6            | 0.17                        | 0.23 | 0.31 | 0.39 | 0.48 | 0.58 | 0.69 | 0.81 | 1.55 |
| 5                 | 778.6            | 0.19                        | 0.26 | 0.34 | 0.43 | 0.53 | 0.64 | 0.76 | 0.89 | 1.70 |
| 10                | 705.6            | 0.21                        | 0.28 | 0.37 | 0.47 | 0.58 | 0.70 | 0.83 | 0.98 | 1.88 |
| 15                | 635.4            | 0.23                        | 0.32 | 0.41 | 0.52 | 0.64 | 0.78 | 0.93 | 1.09 | 2.09 |
| 20                | 569.0            | 0.26                        | 0.35 | 0.46 | 0.58 | 0.72 | 0.87 | 1.04 | 1.22 | 2.33 |
| 25                | 507.3            | 0.29                        | 0.40 | 0.52 | 0.65 | 0.81 | 0.98 | 1.16 | 1.36 | 2.61 |
| 30                | 451.5            | 0.33                        | 0.44 | 0.58 | 0.73 | 0.91 | 1.10 | 1.30 | 1.53 | 2.94 |
| 35                | 402.2            | 0.37                        | 0.50 | 0.65 | 0.82 | 1.02 | 1.23 | 1.46 | 1.72 | 3.30 |
| 40                | 359.8            | 0.41                        | 0.56 | 0.73 | 0.92 | 1.14 | 1.38 | 1.64 | 1.92 | 3.68 |
| 45                | 324.1            | 0.45                        | 0.62 | 0.81 | 1.02 | 1.26 | 1.53 | 1.82 | 2.13 | 4.09 |
| 50                | 294.2            | 0.50                        | 0.68 | 0.89 | 1.13 | 1.39 | 1.68 | 2.00 | 2.35 | 4.51 |
| 55                | 269.5            | 0.55                        | 0.74 | 0.97 | 1.23 | 1.52 | 1.84 | 2.19 | 2.57 | 4.92 |
| 60                | 248.8            | 0.59                        | 0.81 | 1.05 | 1.33 | 1.64 | 1.99 | 2.37 | 2.78 | 5.33 |
| 65                | 231.5            | 0.64                        | 0.87 | 1.13 | 1.43 | 1.77 | 2.14 | 2.55 | 2.99 | 5.73 |
| 70                | 216.8            | 0.68                        | 0.92 | 1.21 | 1.53 | 1.89 | 2.28 | 2.72 | 3.19 | 6.11 |
| 75                | 204.2            | 0.72                        | 0.98 | 1.28 | 1.62 | 2.00 | 2.42 | 2.88 | 3.39 | 6.49 |
| 80                | 193.4            | 0.76                        | 1.04 | 1.35 | 1.71 | 2.12 | 2.56 | 3.05 | 3.58 | 6.85 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 28 | <b>of</b> | 98 |

## Appendix 12 - 175mm AAAC (Elm) - (90m Basic)- High altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | ELM     |
| Greased Conductor Weight (kg/m)                    | 0.5922  |
| Cross Sectional Area of Conductor (mm²)            | 211     |
| Conductor Overall Diameter (mm)                    | 18.8    |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 5710.41 |
| Rated Breaking Strength of Conductor (kgf)         | 6346.7  |
| Basic / Recommended Span (m)                       | 90      |
| Wind Pressure on Conductor (N/m²)                  | 570     |
| Radial Ice Thickness (mm)                          | 12.5    |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 1814.4  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 1269.3  |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1814.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.714  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 2.546  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 749.5  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | ELM | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.14                              | 0.37                                        | 0.38  | 0.40  | 0.42  | 0.43  | 0.45  |
| 80             | 0.56                              | 1.47                                        | 1.54  | 1.60  | 1.66  | 1.72  | 1.78  |
| 120            | 1.26                              | 3.31                                        | 3.46  | 3.60  | 3.74  | 3.88  | 4.01  |
| 160            | 2.23                              | 5.89                                        | 6.15  | 6.40  | 6.65  | 6.89  | 7.12  |
| 200            | 3.49                              | 9.20                                        | 9.61  | 10.01 | 10.39 | 10.76 | 11.13 |
| 240            | 5.02                              | 13.25                                       | 13.84 | 14.41 | 14.96 | 15.50 | 16.02 |
| 280            | 6.84                              | 18.04                                       | 18.84 | 19.61 | 20.37 | 21.10 | 21.81 |
| 320            | 8.93                              | 23.56                                       | 24.60 | 25.62 | 26.60 | 27.56 | 28.49 |
| 360            | 11.30                             | 29.81                                       | 31.14 | 32.42 | 33.67 | 34.88 | 36.06 |
| 400            | 13.96                             | 36.81                                       | 38.44 | 40.02 | 41.56 | 43.06 | 44.51 |
| 440            | 16.89                             | 44.54                                       | 46.51 | 48.43 | 50.29 | 52.10 | 53.86 |
| 480            | 20.10                             | 53.00                                       | 55.35 | 57.63 | 59.85 | 62.00 | 64.10 |
| 520            | 23.58                             | 62.20                                       | 64.96 | 67.64 | 70.24 | 72.77 | 75.23 |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 29 | of | 98 |  |

## Appendix 12a - 175mm AAAC (Elm) - (90m Basic) High altitude

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | ELM                                                               |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 848.7            | 0.31                        | 0.43 | 0.56 | 0.71 | 0.87 | 1.06 | 1.26 | 1.47 | 2.83 |
| 0                 | 749.5            | 0.36                        | 0.48 | 0.63 | 0.80 | 0.99 | 1.20 | 1.42 | 1.67 | 3.20 |
| 5                 | 672.4            | 0.40                        | 0.54 | 0.70 | 0.89 | 1.10 | 1.33 | 1.59 | 1.86 | 3.57 |
| 10                | 606.3            | 0.44                        | 0.60 | 0.78 | 0.99 | 1.22 | 1.48 | 1.76 | 2.06 | 3.96 |
| 15                | 550.5            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.34 | 1.63 | 1.94 | 2.27 | 4.36 |
| 20                | 503.6            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.78 | 2.12 | 2.48 | 4.76 |
| 25                | 464.3            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.59 | 1.93 | 2.30 | 2.69 | 5.17 |
| 30                | 431.2            | 0.62                        | 0.84 | 1.10 | 1.39 | 1.72 | 2.08 | 2.47 | 2.90 | 5.56 |
| 35                | 403.1            | 0.66                        | 0.90 | 1.18 | 1.49 | 1.84 | 2.22 | 2.64 | 3.10 | 5.95 |
| 40                | 379.1            | 0.70                        | 0.96 | 1.25 | 1.58 | 1.95 | 2.36 | 2.81 | 3.30 | 6.33 |
| 45                | 358.4            | 0.74                        | 1.01 | 1.32 | 1.67 | 2.07 | 2.50 | 2.97 | 3.49 | 6.69 |
| 50                | 340.4            | 0.78                        | 1.07 | 1.39 | 1.76 | 2.17 | 2.63 | 3.13 | 3.67 | 7.05 |
| 55                | 324.6            | 0.82                        | 1.12 | 1.46 | 1.85 | 2.28 | 2.76 | 3.28 | 3.85 | 7.39 |
| 60                | 310.6            | 0.86                        | 1.17 | 1.53 | 1.93 | 2.38 | 2.88 | 3.43 | 4.03 | 7.72 |
| 65                | 298.2            | 0.89                        | 1.22 | 1.59 | 2.01 | 2.48 | 3.00 | 3.57 | 4.20 | 8.04 |
| 70                | 287.0            | 0.93                        | 1.26 | 1.65 | 2.09 | 2.58 | 3.12 | 3.71 | 4.36 | 8.36 |
| 75                | 276.9            | 0.96                        | 1.31 | 1.71 | 2.17 | 2.67 | 3.23 | 3.85 | 4.52 | 8.66 |
| 80                | 267.8            | 1.00                        | 1.35 | 1.77 | 2.24 | 2.76 | 3.35 | 3.98 | 4.67 | 8.96 |

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | ELM                                                               |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 946.6            | 0.28                        | 0.38 | 0.50 | 0.63 | 0.78 | 0.95 | 1.13 | 1.32 | 2.53 |
| 0                 | 836.2            | 0.32                        | 0.43 | 0.57 | 0.72 | 0.89 | 1.07 | 1.27 | 1.50 | 2.87 |
| 5                 | 748.4            | 0.36                        | 0.48 | 0.63 | 0.80 | 0.99 | 1.20 | 1.42 | 1.67 | 3.20 |
| 10                | 671.5            | 0.40                        | 0.54 | 0.71 | 0.89 | 1.10 | 1.33 | 1.59 | 1.86 | 3.57 |
| 15                | 605.5            | 0.44                        | 0.60 | 0.78 | 0.99 | 1.22 | 1.48 | 1.76 | 2.07 | 3.96 |
| 20                | 549.8            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.28 | 4.36 |
| 25                | 503.0            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.78 | 2.12 | 2.49 | 4.77 |
| 30                | 463.8            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.60 | 1.93 | 2.30 | 2.70 | 5.17 |
| 35                | 430.8            | 0.62                        | 0.84 | 1.10 | 1.39 | 1.72 | 2.08 | 2.47 | 2.90 | 5.57 |
| 40                | 402.8            | 0.66                        | 0.90 | 1.18 | 1.49 | 1.84 | 2.22 | 2.65 | 3.11 | 5.95 |
| 45                | 378.8            | 0.70                        | 0.96 | 1.25 | 1.58 | 1.95 | 2.36 | 2.81 | 3.30 | 6.33 |
| 50                | 358.2            | 0.74                        | 1.01 | 1.32 | 1.67 | 2.07 | 2.50 | 2.98 | 3.49 | 6.70 |
| 55                | 340.2            | 0.78                        | 1.07 | 1.39 | 1.76 | 2.18 | 2.63 | 3.13 | 3.68 | 7.05 |
| 60                | 324.4            | 0.82                        | 1.12 | 1.46 | 1.85 | 2.28 | 2.76 | 3.29 | 3.86 | 7.39 |
| 65                | 310.5            | 0.86                        | 1.17 | 1.53 | 1.93 | 2.38 | 2.88 | 3.43 | 4.03 | 7.73 |
| 70                | 298.0            | 0.89                        | 1.22 | 1.59 | 2.01 | 2.48 | 3.01 | 3.58 | 4.20 | 8.05 |
| 75                | 286.9            | 0.93                        | 1.26 | 1.65 | 2.09 | 2.58 | 3.12 | 3.72 | 4.36 | 8.36 |
| 80                | 276.8            | 0.96                        | 1.31 | 1.71 | 2.17 | 2.67 | 3.24 | 3.85 | 4.52 | 8.67 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 30 | <b>of</b> | 98 |

## Appendix 13 - 175mm AAAC (Elm)- (100m Basic) - High altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |         |
|------------------------------------------------------|---------|
| Conductor Code Name (if any)                         | ELM     |
| Greased Conductor Weight (kg/m)                      | 0.5922  |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 211     |
| Conductor Overall Diameter (mm)                      | 18.8    |
| Coefficient of Linear Expansion (/Degree C)          | 2.3E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 5710.41 |
| Rated Breaking Strength of Conductor (kgf)           | 6346.7  |
| Basic / Recommended Span (m)                         | 100     |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 380     |
| Radial Ice Thickness (mm)                            | 9.5     |
| Ice Density (kg/m <sup>3</sup> )                     | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 2379.3  |
| Temperature at MWT Limit (Degrees C)                 | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 1269.3  |
| Temperature at EDT Limit (Degrees C)                 | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1971.2 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.363  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.465  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 1389.8 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | ELM | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.24                                        | 0.26  | 0.27  | 0.29  | 0.31  | 0.32  |
| 80             | 0.31                              | 0.96                                        | 1.03  | 1.09  | 1.16  | 1.22  | 1.28  |
| 120            | 0.70                              | 2.16                                        | 2.31  | 2.46  | 2.61  | 2.75  | 2.89  |
| 160            | 1.24                              | 3.84                                        | 4.11  | 4.38  | 4.64  | 4.89  | 5.13  |
| 200            | 1.94                              | 6.00                                        | 6.42  | 6.84  | 7.24  | 7.64  | 8.02  |
| 240            | 2.79                              | 8.64                                        | 9.25  | 9.85  | 10.43 | 11.00 | 11.55 |
| 280            | 3.80                              | 11.76                                       | 12.59 | 13.41 | 14.20 | 14.97 | 15.72 |
| 320            | 4.96                              | 15.36                                       | 16.45 | 17.51 | 18.54 | 19.55 | 20.53 |
| 360            | 6.27                              | 19.44                                       | 20.81 | 22.16 | 23.47 | 24.74 | 25.98 |
| 400            | 7.75                              | 24.00                                       | 25.70 | 27.36 | 28.98 | 30.55 | 32.08 |
| 440            | 9.37                              | 29.04                                       | 31.09 | 33.10 | 35.06 | 36.96 | 38.81 |
| 480            | 11.15                             | 34.56                                       | 37.00 | 39.39 | 41.72 | 43.99 | 46.19 |
| 520            | 13.09                             | 40.56                                       | 43.43 | 46.23 | 48.97 | 51.63 | 54.21 |

|                           |                 |                        |         |             |    |           |    |  |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 31 | <b>of</b> | 98 |  |

## Appendix 13a - 175mm AAAC (Elm)- (100m Basic) Severe altitude

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | ELM                                                               |
| Basic / Recommended Span (m)                  | 100                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 1529.1           | 0.17                        | 0.24 | 0.31 | 0.39 | 0.48 | 0.59 | 0.70 | 0.82 | 0.95 |
| 0                 | 1389.8           | 0.19                        | 0.26 | 0.34 | 0.43 | 0.53 | 0.64 | 0.77 | 0.90 | 1.04 |
| 5                 | 1269.3           | 0.21                        | 0.29 | 0.37 | 0.47 | 0.58 | 0.71 | 0.84 | 0.99 | 1.14 |
| 10                | 1153.8           | 0.23                        | 0.31 | 0.41 | 0.52 | 0.64 | 0.78 | 0.92 | 1.08 | 1.26 |
| 15                | 1044.4           | 0.26                        | 0.35 | 0.45 | 0.57 | 0.71 | 0.86 | 1.02 | 1.20 | 1.39 |
| 20                | 942.6            | 0.28                        | 0.38 | 0.50 | 0.64 | 0.79 | 0.95 | 1.13 | 1.33 | 1.54 |
| 25                | 849.7            | 0.31                        | 0.43 | 0.56 | 0.71 | 0.87 | 1.05 | 1.25 | 1.47 | 1.71 |
| 30                | 766.7            | 0.35                        | 0.47 | 0.62 | 0.78 | 0.97 | 1.17 | 1.39 | 1.63 | 1.89 |
| 35                | 694.1            | 0.38                        | 0.52 | 0.68 | 0.86 | 1.07 | 1.29 | 1.54 | 1.80 | 2.09 |
| 40                | 631.6            | 0.42                        | 0.57 | 0.75 | 0.95 | 1.17 | 1.42 | 1.69 | 1.98 | 2.30 |
| 45                | 578.2            | 0.46                        | 0.63 | 0.82 | 1.04 | 1.28 | 1.55 | 1.84 | 2.16 | 2.51 |
| 50                | 532.9            | 0.50                        | 0.68 | 0.89 | 1.13 | 1.39 | 1.68 | 2.00 | 2.35 | 2.72 |
| 55                | 494.5            | 0.54                        | 0.73 | 0.96 | 1.21 | 1.50 | 1.81 | 2.16 | 2.53 | 2.93 |
| 60                | 461.7            | 0.58                        | 0.79 | 1.03 | 1.30 | 1.60 | 1.94 | 2.31 | 2.71 | 3.14 |
| 65                | 433.6            | 0.61                        | 0.84 | 1.09 | 1.38 | 1.71 | 2.07 | 2.46 | 2.88 | 3.35 |
| 70                | 409.4            | 0.65                        | 0.89 | 1.16 | 1.46 | 1.81 | 2.19 | 2.60 | 3.06 | 3.54 |
| 75                | 388.2            | 0.69                        | 0.93 | 1.22 | 1.54 | 1.91 | 2.31 | 2.75 | 3.22 | 3.74 |
| 80                | 369.7            | 0.72                        | 0.98 | 1.28 | 1.62 | 2.00 | 2.42 | 2.88 | 3.38 | 3.92 |

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | ELM                                                               |
| Basic / Recommended Span (m)                  | 100                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 1650.9           | 0.16                        | 0.22 | 0.29 | 0.36 | 0.45 | 0.54 | 0.65 | 0.76 | 0.88 |
| 0                 | 1508.5           | 0.18                        | 0.24 | 0.31 | 0.40 | 0.49 | 0.59 | 0.71 | 0.83 | 0.96 |
| 5                 | 1384.4           | 0.19                        | 0.26 | 0.34 | 0.43 | 0.53 | 0.65 | 0.77 | 0.90 | 1.05 |
| 10                | 1264.1           | 0.21                        | 0.29 | 0.37 | 0.47 | 0.59 | 0.71 | 0.84 | 0.99 | 1.15 |
| 15                | 1148.8           | 0.23                        | 0.32 | 0.41 | 0.52 | 0.64 | 0.78 | 0.93 | 1.09 | 1.26 |
| 20                | 1039.7           | 0.26                        | 0.35 | 0.46 | 0.58 | 0.71 | 0.86 | 1.03 | 1.20 | 1.40 |
| 25                | 938.3            | 0.28                        | 0.39 | 0.50 | 0.64 | 0.79 | 0.95 | 1.14 | 1.33 | 1.55 |
| 30                | 845.8            | 0.32                        | 0.43 | 0.56 | 0.71 | 0.88 | 1.06 | 1.26 | 1.48 | 1.72 |
| 35                | 763.3            | 0.35                        | 0.48 | 0.62 | 0.79 | 0.97 | 1.17 | 1.40 | 1.64 | 1.90 |
| 40                | 691.2            | 0.39                        | 0.52 | 0.69 | 0.87 | 1.07 | 1.30 | 1.54 | 1.81 | 2.10 |
| 45                | 629.0            | 0.42                        | 0.58 | 0.75 | 0.95 | 1.18 | 1.42 | 1.69 | 1.99 | 2.31 |
| 50                | 576.0            | 0.46                        | 0.63 | 0.82 | 1.04 | 1.29 | 1.55 | 1.85 | 2.17 | 2.52 |
| 55                | 531.1            | 0.50                        | 0.68 | 0.89 | 1.13 | 1.39 | 1.69 | 2.01 | 2.36 | 2.73 |
| 60                | 492.9            | 0.54                        | 0.74 | 0.96 | 1.22 | 1.50 | 1.82 | 2.16 | 2.54 | 2.94 |
| 65                | 460.4            | 0.58                        | 0.79 | 1.03 | 1.30 | 1.61 | 1.95 | 2.32 | 2.72 | 3.15 |
| 70                | 432.5            | 0.62                        | 0.84 | 1.10 | 1.39 | 1.71 | 2.07 | 2.46 | 2.89 | 3.35 |
| 75                | 408.4            | 0.65                        | 0.89 | 1.16 | 1.47 | 1.81 | 2.19 | 2.61 | 3.06 | 3.55 |
| 80                | 387.4            | 0.69                        | 0.94 | 1.22 | 1.55 | 1.91 | 2.31 | 2.75 | 3.23 | 3.75 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 32 | <b>of</b> | 98 |

## Appendix 14 - 200mm AAAC (Poplar) - (90m Basic)- High altitude

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | POPLAR  |
| Greased Conductor Weight (kg/m)                    | 0.68992 |
| Cross Sectional Area of Conductor (mm²)            | 239.4   |
| Conductor Overall Diameter (mm)                    | 20.1    |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 5700    |
| Rated Breaking Strength of Conductor (kgf)         | 7200.2  |
| Basic / Recommended Span (m)                       | 90      |
| Wind Pressure on Conductor (N/m²)                  | 570     |
| Radial Ice Thickness (mm)                          | 12.5    |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 1814.4  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 1440.04 |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1814.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.859  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 2.621  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 675.0  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |        |                               |    |
|------------|--------|-------------------------------|----|
| Conductor: | POPLAR | Temperature Shift (Deg. C)    | 0  |
|            |        | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.18                              | 0.41                                        | 0.43  | 0.44  | 0.46  | 0.47  | 0.48  |
| 80             | 0.73                              | 1.65                                        | 1.71  | 1.77  | 1.82  | 1.88  | 1.93  |
| 120            | 1.65                              | 3.70                                        | 3.84  | 3.97  | 4.10  | 4.22  | 4.35  |
| 160            | 2.93                              | 6.59                                        | 6.83  | 7.06  | 7.29  | 7.51  | 7.73  |
| 200            | 4.57                              | 10.29                                       | 10.67 | 11.03 | 11.39 | 11.74 | 12.07 |
| 240            | 6.58                              | 14.82                                       | 15.36 | 15.89 | 16.40 | 16.90 | 17.39 |
| 280            | 8.96                              | 20.17                                       | 20.91 | 21.62 | 22.32 | 23.00 | 23.67 |
| 320            | 11.70                             | 26.34                                       | 27.30 | 28.24 | 29.15 | 30.04 | 30.91 |
| 360            | 14.81                             | 33.34                                       | 34.56 | 35.74 | 36.90 | 38.02 | 39.12 |
| 400            | 18.28                             | 41.16                                       | 42.66 | 44.13 | 45.55 | 46.94 | 48.30 |
| 440            | 22.12                             | 49.80                                       | 51.62 | 53.39 | 55.12 | 56.80 | 58.44 |
| 480            | 26.33                             | 59.27                                       | 61.44 | 63.54 | 65.59 | 67.59 | 69.55 |
| 520            | 30.90                             | 69.56                                       | 72.10 | 74.57 | 76.98 | 79.33 | 81.62 |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 33 | of | 98 |  |

## Appendix 14a - 200mm AAAC (Poplar) - (90m Basic)- High altitude

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | POPLAR                                                            |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 754.7            | 0.41                        | 0.56 | 0.73 | 0.93 | 1.14 | 1.38 | 1.65 | 1.93 | 3.70 |
| 0                 | 675.0            | 0.46                        | 0.63 | 0.82 | 1.03 | 1.28 | 1.55 | 1.84 | 2.16 | 4.14 |
| 5                 | 615.6            | 0.50                        | 0.69 | 0.90 | 1.13 | 1.40 | 1.70 | 2.02 | 2.37 | 4.54 |
| 10                | 565.5            | 0.55                        | 0.75 | 0.98 | 1.24 | 1.52 | 1.85 | 2.20 | 2.58 | 4.94 |
| 15                | 523.4            | 0.59                        | 0.81 | 1.05 | 1.33 | 1.65 | 1.99 | 2.37 | 2.78 | 5.34 |
| 20                | 487.7            | 0.64                        | 0.87 | 1.13 | 1.43 | 1.77 | 2.14 | 2.55 | 2.99 | 5.73 |
| 25                | 457.4            | 0.68                        | 0.92 | 1.21 | 1.53 | 1.89 | 2.28 | 2.72 | 3.19 | 6.11 |
| 30                | 431.2            | 0.72                        | 0.98 | 1.28 | 1.62 | 2.00 | 2.42 | 2.88 | 3.38 | 6.48 |
| 35                | 408.6            | 0.76                        | 1.03 | 1.35 | 1.71 | 2.11 | 2.55 | 3.04 | 3.57 | 6.84 |
| 40                | 388.8            | 0.80                        | 1.09 | 1.42 | 1.80 | 2.22 | 2.68 | 3.19 | 3.75 | 7.19 |
| 45                | 371.3            | 0.84                        | 1.14 | 1.49 | 1.88 | 2.32 | 2.81 | 3.34 | 3.92 | 7.52 |
| 50                | 355.9            | 0.87                        | 1.19 | 1.55 | 1.96 | 2.42 | 2.93 | 3.49 | 4.10 | 7.85 |
| 55                | 342.0            | 0.91                        | 1.24 | 1.61 | 2.04 | 2.52 | 3.05 | 3.63 | 4.26 | 8.17 |
| 60                | 329.5            | 0.94                        | 1.28 | 1.67 | 2.12 | 2.62 | 3.17 | 3.77 | 4.42 | 8.48 |
| 65                | 318.3            | 0.98                        | 1.33 | 1.73 | 2.19 | 2.71 | 3.28 | 3.90 | 4.58 | 8.78 |
| 70                | 308.0            | 1.01                        | 1.37 | 1.79 | 2.27 | 2.80 | 3.39 | 4.03 | 4.73 | 9.07 |
| 75                | 298.6            | 1.04                        | 1.42 | 1.85 | 2.34 | 2.89 | 3.49 | 4.16 | 4.88 | 9.36 |
| 80                | 290.0            | 1.07                        | 1.46 | 1.90 | 2.41 | 2.97 | 3.60 | 4.28 | 5.03 | 9.64 |

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | POPLAR                                                            |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 858.4            | 0.36                        | 0.49 | 0.64 | 0.81 | 1.00 | 1.22 | 1.45 | 1.70 | 3.25 |
| 0                 | 762.4            | 0.41                        | 0.55 | 0.72 | 0.92 | 1.13 | 1.37 | 1.63 | 1.91 | 3.67 |
| 5                 | 689.5            | 0.45                        | 0.61 | 0.80 | 1.01 | 1.25 | 1.51 | 1.80 | 2.11 | 4.05 |
| 10                | 627.7            | 0.49                        | 0.67 | 0.88 | 1.11 | 1.37 | 1.66 | 1.98 | 2.32 | 4.45 |
| 15                | 575.7            | 0.54                        | 0.73 | 0.96 | 1.21 | 1.50 | 1.81 | 2.16 | 2.53 | 4.85 |
| 20                | 532.0            | 0.58                        | 0.79 | 1.04 | 1.31 | 1.62 | 1.96 | 2.33 | 2.74 | 5.25 |
| 25                | 495.1            | 0.63                        | 0.85 | 1.11 | 1.41 | 1.74 | 2.11 | 2.51 | 2.94 | 5.64 |
| 30                | 463.6            | 0.67                        | 0.91 | 1.19 | 1.51 | 1.86 | 2.25 | 2.68 | 3.14 | 6.03 |
| 35                | 436.6            | 0.71                        | 0.97 | 1.26 | 1.60 | 1.98 | 2.39 | 2.84 | 3.34 | 6.40 |
| 40                | 413.3            | 0.75                        | 1.02 | 1.34 | 1.69 | 2.09 | 2.52 | 3.00 | 3.53 | 6.76 |
| 45                | 392.9            | 0.79                        | 1.08 | 1.40 | 1.78 | 2.19 | 2.66 | 3.16 | 3.71 | 7.11 |
| 50                | 375.0            | 0.83                        | 1.13 | 1.47 | 1.86 | 2.30 | 2.78 | 3.31 | 3.89 | 7.45 |
| 55                | 359.1            | 0.86                        | 1.18 | 1.54 | 1.95 | 2.40 | 2.91 | 3.46 | 4.06 | 7.78 |
| 60                | 344.9            | 0.90                        | 1.23 | 1.60 | 2.03 | 2.50 | 3.03 | 3.60 | 4.23 | 8.10 |
| 65                | 332.2            | 0.93                        | 1.27 | 1.66 | 2.10 | 2.60 | 3.14 | 3.74 | 4.39 | 8.41 |
| 70                | 320.6            | 0.97                        | 1.32 | 1.72 | 2.18 | 2.69 | 3.25 | 3.87 | 4.55 | 8.71 |
| 75                | 310.2            | 1.00                        | 1.36 | 1.78 | 2.25 | 2.78 | 3.36 | 4.00 | 4.70 | 9.01 |
| 80                | 300.6            | 1.03                        | 1.41 | 1.84 | 2.32 | 2.87 | 3.47 | 4.13 | 4.85 | 9.30 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 34 | <b>of</b> | 98 |

## Appendix 15 - 200mm AAAC (Poplar) - (100m Basic)- High altitude

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | POPLAR  |
| Greased Conductor Weight (kg/m)                    | 0.68992 |
| Cross Sectional Area of Conductor (mm²)            | 239.4   |
| Conductor Overall Diameter (mm)                    | 20.1    |
| Coefficient of Linear Expansion (/Degree C)        | 2.3E-05 |
| Modulus of Elasticity (kg/mm²)                     | 5700    |
| Rated Breaking Strength of Conductor (kgf)         | 7200.2  |
| Basic / Recommended Span (m)                       | 100     |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 9.5     |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 2379.3  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 1440.04 |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 2183.2 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.496  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.515  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 1575.5 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |        |                               |    |
|------------|--------|-------------------------------|----|
| Conductor: | POPLAR | Temperature Shift (Deg. C)    | 0  |
|            |        | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.24                                        | 0.26  | 0.28  | 0.29  | 0.31  | 0.32  |
| 80             | 0.32                              | 0.97                                        | 1.04  | 1.10  | 1.17  | 1.23  | 1.29  |
| 120            | 0.72                              | 2.18                                        | 2.33  | 2.48  | 2.63  | 2.77  | 2.90  |
| 160            | 1.27                              | 3.88                                        | 4.15  | 4.41  | 4.67  | 4.92  | 5.16  |
| 200            | 1.99                              | 6.06                                        | 6.48  | 6.89  | 7.29  | 7.68  | 8.06  |
| 240            | 2.87                              | 8.73                                        | 9.33  | 9.93  | 10.50 | 11.07 | 11.61 |
| 280            | 3.90                              | 11.88                                       | 12.71 | 13.51 | 14.30 | 15.06 | 15.81 |
| 320            | 5.10                              | 15.52                                       | 16.59 | 17.65 | 18.67 | 19.67 | 20.64 |
| 360            | 6.45                              | 19.64                                       | 21.00 | 22.34 | 23.64 | 24.90 | 26.13 |
| 400            | 7.96                              | 24.25                                       | 25.93 | 27.57 | 29.18 | 30.74 | 32.26 |
| 440            | 9.64                              | 29.34                                       | 31.37 | 33.37 | 35.31 | 37.20 | 39.03 |
| 480            | 11.47                             | 34.92                                       | 37.34 | 39.71 | 42.02 | 44.27 | 46.45 |
| 520            | 13.46                             | 40.98                                       | 43.82 | 46.60 | 49.31 | 51.95 | 54.51 |

|                           |                 |                        |         |             |    |           |    |  |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 35 | <b>of</b> | 98 |  |

## Appendix 15a - 200mm AAAC (Poplar) - (100m Basic)- High altitude

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | POPLAR                                                            |
| Basic / Recommended Span (m)                  | 100                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 1732.4           | 0.18                        | 0.24 | 0.32 | 0.40 | 0.50 | 0.60 | 0.72 | 0.84 | 0.98 |
| 0                 | 1575.5           | 0.20                        | 0.27 | 0.35 | 0.44 | 0.55 | 0.66 | 0.79 | 0.93 | 1.07 |
| 5                 | 1440.0           | 0.22                        | 0.29 | 0.38 | 0.49 | 0.60 | 0.72 | 0.86 | 1.01 | 1.17 |
| 10                | 1310.2           | 0.24                        | 0.32 | 0.42 | 0.53 | 0.66 | 0.80 | 0.95 | 1.11 | 1.29 |
| 15                | 1187.6           | 0.26                        | 0.36 | 0.46 | 0.59 | 0.73 | 0.88 | 1.05 | 1.23 | 1.42 |
| 20                | 1073.6           | 0.29                        | 0.39 | 0.51 | 0.65 | 0.80 | 0.97 | 1.16 | 1.36 | 1.57 |
| 25                | 969.7            | 0.32                        | 0.44 | 0.57 | 0.72 | 0.89 | 1.08 | 1.28 | 1.50 | 1.74 |
| 30                | 876.9            | 0.35                        | 0.48 | 0.63 | 0.80 | 0.98 | 1.19 | 1.42 | 1.66 | 1.93 |
| 35                | 795.6            | 0.39                        | 0.53 | 0.69 | 0.88 | 1.08 | 1.31 | 1.56 | 1.83 | 2.12 |
| 40                | 725.4            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.19 | 1.44 | 1.71 | 2.01 | 2.33 |
| 45                | 665.4            | 0.47                        | 0.64 | 0.83 | 1.05 | 1.30 | 1.57 | 1.87 | 2.19 | 2.54 |
| 50                | 614.3            | 0.51                        | 0.69 | 0.90 | 1.14 | 1.40 | 1.70 | 2.02 | 2.37 | 2.75 |
| 55                | 570.8            | 0.54                        | 0.74 | 0.97 | 1.22 | 1.51 | 1.83 | 2.18 | 2.55 | 2.96 |
| 60                | 533.7            | 0.58                        | 0.79 | 1.03 | 1.31 | 1.62 | 1.96 | 2.33 | 2.73 | 3.17 |
| 65                | 501.7            | 0.62                        | 0.84 | 1.10 | 1.39 | 1.72 | 2.08 | 2.48 | 2.91 | 3.37 |
| 70                | 474.0            | 0.65                        | 0.89 | 1.16 | 1.47 | 1.82 | 2.20 | 2.62 | 3.07 | 3.57 |
| 75                | 449.9            | 0.69                        | 0.94 | 1.23 | 1.55 | 1.92 | 2.32 | 2.76 | 3.24 | 3.76 |
| 80                | 428.6            | 0.72                        | 0.99 | 1.29 | 1.63 | 2.01 | 2.43 | 2.90 | 3.40 | 3.94 |

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | POPLAR                                                            |
| Basic / Recommended Span (m)                  | 100                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 140  |
| -5.6              | 1871.5           | 0.17                        | 0.23 | 0.29 | 0.37 | 0.46 | 0.56 | 0.66 | 0.78 | 0.90 |
| 0                 | 1711.0           | 0.18                        | 0.25 | 0.32 | 0.41 | 0.50 | 0.61 | 0.73 | 0.85 | 0.99 |
| 5                 | 1571.2           | 0.20                        | 0.27 | 0.35 | 0.44 | 0.55 | 0.66 | 0.79 | 0.93 | 1.08 |
| 10                | 1435.9           | 0.22                        | 0.29 | 0.38 | 0.49 | 0.60 | 0.73 | 0.86 | 1.01 | 1.18 |
| 15                | 1306.3           | 0.24                        | 0.32 | 0.42 | 0.53 | 0.66 | 0.80 | 0.95 | 1.12 | 1.29 |
| 20                | 1183.9           | 0.26                        | 0.36 | 0.47 | 0.59 | 0.73 | 0.88 | 1.05 | 1.23 | 1.43 |
| 25                | 1070.2           | 0.29                        | 0.39 | 0.52 | 0.65 | 0.81 | 0.98 | 1.16 | 1.36 | 1.58 |
| 30                | 966.6            | 0.32                        | 0.44 | 0.57 | 0.72 | 0.89 | 1.08 | 1.28 | 1.51 | 1.75 |
| 35                | 874.2            | 0.36                        | 0.48 | 0.63 | 0.80 | 0.99 | 1.19 | 1.42 | 1.67 | 1.93 |
| 40                | 793.2            | 0.39                        | 0.53 | 0.70 | 0.88 | 1.09 | 1.32 | 1.57 | 1.84 | 2.13 |
| 45                | 723.4            | 0.43                        | 0.58 | 0.76 | 0.97 | 1.19 | 1.44 | 1.72 | 2.01 | 2.34 |
| 50                | 663.7            | 0.47                        | 0.64 | 0.83 | 1.05 | 1.30 | 1.57 | 1.87 | 2.20 | 2.55 |
| 55                | 612.9            | 0.51                        | 0.69 | 0.90 | 1.14 | 1.41 | 1.70 | 2.03 | 2.38 | 2.76 |
| 60                | 569.6            | 0.55                        | 0.74 | 0.97 | 1.23 | 1.51 | 1.83 | 2.18 | 2.56 | 2.97 |
| 65                | 532.6            | 0.58                        | 0.79 | 1.04 | 1.31 | 1.62 | 1.96 | 2.33 | 2.74 | 3.17 |
| 70                | 500.8            | 0.62                        | 0.84 | 1.10 | 1.39 | 1.72 | 2.08 | 2.48 | 2.91 | 3.38 |
| 75                | 473.2            | 0.66                        | 0.89 | 1.17 | 1.48 | 1.82 | 2.21 | 2.62 | 3.08 | 3.57 |
| 80                | 449.2            | 0.69                        | 0.94 | 1.23 | 1.56 | 1.92 | 2.32 | 2.76 | 3.24 | 3.76 |

|                           |     |                        |         |             |          |
|---------------------------|-----|------------------------|---------|-------------|----------|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |          |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 36 of 98 |

## Appendix 16 - Section 2 - Sag charts for the renovation of existing overhead lines

| Conductor Type               | Basic Span   | Altitude or Loading                                 | Appendix No. |
|------------------------------|--------------|-----------------------------------------------------|--------------|
| .017" / 13mm HDBC            | 80           | All Altitudes                                       | 17           |
|                              | 90           |                                                     | 18           |
|                              | 100          |                                                     | 19           |
|                              | 120          |                                                     | 20           |
| .05" / 32mm HDBC             | 80           | Wind = 760(N/m <sup>2</sup> ),<br>Ice = 0mm         | 21           |
|                              | 90           |                                                     | 22           |
|                              | 100          |                                                     | 23           |
|                              | 120          |                                                     | 24           |
| .05" / 50mm AAAC             | 80           |                                                     | 25           |
|                              | 100          |                                                     | 26           |
|                              | 120          |                                                     | 27           |
| .25" / 16mm HDBC             | 90           |                                                     | 28           |
|                              | 100          |                                                     | 29           |
| 3 swg HDBC                   | 90           |                                                     | 30           |
|                              | 100          |                                                     | 31           |
| 5 swg HDBC                   | 90           |                                                     | 32           |
|                              | 100          |                                                     | 33           |
| 6 swg HDBC                   | 90           |                                                     | 34           |
|                              | 100          |                                                     | 35           |
| 7swg HDBC                    | 90           |                                                     | 36           |
|                              | 100          |                                                     | 37           |
| .175" / 175mm ACSR (CE/C/36) | 76.2 (250ft) | Normal Altitude                                     | 38           |
| .1" / 100mm ACSR             | 90           |                                                     | 39           |
| .1" / 100mm ACSR             | 100          | Wind = 380(N/m <sup>2</sup> ),<br>Ice = 19mm (Dia.) | 40           |
| .1" / 70mm HDBC              | 90           |                                                     | 41           |
| .15" / 100mm HDBC            | 90           |                                                     | 42           |
| .2" / 125mm HDBC             | 100          |                                                     | 43           |
| .1" / 100mm ACSR             | 90           | High Altitude                                       | 44           |
| .1" / 70mm HDBC              | 90           |                                                     | 45           |
| .15" / 100mm HDBC            | 90           | Wind = 570(N/m <sup>2</sup> ),<br>Ice = 25mm (Dia.) | 46           |
| .2" / 125mm HDBC             | 100          |                                                     | 47           |
|                              |              |                                                     |              |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 37 | <b>of</b> | 98 |

## Appendix 17 - .017" or 13mm<sup>2</sup> CACU - (All altitudes ) - 80m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |           |
|------------------------------------------------------|-----------|
| Conductor Code Name (if any)                         | 13mm CACU |
| Greased Conductor Weight (kg/m)                      | 0.11856   |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 13.1475   |
| Conductor Overall Diameter (mm)                      | 5.08      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3   |
| Rated Breaking Strength of Conductor (kgf)           | 807.4     |
| Basic / Recommended Span (m)                         | 80        |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760       |
| Radial Ice Thickness (mm)                            | 0         |
| Ice Density (kg/m <sup>3</sup> )                     | 913       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 323.0     |
| Temperature at MWT Limit (Degrees C)                 | -5.6      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 269.1     |
| Temperature at EDT Limit (Degrees C)                 | 5         |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 323.0 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.119 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.394 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 245.6 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                      |                               |   |
|----------------------|-------------------------------|---|
| Conductor: 13mm CACU | Temperature Shift (Deg. C)    | 0 |
|                      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.09                              | 0.19                                        | 0.21  | 0.22  | 0.24  | 0.25  | 0.27  |
| 80             | 0.36                              | 0.77                                        | 0.83  | 0.88  | 0.94  | 1.00  | 1.06  |
| 120            | 0.82                              | 1.73                                        | 1.86  | 1.98  | 2.12  | 2.25  | 2.39  |
| 160            | 1.46                              | 3.08                                        | 3.30  | 3.53  | 3.76  | 4.00  | 4.25  |
| 200            | 2.28                              | 4.82                                        | 5.16  | 5.51  | 5.88  | 6.26  | 6.64  |
| 240            | 3.28                              | 6.94                                        | 7.43  | 7.94  | 8.47  | 9.01  | 9.56  |
| 280            | 4.46                              | 9.44                                        | 10.11 | 10.81 | 11.53 | 12.26 | 13.01 |
| 320            | 5.83                              | 12.33                                       | 13.20 | 14.11 | 15.06 | 16.02 | 16.99 |
| 360            | 7.38                              | 15.61                                       | 16.71 | 17.86 | 19.05 | 20.27 | 21.50 |
| 400            | 9.11                              | 19.27                                       | 20.63 | 22.05 | 23.52 | 25.03 | 26.55 |
| 440            | 11.02                             | 23.32                                       | 24.97 | 26.69 | 28.46 | 30.28 | 32.12 |
| 480            | 13.12                             | 27.75                                       | 29.71 | 31.76 | 33.87 | 36.04 | 38.23 |
| 520            | 15.40                             | 32.57                                       | 34.87 | 37.27 | 39.76 | 42.29 | 44.87 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 38 | <b>of</b> | 98 |  |

## Appendix 17a - .017" or 13mm<sup>2</sup> CACU - (All altitudes ) - 80m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | 13mm CACU                                                         |
| Basic / Recommended Span (m)                  | 80                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 260.3            | 0.20                        | 0.28 | 0.36 | 0.46 | 0.57 | 0.69 | 0.82 | 0.96 | 1.84 |
| 0                 | 245.6            | 0.22                        | 0.30 | 0.39 | 0.49 | 0.60 | 0.73 | 0.87 | 1.02 | 1.96 |
| 5                 | 232.6            | 0.23                        | 0.31 | 0.41 | 0.52 | 0.64 | 0.77 | 0.92 | 1.08 | 2.06 |
| 10                | 219.8            | 0.24                        | 0.33 | 0.43 | 0.55 | 0.67 | 0.82 | 0.97 | 1.14 | 2.18 |
| 15                | 207.3            | 0.26                        | 0.35 | 0.46 | 0.58 | 0.71 | 0.87 | 1.03 | 1.21 | 2.32 |
| 20                | 195.0            | 0.27                        | 0.37 | 0.49 | 0.62 | 0.76 | 0.92 | 1.09 | 1.28 | 2.46 |
| 25                | 183.1            | 0.29                        | 0.40 | 0.52 | 0.66 | 0.81 | 0.98 | 1.17 | 1.37 | 2.62 |
| 30                | 171.5            | 0.31                        | 0.42 | 0.55 | 0.70 | 0.86 | 1.05 | 1.24 | 1.46 | 2.80 |
| 35                | 160.4            | 0.33                        | 0.45 | 0.59 | 0.75 | 0.92 | 1.12 | 1.33 | 1.56 | 2.99 |
| 40                | 149.8            | 0.36                        | 0.48 | 0.63 | 0.80 | 0.99 | 1.20 | 1.42 | 1.67 | 3.20 |
| 45                | 139.8            | 0.38                        | 0.52 | 0.68 | 0.86 | 1.06 | 1.28 | 1.53 | 1.79 | 3.43 |
| 50                | 130.4            | 0.41                        | 0.56 | 0.73 | 0.92 | 1.14 | 1.37 | 1.64 | 1.92 | 3.68 |
| 55                | 121.7            | 0.44                        | 0.60 | 0.78 | 0.99 | 1.22 | 1.47 | 1.75 | 2.06 | 3.94 |
| 60                | 113.7            | 0.47                        | 0.64 | 0.83 | 1.06 | 1.30 | 1.58 | 1.88 | 2.20 | 4.22 |
| 65                | 106.4            | 0.50                        | 0.68 | 0.89 | 1.13 | 1.39 | 1.69 | 2.01 | 2.35 | 4.51 |
| 70                | 99.8             | 0.53                        | 0.73 | 0.95 | 1.20 | 1.48 | 1.80 | 2.14 | 2.51 | 4.81 |
| 75                | 93.9             | 0.57                        | 0.77 | 1.01 | 1.28 | 1.58 | 1.91 | 2.27 | 2.67 | 5.12 |
| 80                | 88.5             | 0.60                        | 0.82 | 1.07 | 1.36 | 1.67 | 2.03 | 2.41 | 2.83 | 5.42 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 39 | <b>of</b> | 98 |

Appendix 18 - .017" or 13mm<sup>2</sup> CACU - (All altitudes ) - 90m Basic

## CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |           |
|------------------------------------------------------|-----------|
| Conductor Code Name (if any)                         | 13mm CACU |
| Greased Conductor Weight (kg/m)                      | 0.11856   |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 13.1475   |
| Conductor Overall Diameter (mm)                      | 5.08      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3   |
| Rated Breaking Strength of Conductor (kgf)           | 807.4     |
| Basic / Recommended Span (m)                         | 90        |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760       |
| Radial Ice Thickness (mm)                            | 0         |
| Ice Density (kg/m <sup>3</sup> )                     | 913       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 323.0     |
| Temperature at MWT Limit (Degrees C)                 | -5.6      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 269.1     |
| Temperature at EDT Limit (Degrees C)                 | 5         |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 323.0 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.119 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.394 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 230.9 |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                      |                               |   |
|----------------------|-------------------------------|---|
| Conductor: 13mm CACU | Temperature Shift (Deg. C)    | 0 |
|                      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.10                              | 0.20                                        | 0.21  | 0.22  | 0.24  | 0.25  | 0.26  |
| 80             | 0.39                              | 0.80                                        | 0.85  | 0.90  | 0.95  | 1.00  | 1.06  |
| 120            | 0.87                              | 1.80                                        | 1.91  | 2.02  | 2.14  | 2.26  | 2.38  |
| 160            | 1.55                              | 3.20                                        | 3.39  | 3.60  | 3.81  | 4.01  | 4.22  |
| 200            | 2.42                              | 4.99                                        | 5.30  | 5.62  | 5.95  | 6.27  | 6.60  |
| 240            | 3.48                              | 7.19                                        | 7.64  | 8.10  | 8.56  | 9.03  | 9.50  |
| 280            | 4.74                              | 9.79                                        | 10.40 | 11.02 | 11.65 | 12.29 | 12.94 |
| 320            | 6.19                              | 12.78                                       | 13.58 | 14.39 | 15.22 | 16.06 | 16.90 |
| 360            | 7.84                              | 16.18                                       | 17.19 | 18.22 | 19.27 | 20.32 | 21.38 |
| 400            | 9.67                              | 19.97                                       | 21.22 | 22.49 | 23.79 | 25.09 | 26.40 |
| 440            | 11.71                             | 24.17                                       | 25.67 | 27.21 | 28.78 | 30.36 | 31.94 |
| 480            | 13.93                             | 28.76                                       | 30.55 | 32.39 | 34.25 | 36.13 | 38.01 |
| 520            | 16.35                             | 33.76                                       | 35.86 | 38.01 | 40.20 | 42.40 | 44.61 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 40 | <b>of</b> | 98 |  |

## Appendix 18a - .017" or 13mm<sup>2</sup> CACU - (All altitudes ) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                     |
|-----------------------------------------------|---------------------------------------------------------------------|
| Conductor Code Name                           | 13mm CACU                                                           |
| Basic / Recommended Span (m)                  | 90                                                                  |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">` Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                         |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                           |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 245.1            | 0.22                        | 0.30 | 0.39 | 0.49 | 0.60 | 0.73 | 0.87 | 1.02 | 1.96 |
| 0                 | 230.9            | 0.23                        | 0.31 | 0.41 | 0.52 | 0.64 | 0.78 | 0.92 | 1.08 | 2.08 |
| 5                 | 218.5            | 0.24                        | 0.33 | 0.43 | 0.55 | 0.68 | 0.82 | 0.98 | 1.15 | 2.20 |
| 10                | 206.4            | 0.26                        | 0.35 | 0.46 | 0.58 | 0.72 | 0.87 | 1.03 | 1.21 | 2.33 |
| 15                | 194.5            | 0.27                        | 0.37 | 0.49 | 0.62 | 0.76 | 0.92 | 1.10 | 1.29 | 2.47 |
| 20                | 183.1            | 0.29                        | 0.40 | 0.52 | 0.66 | 0.81 | 0.98 | 1.17 | 1.37 | 2.62 |
| 25                | 172.1            | 0.31                        | 0.42 | 0.55 | 0.70 | 0.86 | 1.04 | 1.24 | 1.46 | 2.79 |
| 30                | 161.5            | 0.33                        | 0.45 | 0.59 | 0.74 | 0.92 | 1.11 | 1.32 | 1.55 | 2.97 |
| 35                | 151.5            | 0.35                        | 0.48 | 0.63 | 0.79 | 0.98 | 1.18 | 1.41 | 1.65 | 3.17 |
| 40                | 142.1            | 0.38                        | 0.51 | 0.67 | 0.84 | 1.04 | 1.26 | 1.50 | 1.76 | 3.38 |
| 45                | 133.3            | 0.40                        | 0.54 | 0.71 | 0.90 | 1.11 | 1.35 | 1.60 | 1.88 | 3.60 |
| 50                | 125.1            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.18 | 1.43 | 1.71 | 2.00 | 3.84 |
| 55                | 117.6            | 0.45                        | 0.62 | 0.81 | 1.02 | 1.26 | 1.52 | 1.81 | 2.13 | 4.08 |
| 60                | 110.7            | 0.48                        | 0.66 | 0.86 | 1.08 | 1.34 | 1.62 | 1.93 | 2.26 | 4.34 |
| 65                | 104.5            | 0.51                        | 0.69 | 0.91 | 1.15 | 1.42 | 1.72 | 2.04 | 2.40 | 4.59 |
| 70                | 98.9             | 0.54                        | 0.73 | 0.96 | 1.21 | 1.50 | 1.81 | 2.16 | 2.53 | 4.86 |
| 75                | 93.8             | 0.57                        | 0.77 | 1.01 | 1.28 | 1.58 | 1.91 | 2.28 | 2.67 | 5.12 |
| 80                | 89.1             | 0.60                        | 0.81 | 1.06 | 1.35 | 1.66 | 2.01 | 2.39 | 2.81 | 5.39 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 41 | <b>of</b> | 98 |

## Appendix 19 - .017" or 13mm<sup>2</sup> CACU - (All altitudes) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |           |
|------------------------------------------------------|-----------|
| Conductor Code Name (if any)                         | 13mm CACU |
| Greased Conductor Weight (kg/m)                      | 0.11856   |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 13.1475   |
| Conductor Overall Diameter (mm)                      | 5.08      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3   |
| Rated Breaking Strength of Conductor (kgf)           | 807.4     |
| Basic / Recommended Span (m)                         | 100       |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760       |
| Radial Ice Thickness (mm)                            | 0         |
| Ice Density (kg/m <sup>3</sup> )                     | 913       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 323.0     |
| Temperature at MWT Limit (Degrees C)                 | -5.6      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 269.1     |
| Temperature at EDT Limit (Degrees C)                 | 5         |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 323.0 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.119 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.394 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 215.7 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                      |                               |   |
|----------------------|-------------------------------|---|
| Conductor: 13mm CACU | Temperature Shift (Deg. C)    | 0 |
|                      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.10                              | 0.21                                        | 0.22  | 0.23  | 0.24  | 0.25  | 0.26  |
| 80             | 0.41                              | 0.83                                        | 0.87  | 0.92  | 0.96  | 1.01  | 1.05  |
| 120            | 0.93                              | 1.86                                        | 1.96  | 2.06  | 2.16  | 2.26  | 2.37  |
| 160            | 1.66                              | 3.30                                        | 3.48  | 3.66  | 3.84  | 4.03  | 4.21  |
| 200            | 2.59                              | 5.16                                        | 5.44  | 5.72  | 6.01  | 6.29  | 6.57  |
| 240            | 3.73                              | 7.43                                        | 7.83  | 8.24  | 8.65  | 9.06  | 9.47  |
| 280            | 5.07                              | 10.11                                       | 10.66 | 11.22 | 11.77 | 12.33 | 12.88 |
| 320            | 6.62                              | 13.21                                       | 13.93 | 14.65 | 15.38 | 16.10 | 16.83 |
| 360            | 8.38                              | 16.72                                       | 17.63 | 18.54 | 19.46 | 20.38 | 21.30 |
| 400            | 10.35                             | 20.64                                       | 21.76 | 22.89 | 24.03 | 25.16 | 26.29 |
| 440            | 12.52                             | 24.97                                       | 26.33 | 27.70 | 29.07 | 30.45 | 31.82 |
| 480            | 14.90                             | 29.72                                       | 31.33 | 32.96 | 34.60 | 36.24 | 37.86 |
| 520            | 17.49                             | 34.88                                       | 36.77 | 38.69 | 40.61 | 42.53 | 44.44 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 42 | <b>of</b> | 98 |  |

## Appendix 19a - .017" or 13mm<sup>2</sup> CACU - (All altitudes) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | 13mm CACU                                                         |
| Basic / Recommended Span (m)                  | 100                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 229.2            | 0.23                        | 0.32 | 0.41 | 0.52 | 0.65 | 0.78 | 0.93 | 1.09 | 2.10 |
| 0                 | 215.7            | 0.25                        | 0.34 | 0.44 | 0.56 | 0.69 | 0.83 | 0.99 | 1.16 | 2.23 |
| 5                 | 204.0            | 0.26                        | 0.36 | 0.46 | 0.59 | 0.73 | 0.88 | 1.05 | 1.23 | 2.35 |
| 10                | 192.7            | 0.28                        | 0.38 | 0.49 | 0.62 | 0.77 | 0.93 | 1.11 | 1.30 | 2.49 |
| 15                | 181.8            | 0.29                        | 0.40 | 0.52 | 0.66 | 0.82 | 0.99 | 1.17 | 1.38 | 2.64 |
| 20                | 171.4            | 0.31                        | 0.42 | 0.55 | 0.70 | 0.86 | 1.05 | 1.25 | 1.46 | 2.80 |
| 25                | 161.4            | 0.33                        | 0.45 | 0.59 | 0.74 | 0.92 | 1.11 | 1.32 | 1.55 | 2.97 |
| 30                | 152.1            | 0.35                        | 0.48 | 0.62 | 0.79 | 0.97 | 1.18 | 1.40 | 1.65 | 3.16 |
| 35                | 143.2            | 0.37                        | 0.51 | 0.66 | 0.84 | 1.03 | 1.25 | 1.49 | 1.75 | 3.35 |
| 40                | 135.0            | 0.40                        | 0.54 | 0.70 | 0.89 | 1.10 | 1.33 | 1.58 | 1.85 | 3.56 |
| 45                | 127.5            | 0.42                        | 0.57 | 0.74 | 0.94 | 1.16 | 1.41 | 1.67 | 1.97 | 3.77 |
| 50                | 120.5            | 0.44                        | 0.60 | 0.79 | 1.00 | 1.23 | 1.49 | 1.77 | 2.08 | 3.99 |
| 55                | 114.1            | 0.47                        | 0.64 | 0.83 | 1.05 | 1.30 | 1.57 | 1.87 | 2.20 | 4.21 |
| 60                | 108.2            | 0.49                        | 0.67 | 0.88 | 1.11 | 1.37 | 1.66 | 1.97 | 2.31 | 4.44 |
| 65                | 102.9            | 0.52                        | 0.71 | 0.92 | 1.17 | 1.44 | 1.74 | 2.07 | 2.43 | 4.67 |
| 70                | 98.1             | 0.54                        | 0.74 | 0.97 | 1.22 | 1.51 | 1.83 | 2.18 | 2.55 | 4.90 |
| 75                | 93.7             | 0.57                        | 0.78 | 1.01 | 1.28 | 1.58 | 1.91 | 2.28 | 2.67 | 5.13 |
| 80                | 89.7             | 0.59                        | 0.81 | 1.06 | 1.34 | 1.65 | 2.00 | 2.38 | 2.79 | 5.35 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 43 | <b>of</b> | 98 |

Appendix 20 - .017" or 13mm<sup>2</sup> CACU - (All altitudes) - 120m Basic

## CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |           |
|------------------------------------------------------|-----------|
| Conductor Code Name (if any)                         | 13mm CACU |
| Greased Conductor Weight (kg/m)                      | 0.11856   |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 13.1475   |
| Conductor Overall Diameter (mm)                      | 5.08      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3   |
| Rated Breaking Strength of Conductor (kgf)           | 807.4     |
| Basic / Recommended Span (m)                         | 120       |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760       |
| Radial Ice Thickness (mm)                            | 0         |
| Ice Density (kg/m <sup>3</sup> )                     | 913       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 323.0     |
| Temperature at MWT Limit (Degrees C)                 | -5.6      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 269.1     |
| Temperature at EDT Limit (Degrees C)                 | 5         |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 323.0 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.119 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.394 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 185.9 |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                      |                               |   |
|----------------------|-------------------------------|---|
| Conductor: 13mm CACU | Temperature Shift (Deg. C)    | 0 |
|                      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.12                              | 0.22                                        | 0.23  | 0.24  | 0.24  | 0.25  | 0.26  |
| 80             | 0.48                              | 0.87                                        | 0.91  | 0.94  | 0.98  | 1.01  | 1.05  |
| 120            | 1.08                              | 1.96                                        | 2.04  | 2.12  | 2.20  | 2.28  | 2.36  |
| 160            | 1.92                              | 3.49                                        | 3.63  | 3.77  | 3.91  | 4.05  | 4.19  |
| 200            | 3.01                              | 5.45                                        | 5.68  | 5.90  | 6.12  | 6.33  | 6.55  |
| 240            | 4.33                              | 7.85                                        | 8.17  | 8.49  | 8.81  | 9.12  | 9.43  |
| 280            | 5.89                              | 10.69                                       | 11.12 | 11.56 | 11.99 | 12.42 | 12.84 |
| 320            | 7.69                              | 13.96                                       | 14.53 | 15.10 | 15.66 | 16.22 | 16.77 |
| 360            | 9.74                              | 17.67                                       | 18.39 | 19.11 | 19.82 | 20.52 | 21.22 |
| 400            | 12.02                             | 21.81                                       | 22.70 | 23.59 | 24.47 | 25.34 | 26.20 |
| 440            | 14.55                             | 26.40                                       | 27.47 | 28.54 | 29.61 | 30.66 | 31.70 |
| 480            | 17.31                             | 31.41                                       | 32.69 | 33.97 | 35.23 | 36.49 | 37.72 |
| 520            | 20.32                             | 36.87                                       | 38.37 | 39.87 | 41.35 | 42.82 | 44.27 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 44 | <b>of</b> | 98 |

## Appendix 20a - .017" or 13mm<sup>2</sup> CACU - (All altitudes) - 120m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | 13mm CACU                                                         |
| Basic / Recommended Span (m)                  | 120                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 197.2            | 0.27                        | 0.37 | 0.48 | 0.61 | 0.75 | 0.91 | 1.08 | 1.27 | 2.43 |
| 0                 | 185.9            | 0.29                        | 0.39 | 0.51 | 0.65 | 0.80 | 0.96 | 1.15 | 1.35 | 2.58 |
| 5                 | 176.3            | 0.30                        | 0.41 | 0.54 | 0.68 | 0.84 | 1.02 | 1.21 | 1.42 | 2.72 |
| 10                | 167.2            | 0.32                        | 0.43 | 0.57 | 0.72 | 0.89 | 1.07 | 1.28 | 1.50 | 2.87 |
| 15                | 158.6            | 0.34                        | 0.46 | 0.60 | 0.76 | 0.93 | 1.13 | 1.35 | 1.58 | 3.03 |
| 20                | 150.6            | 0.35                        | 0.48 | 0.63 | 0.80 | 0.98 | 1.19 | 1.42 | 1.66 | 3.19 |
| 25                | 143.1            | 0.37                        | 0.51 | 0.66 | 0.84 | 1.04 | 1.25 | 1.49 | 1.75 | 3.36 |
| 30                | 136.2            | 0.39                        | 0.53 | 0.70 | 0.88 | 1.09 | 1.32 | 1.57 | 1.84 | 3.53 |
| 35                | 129.7            | 0.41                        | 0.56 | 0.73 | 0.93 | 1.14 | 1.38 | 1.65 | 1.93 | 3.70 |
| 40                | 123.8            | 0.43                        | 0.59 | 0.77 | 0.97 | 1.20 | 1.45 | 1.72 | 2.02 | 3.88 |
| 45                | 118.3            | 0.45                        | 0.61 | 0.80 | 1.01 | 1.25 | 1.52 | 1.80 | 2.12 | 4.06 |
| 50                | 113.3            | 0.47                        | 0.64 | 0.84 | 1.06 | 1.31 | 1.58 | 1.88 | 2.21 | 4.24 |
| 55                | 108.6            | 0.49                        | 0.67 | 0.87 | 1.10 | 1.36 | 1.65 | 1.96 | 2.31 | 4.42 |
| 60                | 104.4            | 0.51                        | 0.70 | 0.91 | 1.15 | 1.42 | 1.72 | 2.04 | 2.40 | 4.60 |
| 65                | 100.5            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.48 | 1.78 | 2.12 | 2.49 | 4.78 |
| 70                | 96.9             | 0.55                        | 0.75 | 0.98 | 1.24 | 1.53 | 1.85 | 2.20 | 2.59 | 4.96 |
| 75                | 93.5             | 0.57                        | 0.78 | 1.01 | 1.28 | 1.58 | 1.92 | 2.28 | 2.68 | 5.13 |
| 80                | 90.5             | 0.59                        | 0.80 | 1.05 | 1.33 | 1.64 | 1.98 | 2.36 | 2.77 | 5.31 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 45 | <b>of</b> | 98 |

## Appendix 21 - .05" or 32mm<sup>2</sup> HDBC - (All altitudes) - 80m Basic

|                                                      |           |
|------------------------------------------------------|-----------|
| Conductor Code Name (if any)                         | 32mm HDBC |
| Greased Conductor Weight (kg/m)                      | 0.2969    |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 33.134    |
| Conductor Overall Diameter (mm)                      | 8.05      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3   |
| Rated Breaking Strength of Conductor (kgf)           | 1296.05   |
| Basic / Recommended Span (m)                         | 80        |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760       |
| Radial Ice Thickness (mm)                            | 0         |
| Ice Density (kg/m <sup>3</sup> )                     | 913       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 518.4     |
| Temperature at MWT Limit (Degrees C)                 | -5.6      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 432.0     |
| Temperature at EDT Limit (Degrees C)                 | 5         |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 518.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.297 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.624 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 357.2 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                      |                               |   |
|----------------------|-------------------------------|---|
| Conductor: 32mm HDBC | Temperature Shift (Deg. C)    | 0 |
|                      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.15                              | 0.32                                        | 0.34  | 0.35  | 0.37  | 0.38  | 0.39  |
| 80             | 0.62                              | 1.29                                        | 1.34  | 1.40  | 1.46  | 1.52  | 1.57  |
| 120            | 1.38                              | 2.89                                        | 3.03  | 3.16  | 3.29  | 3.42  | 3.54  |
| 160            | 2.46                              | 5.14                                        | 5.38  | 5.61  | 5.85  | 6.07  | 6.30  |
| 200            | 3.85                              | 8.03                                        | 8.40  | 8.77  | 9.13  | 9.49  | 9.84  |
| 240            | 5.54                              | 11.57                                       | 12.10 | 12.63 | 13.15 | 13.66 | 14.17 |
| 280            | 7.54                              | 15.74                                       | 16.47 | 17.19 | 17.90 | 18.60 | 19.28 |
| 320            | 9.85                              | 20.56                                       | 21.52 | 22.46 | 23.38 | 24.29 | 25.18 |
| 360            | 12.46                             | 26.02                                       | 27.23 | 28.42 | 29.59 | 30.74 | 31.87 |
| 400            | 15.38                             | 32.13                                       | 33.62 | 35.09 | 36.54 | 37.95 | 39.35 |
| 440            | 18.62                             | 38.87                                       | 40.68 | 42.46 | 44.21 | 45.93 | 47.61 |
| 480            | 22.15                             | 46.26                                       | 48.41 | 50.53 | 52.61 | 54.65 | 56.66 |
| 520            | 26.00                             | 54.29                                       | 56.82 | 59.30 | 61.74 | 64.14 | 66.50 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 46 | <b>of</b> | 98 |  |

## Appendix 21a - .05" or 32mm<sup>2</sup> HDBC - (All altitudes) - 80m Basic

### Common Design & Erection Tables

#### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | 32mm HDBC                                                         |
| Basic / Recommended Span (m)                  | 80                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 386.0            | 0.35                        | 0.47 | 0.62 | 0.78 | 0.96 | 1.16 | 1.38 | 1.62 | 3.12 |
| 0                 | 357.2            | 0.37                        | 0.51 | 0.67 | 0.84 | 1.04 | 1.26 | 1.50 | 1.76 | 3.37 |
| 5                 | 333.1            | 0.40                        | 0.55 | 0.71 | 0.90 | 1.11 | 1.35 | 1.60 | 1.88 | 3.61 |
| 10                | 310.7            | 0.43                        | 0.59 | 0.76 | 0.97 | 1.19 | 1.45 | 1.72 | 2.02 | 3.87 |
| 15                | 290.1            | 0.46                        | 0.63 | 0.82 | 1.04 | 1.28 | 1.55 | 1.84 | 2.16 | 4.15 |
| 20                | 271.3            | 0.49                        | 0.67 | 0.88 | 1.11 | 1.37 | 1.66 | 1.97 | 2.31 | 4.43 |
| 25                | 254.2            | 0.53                        | 0.72 | 0.93 | 1.18 | 1.46 | 1.77 | 2.10 | 2.47 | 4.73 |
| 30                | 238.8            | 0.56                        | 0.76 | 0.99 | 1.26 | 1.55 | 1.88 | 2.24 | 2.63 | 5.03 |
| 35                | 225.0            | 0.59                        | 0.81 | 1.06 | 1.34 | 1.65 | 2.00 | 2.37 | 2.79 | 5.34 |
| 40                | 212.7            | 0.63                        | 0.86 | 1.12 | 1.41 | 1.75 | 2.11 | 2.51 | 2.95 | 5.65 |
| 45                | 201.6            | 0.66                        | 0.90 | 1.18 | 1.49 | 1.84 | 2.23 | 2.65 | 3.11 | 5.96 |
| 50                | 191.7            | 0.70                        | 0.95 | 1.24 | 1.57 | 1.94 | 2.34 | 2.79 | 3.27 | 6.27 |
| 55                | 182.8            | 0.73                        | 0.99 | 1.30 | 1.64 | 2.03 | 2.46 | 2.92 | 3.43 | 6.58 |
| 60                | 174.8            | 0.76                        | 1.04 | 1.36 | 1.72 | 2.12 | 2.57 | 3.06 | 3.59 | 6.88 |
| 65                | 167.6            | 0.80                        | 1.09 | 1.42 | 1.79 | 2.21 | 2.68 | 3.19 | 3.74 | 7.18 |
| 70                | 161.0            | 0.83                        | 1.13 | 1.48 | 1.87 | 2.30 | 2.79 | 3.32 | 3.89 | 7.47 |
| 75                | 155.1            | 0.86                        | 1.17 | 1.53 | 1.94 | 2.39 | 2.90 | 3.45 | 4.04 | 7.75 |
| 80                | 149.7            | 0.89                        | 1.22 | 1.59 | 2.01 | 2.48 | 3.00 | 3.57 | 4.19 | 8.03 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 47 | <b>of</b> | 98 |

## Appendix 22 - .05" or 32mm<sup>2</sup> HDBC - (All altitudes) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |           |
|------------------------------------------------------|-----------|
| Conductor Code Name (if any)                         | 32mm HDBC |
| Greased Conductor Weight (kg/m)                      | 0.2969    |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 33.134    |
| Conductor Overall Diameter (mm)                      | 8.05      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3   |
| Rated Breaking Strength of Conductor (kgf)           | 1296.05   |
| Basic / Recommended Span (m)                         | 90        |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760       |
| Radial Ice Thickness (mm)                            | 0         |
| Ice Density (kg/m <sup>3</sup> )                     | 913       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 518.4     |
| Temperature at MWT Limit (Degrees C)                 | -5.6      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 432.0     |
| Temperature at EDT Limit (Degrees C)                 | 5         |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 518.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.297 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.624 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 337.0 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                      |                               |   |
|----------------------|-------------------------------|---|
| Conductor: 32mm HDBC | Temperature Shift (Deg. C)    | 0 |
|                      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.16                              | 0.31                                        | 0.33  | 0.34  | 0.35  | 0.36  | 0.37  |
| 80             | 0.66                              | 1.26                                        | 1.31  | 1.36  | 1.40  | 1.45  | 1.50  |
| 120            | 1.48                              | 2.83                                        | 2.94  | 3.05  | 3.16  | 3.27  | 3.37  |
| 160            | 2.62                              | 5.03                                        | 5.23  | 5.43  | 5.62  | 5.81  | 6.00  |
| 200            | 4.10                              | 7.86                                        | 8.17  | 8.48  | 8.78  | 9.08  | 9.37  |
| 240            | 5.90                              | 11.31                                       | 11.76 | 12.21 | 12.64 | 13.07 | 13.49 |
| 280            | 8.03                              | 15.40                                       | 16.01 | 16.62 | 17.21 | 17.79 | 18.37 |
| 320            | 10.49                             | 20.11                                       | 20.91 | 21.70 | 22.48 | 23.24 | 23.99 |
| 360            | 13.28                             | 25.46                                       | 26.47 | 27.47 | 28.45 | 29.41 | 30.36 |
| 400            | 16.40                             | 31.43                                       | 32.68 | 33.91 | 35.12 | 36.31 | 37.48 |
| 440            | 19.84                             | 38.03                                       | 39.54 | 41.03 | 42.50 | 43.94 | 45.36 |
| 480            | 23.61                             | 45.26                                       | 47.06 | 48.83 | 50.58 | 52.29 | 53.98 |
| 520            | 27.71                             | 53.11                                       | 55.23 | 57.31 | 59.36 | 61.37 | 63.35 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 48 | <b>of</b> | 98 |  |

## Appendix 22a - .05" or 32mm<sup>2</sup> HDBC - (All altitudes) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | 32mm HDBC                                                         |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 362.2            | 0.37                        | 0.50 | 0.66 | 0.83 | 1.02 | 1.24 | 1.48 | 1.73 | 3.32 |
| 0                 | 337.0            | 0.40                        | 0.54 | 0.70 | 0.89 | 1.10 | 1.33 | 1.59 | 1.86 | 3.57 |
| 5                 | 316.2            | 0.42                        | 0.58 | 0.75 | 0.95 | 1.17 | 1.42 | 1.69 | 1.98 | 3.80 |
| 10                | 297.1            | 0.45                        | 0.61 | 0.80 | 1.01 | 1.25 | 1.51 | 1.80 | 2.11 | 4.05 |
| 15                | 279.7            | 0.48                        | 0.65 | 0.85 | 1.07 | 1.33 | 1.61 | 1.91 | 2.24 | 4.30 |
| 20                | 263.8            | 0.51                        | 0.69 | 0.90 | 1.14 | 1.41 | 1.70 | 2.03 | 2.38 | 4.56 |
| 25                | 249.4            | 0.54                        | 0.73 | 0.95 | 1.21 | 1.49 | 1.80 | 2.14 | 2.51 | 4.82 |
| 30                | 236.4            | 0.57                        | 0.77 | 1.00 | 1.27 | 1.57 | 1.90 | 2.26 | 2.65 | 5.09 |
| 35                | 224.7            | 0.59                        | 0.81 | 1.06 | 1.34 | 1.65 | 2.00 | 2.38 | 2.79 | 5.35 |
| 40                | 214.1            | 0.62                        | 0.85 | 1.11 | 1.40 | 1.73 | 2.10 | 2.50 | 2.93 | 5.62 |
| 45                | 204.5            | 0.65                        | 0.89 | 1.16 | 1.47 | 1.81 | 2.20 | 2.61 | 3.07 | 5.88 |
| 50                | 195.9            | 0.68                        | 0.93 | 1.21 | 1.53 | 1.89 | 2.29 | 2.73 | 3.20 | 6.14 |
| 55                | 188.0            | 0.71                        | 0.97 | 1.26 | 1.60 | 1.97 | 2.39 | 2.84 | 3.34 | 6.40 |
| 60                | 180.9            | 0.74                        | 1.01 | 1.31 | 1.66 | 2.05 | 2.48 | 2.96 | 3.47 | 6.65 |
| 65                | 174.3            | 0.77                        | 1.04 | 1.36 | 1.72 | 2.13 | 2.58 | 3.07 | 3.60 | 6.90 |
| 70                | 168.3            | 0.79                        | 1.08 | 1.41 | 1.79 | 2.20 | 2.67 | 3.17 | 3.73 | 7.14 |
| 75                | 162.9            | 0.82                        | 1.12 | 1.46 | 1.85 | 2.28 | 2.76 | 3.28 | 3.85 | 7.38 |
| 80                | 157.8            | 0.85                        | 1.15 | 1.51 | 1.90 | 2.35 | 2.85 | 3.39 | 3.97 | 7.62 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 49 | <b>of</b> | 98 |

## Appendix 23 - .05" or 32mm<sup>2</sup> HDBC - (All altitudes) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |           |
|------------------------------------------------------|-----------|
| Conductor Code Name (if any)                         | 32mm HDBC |
| Greased Conductor Weight (kg/m)                      | 0.2969    |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 33.134    |
| Conductor Overall Diameter (mm)                      | 8.05      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3   |
| Rated Breaking Strength of Conductor (kgf)           | 1296.05   |
| Basic / Recommended Span (m)                         | 100       |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760       |
| Radial Ice Thickness (mm)                            | 0         |
| Ice Density (kg/m <sup>3</sup> )                     | 913       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 518.4     |
| Temperature at MWT Limit (Degrees C)                 | -5.6      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 432.0     |
| Temperature at EDT Limit (Degrees C)                 | 5         |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 518.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.297 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.624 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 319.3 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                      |                               |   |
|----------------------|-------------------------------|---|
| Conductor: 32mm HDBC | Temperature Shift (Deg. C)    | 0 |
|                      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.17                              | 0.31                                        | 0.32  | 0.33  | 0.34  | 0.35  | 0.36  |
| 80             | 0.70                              | 1.24                                        | 1.28  | 1.32  | 1.36  | 1.40  | 1.44  |
| 120            | 1.57                              | 2.78                                        | 2.88  | 2.97  | 3.06  | 3.15  | 3.24  |
| 160            | 2.79                              | 4.94                                        | 5.11  | 5.28  | 5.45  | 5.61  | 5.77  |
| 200            | 4.36                              | 7.72                                        | 7.99  | 8.25  | 8.51  | 8.76  | 9.01  |
| 240            | 6.27                              | 11.12                                       | 11.50 | 11.88 | 12.25 | 12.62 | 12.98 |
| 280            | 8.54                              | 15.13                                       | 15.66 | 16.17 | 16.68 | 17.17 | 17.66 |
| 320            | 11.15                             | 19.77                                       | 20.45 | 21.12 | 21.78 | 22.43 | 23.07 |
| 360            | 14.12                             | 25.02                                       | 25.88 | 26.73 | 27.57 | 28.39 | 29.19 |
| 400            | 17.43                             | 30.89                                       | 31.95 | 33.00 | 34.03 | 35.05 | 36.04 |
| 440            | 21.09                             | 37.37                                       | 38.66 | 39.93 | 41.18 | 42.40 | 43.61 |
| 480            | 25.09                             | 44.48                                       | 46.01 | 47.52 | 49.01 | 50.47 | 51.90 |
| 520            | 29.45                             | 52.20                                       | 54.00 | 55.77 | 57.51 | 59.23 | 60.91 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 50 | <b>of</b> | 98 |  |

## Appendix 23a - .05" or 32mm<sup>2</sup> HDBC - (All altitudes) - 100m Basic

### Common Design & Erection Tables

#### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | 32mm HDBC                                                         |
| Basic / Recommended Span (m)                  | 100                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 340.7            | 0.39                        | 0.53 | 0.70 | 0.88 | 1.09 | 1.32 | 1.57 | 1.84 | 3.53 |
| 0                 | 319.3            | 0.42                        | 0.57 | 0.74 | 0.94 | 1.16 | 1.41 | 1.67 | 1.96 | 3.77 |
| 5                 | 301.7            | 0.44                        | 0.60 | 0.79 | 1.00 | 1.23 | 1.49 | 1.77 | 2.08 | 3.99 |
| 10                | 285.6            | 0.47                        | 0.64 | 0.83 | 1.05 | 1.30 | 1.57 | 1.87 | 2.20 | 4.21 |
| 15                | 271.0            | 0.49                        | 0.67 | 0.88 | 1.11 | 1.37 | 1.66 | 1.97 | 2.31 | 4.44 |
| 20                | 257.6            | 0.52                        | 0.71 | 0.92 | 1.17 | 1.44 | 1.74 | 2.07 | 2.43 | 4.67 |
| 25                | 245.5            | 0.54                        | 0.74 | 0.97 | 1.22 | 1.51 | 1.83 | 2.18 | 2.55 | 4.90 |
| 30                | 234.5            | 0.57                        | 0.78 | 1.01 | 1.28 | 1.58 | 1.92 | 2.28 | 2.67 | 5.13 |
| 35                | 224.4            | 0.60                        | 0.81 | 1.06 | 1.34 | 1.65 | 2.00 | 2.38 | 2.79 | 5.36 |
| 40                | 215.3            | 0.62                        | 0.84 | 1.10 | 1.40 | 1.72 | 2.09 | 2.48 | 2.91 | 5.59 |
| 45                | 207.0            | 0.65                        | 0.88 | 1.15 | 1.45 | 1.79 | 2.17 | 2.58 | 3.03 | 5.81 |
| 50                | 199.3            | 0.67                        | 0.91 | 1.19 | 1.51 | 1.86 | 2.25 | 2.68 | 3.15 | 6.03 |
| 55                | 192.4            | 0.69                        | 0.95 | 1.23 | 1.56 | 1.93 | 2.33 | 2.78 | 3.26 | 6.25 |
| 60                | 185.9            | 0.72                        | 0.98 | 1.28 | 1.62 | 2.00 | 2.42 | 2.87 | 3.37 | 6.47 |
| 65                | 180.0            | 0.74                        | 1.01 | 1.32 | 1.67 | 2.06 | 2.49 | 2.97 | 3.48 | 6.68 |
| 70                | 174.6            | 0.77                        | 1.04 | 1.36 | 1.72 | 2.13 | 2.57 | 3.06 | 3.59 | 6.89 |
| 75                | 169.5            | 0.79                        | 1.07 | 1.40 | 1.77 | 2.19 | 2.65 | 3.15 | 3.70 | 7.09 |
| 80                | 164.8            | 0.81                        | 1.10 | 1.44 | 1.82 | 2.25 | 2.72 | 3.24 | 3.81 | 7.30 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 51 | <b>of</b> | 98 |

Appendix 24 - .05" or 32mm<sup>2</sup> HDBC - (All altitudes) - 120m Basic

## CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |           |
|------------------------------------------------------|-----------|
| Conductor Code Name (if any)                         | 32mm HDBC |
| Greased Conductor Weight (kg/m)                      | 0.2969    |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 33.134    |
| Conductor Overall Diameter (mm)                      | 8.05      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3   |
| Rated Breaking Strength of Conductor (kgf)           | 1296.05   |
| Basic / Recommended Span (m)                         | 120       |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760       |
| Radial Ice Thickness (mm)                            | 0         |
| Ice Density (kg/m <sup>3</sup> )                     | 913       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 518.4     |
| Temperature at MWT Limit (Degrees C)                 | -5.6      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 432.0     |
| Temperature at EDT Limit (Degrees C)                 | 5         |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 518.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.297 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.624 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 291.9 |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                      |                               |   |
|----------------------|-------------------------------|---|
| Conductor: 32mm HDBC | Temperature Shift (Deg. C)    | 0 |
|                      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.19                              | 0.30                                        | 0.31  | 0.32  | 0.32  | 0.33  | 0.34  |
| 80             | 0.77                              | 1.20                                        | 1.24  | 1.27  | 1.30  | 1.33  | 1.36  |
| 120            | 1.74                              | 2.71                                        | 2.78  | 2.85  | 2.92  | 2.99  | 3.06  |
| 160            | 3.10                              | 4.82                                        | 4.95  | 5.07  | 5.20  | 5.32  | 5.44  |
| 200            | 4.84                              | 7.53                                        | 7.73  | 7.93  | 8.12  | 8.31  | 8.50  |
| 240            | 6.96                              | 10.84                                       | 11.13 | 11.41 | 11.69 | 11.97 | 12.24 |
| 280            | 9.48                              | 14.76                                       | 15.15 | 15.54 | 15.91 | 16.29 | 16.65 |
| 320            | 12.38                             | 19.28                                       | 19.79 | 20.29 | 20.79 | 21.27 | 21.75 |
| 360            | 15.67                             | 24.40                                       | 25.05 | 25.68 | 26.31 | 26.92 | 27.53 |
| 400            | 19.35                             | 30.12                                       | 30.92 | 31.71 | 32.48 | 33.24 | 33.99 |
| 440            | 23.41                             | 36.45                                       | 37.41 | 38.36 | 39.30 | 40.22 | 41.13 |
| 480            | 27.86                             | 43.38                                       | 44.53 | 45.66 | 46.77 | 47.86 | 48.94 |
| 520            | 32.69                             | 50.91                                       | 52.26 | 53.58 | 54.89 | 56.17 | 57.44 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 52 | <b>of</b> | 98 |  |

## Appendix 24a - .05" or 32mm<sup>2</sup> HDBC - (All altitudes) - 120m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | 32mm HDBC                                                         |
| Basic / Recommended Span (m)                  | 120                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 306.9            | 0.44                        | 0.59 | 0.77 | 0.98 | 1.21 | 1.46 | 1.74 | 2.04 | 3.92 |
| 0                 | 291.9            | 0.46                        | 0.62 | 0.81 | 1.03 | 1.27 | 1.54 | 1.83 | 2.15 | 4.12 |
| 5                 | 279.6            | 0.48                        | 0.65 | 0.85 | 1.08 | 1.33 | 1.61 | 1.91 | 2.24 | 4.30 |
| 10                | 268.3            | 0.50                        | 0.68 | 0.89 | 1.12 | 1.38 | 1.67 | 1.99 | 2.34 | 4.48 |
| 15                | 258.0            | 0.52                        | 0.70 | 0.92 | 1.17 | 1.44 | 1.74 | 2.07 | 2.43 | 4.66 |
| 20                | 248.4            | 0.54                        | 0.73 | 0.96 | 1.21 | 1.49 | 1.81 | 2.15 | 2.52 | 4.84 |
| 25                | 239.6            | 0.56                        | 0.76 | 0.99 | 1.25 | 1.55 | 1.87 | 2.23 | 2.62 | 5.02 |
| 30                | 231.5            | 0.58                        | 0.79 | 1.03 | 1.30 | 1.60 | 1.94 | 2.31 | 2.71 | 5.19 |
| 35                | 224.0            | 0.60                        | 0.81 | 1.06 | 1.34 | 1.66 | 2.00 | 2.39 | 2.80 | 5.37 |
| 40                | 217.1            | 0.62                        | 0.84 | 1.09 | 1.38 | 1.71 | 2.07 | 2.46 | 2.89 | 5.54 |
| 45                | 210.7            | 0.63                        | 0.86 | 1.13 | 1.43 | 1.76 | 2.13 | 2.54 | 2.98 | 5.71 |
| 50                | 204.7            | 0.65                        | 0.89 | 1.16 | 1.47 | 1.81 | 2.19 | 2.61 | 3.06 | 5.88 |
| 55                | 199.1            | 0.67                        | 0.91 | 1.19 | 1.51 | 1.86 | 2.26 | 2.68 | 3.15 | 6.04 |
| 60                | 193.9            | 0.69                        | 0.94 | 1.23 | 1.55 | 1.91 | 2.32 | 2.76 | 3.23 | 6.20 |
| 65                | 189.0            | 0.71                        | 0.96 | 1.26 | 1.59 | 1.96 | 2.38 | 2.83 | 3.32 | 6.36 |
| 70                | 184.5            | 0.72                        | 0.99 | 1.29 | 1.63 | 2.01 | 2.43 | 2.90 | 3.40 | 6.52 |
| 75                | 180.2            | 0.74                        | 1.01 | 1.32 | 1.67 | 2.06 | 2.49 | 2.97 | 3.48 | 6.67 |
| 80                | 176.1            | 0.76                        | 1.03 | 1.35 | 1.71 | 2.11 | 2.55 | 3.03 | 3.56 | 6.83 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 53 | <b>of</b> | 98 |

## Appendix 25 - .05" or 50mm<sup>2</sup> AAAC - (All altitudes) - 80m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |         |
|------------------------------------------------------|---------|
| Conductor Code Name (if any)                         | HAZEL   |
| Greased Conductor Weight (kg/m)                      | 0.1648  |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 59.9    |
| Conductor Overall Diameter (mm)                      | 9.9     |
| Coefficient of Linear Expansion (/Degree C)          | 2.3E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)           | 1800.8  |
| Basic / Recommended Span (m)                         | 80      |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760     |
| Radial Ice Thickness (mm)                            | 0       |
| Ice Density (kg/m <sup>3</sup> )                     | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 720.3   |
| Temperature at MWT Limit (Degrees C)                 | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 360.2   |
| Temperature at EDT Limit (Degrees C)                 | 5       |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 595.0 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.165 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.767 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 398.0 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |       |                               |   |
|------------|-------|-------------------------------|---|
| Conductor: | HAZEL | Temperature Shift (Deg. C)    | 0 |
|            |       | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.07                              | 0.27                                        | 0.29  | 0.31  | 0.34  | 0.36  | 0.38  |
| 80             | 0.30                              | 1.07                                        | 1.17  | 1.26  | 1.35  | 1.43  | 1.52  |
| 120            | 0.67                              | 2.42                                        | 2.63  | 2.83  | 3.03  | 3.23  | 3.41  |
| 160            | 1.20                              | 4.30                                        | 4.67  | 5.04  | 5.39  | 5.73  | 6.06  |
| 200            | 1.87                              | 6.72                                        | 7.30  | 7.87  | 8.43  | 8.96  | 9.47  |
| 240            | 2.69                              | 9.67                                        | 10.52 | 11.34 | 12.13 | 12.90 | 13.64 |
| 280            | 3.66                              | 13.17                                       | 14.31 | 15.43 | 16.51 | 17.56 | 18.57 |
| 320            | 4.78                              | 17.20                                       | 18.70 | 20.16 | 21.57 | 22.93 | 24.25 |
| 360            | 6.05                              | 21.76                                       | 23.66 | 25.51 | 27.30 | 29.03 | 30.69 |
| 400            | 7.47                              | 26.87                                       | 29.21 | 31.49 | 33.70 | 35.83 | 37.89 |
| 440            | 9.04                              | 32.51                                       | 35.35 | 38.11 | 40.78 | 43.36 | 45.85 |
| 480            | 10.76                             | 38.69                                       | 42.07 | 45.35 | 48.53 | 51.60 | 54.57 |
| 520            | 12.62                             | 45.41                                       | 49.37 | 53.23 | 56.96 | 60.56 | 64.04 |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 54 | of | 98 |  |

## Appendix 25a - .05" or 50mm<sup>2</sup> AAAC - (All altitudes) - 80m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | HAZEL                                                             |
| Basic / Recommended Span (m)                  | 80                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 441.3            | 0.17                        | 0.23 | 0.30 | 0.38 | 0.47 | 0.56 | 0.67 | 0.79 | 1.51 |
| 0                 | 398.0            | 0.19                        | 0.25 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.87 | 1.68 |
| 5                 | 360.2            | 0.21                        | 0.28 | 0.37 | 0.46 | 0.57 | 0.69 | 0.82 | 0.97 | 1.85 |
| 10                | 323.5            | 0.23                        | 0.31 | 0.41 | 0.52 | 0.64 | 0.77 | 0.92 | 1.08 | 2.06 |
| 15                | 288.5            | 0.26                        | 0.35 | 0.46 | 0.58 | 0.71 | 0.86 | 1.03 | 1.21 | 2.31 |
| 20                | 255.6            | 0.29                        | 0.39 | 0.52 | 0.65 | 0.81 | 0.98 | 1.16 | 1.36 | 2.61 |
| 25                | 225.6            | 0.33                        | 0.45 | 0.58 | 0.74 | 0.91 | 1.11 | 1.32 | 1.54 | 2.96 |
| 30                | 198.8            | 0.37                        | 0.51 | 0.66 | 0.84 | 1.04 | 1.25 | 1.49 | 1.75 | 3.36 |
| 35                | 175.8            | 0.42                        | 0.57 | 0.75 | 0.95 | 1.17 | 1.42 | 1.69 | 1.98 | 3.80 |
| 40                | 156.5            | 0.47                        | 0.64 | 0.84 | 1.07 | 1.32 | 1.59 | 1.90 | 2.22 | 4.26 |
| 45                | 140.6            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.77 | 2.11 | 2.48 | 4.75 |
| 50                | 127.5            | 0.58                        | 0.79 | 1.03 | 1.31 | 1.62 | 1.95 | 2.33 | 2.73 | 5.23 |
| 55                | 116.8            | 0.63                        | 0.86 | 1.13 | 1.43 | 1.76 | 2.13 | 2.54 | 2.98 | 5.71 |
| 60                | 108.0            | 0.69                        | 0.93 | 1.22 | 1.55 | 1.91 | 2.31 | 2.75 | 3.22 | 6.18 |
| 65                | 100.6            | 0.74                        | 1.00 | 1.31 | 1.66 | 2.05 | 2.48 | 2.95 | 3.46 | 6.63 |
| 70                | 94.4             | 0.79                        | 1.07 | 1.40 | 1.77 | 2.18 | 2.64 | 3.14 | 3.69 | 7.07 |
| 75                | 89.0             | 0.83                        | 1.13 | 1.48 | 1.87 | 2.31 | 2.80 | 3.33 | 3.91 | 7.50 |
| 80                | 84.4             | 0.88                        | 1.20 | 1.56 | 1.98 | 2.44 | 2.95 | 3.51 | 4.12 | 7.90 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 55 | <b>of</b> | 98 |

## Appendix 26 - .05" or 50mm<sup>2</sup> AAAC - (All altitudes) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |         |
|------------------------------------------------------|---------|
| Conductor Code Name (if any)                         | HAZEL   |
| Greased Conductor Weight (kg/m)                      | 0.1648  |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 59.9    |
| Conductor Overall Diameter (mm)                      | 9.9     |
| Coefficient of Linear Expansion (/Degree C)          | 2.3E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)           | 1800.8  |
| Basic / Recommended Span (m)                         | 100     |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760     |
| Radial Ice Thickness (mm)                            | 0       |
| Ice Density (kg/m <sup>3</sup> )                     | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 720.3   |
| Temperature at MWT Limit (Degrees C)                 | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 360.2   |
| Temperature at EDT Limit (Degrees C)                 | 5       |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 641.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.165 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.767 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 396.2 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |       |                               |   |
|------------|-------|-------------------------------|---|
| Conductor: | HAZEL | Temperature Shift (Deg. C)    | 0 |
|            |       | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.23                                        | 0.25  | 0.27  | 0.28  | 0.30  | 0.32  |
| 80             | 0.30                              | 0.93                                        | 1.00  | 1.07  | 1.14  | 1.20  | 1.27  |
| 120            | 0.68                              | 2.10                                        | 2.26  | 2.41  | 2.56  | 2.71  | 2.85  |
| 160            | 1.20                              | 3.74                                        | 4.02  | 4.29  | 4.55  | 4.81  | 5.06  |
| 200            | 1.88                              | 5.84                                        | 6.28  | 6.70  | 7.12  | 7.52  | 7.91  |
| 240            | 2.71                              | 8.41                                        | 9.04  | 9.65  | 10.25 | 10.83 | 11.39 |
| 280            | 3.69                              | 11.45                                       | 12.31 | 13.14 | 13.95 | 14.73 | 15.50 |
| 320            | 4.82                              | 14.96                                       | 16.07 | 17.16 | 18.22 | 19.24 | 20.24 |
| 360            | 6.10                              | 18.93                                       | 20.34 | 21.72 | 23.06 | 24.36 | 25.62 |
| 400            | 7.53                              | 23.37                                       | 25.11 | 26.81 | 28.47 | 30.07 | 31.63 |
| 440            | 9.11                              | 28.28                                       | 30.39 | 32.44 | 34.44 | 36.38 | 38.27 |
| 480            | 10.84                             | 33.66                                       | 36.16 | 38.61 | 40.99 | 43.30 | 45.54 |
| 520            | 12.72                             | 39.50                                       | 42.44 | 45.31 | 48.11 | 50.82 | 53.45 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 56 | <b>of</b> | 98 |  |

## Appendix 26a - .05" or 50mm<sup>2</sup> AAAC - (All altitudes) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |       |                                |
|-----------------------------------------------|-------|--------------------------------|
| Conductor Code Name                           | HAZEL |                                |
| Basic / Recommended Span (m)                  | 100   |                                |
| Temperature Shift for Creep (Deg. C)          | 0     | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 0.0   | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 0     | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 437.9            | 0.17                        | 0.23 | 0.30 | 0.38 | 0.47 | 0.57 | 0.68 | 0.80 | 1.52 |
| 0                 | 396.2            | 0.19                        | 0.25 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.88 | 1.68 |
| 5                 | 360.2            | 0.21                        | 0.28 | 0.37 | 0.46 | 0.57 | 0.69 | 0.82 | 0.97 | 1.85 |
| 10                | 325.7            | 0.23                        | 0.31 | 0.40 | 0.51 | 0.63 | 0.77 | 0.91 | 1.07 | 2.05 |
| 15                | 293.3            | 0.25                        | 0.34 | 0.45 | 0.57 | 0.70 | 0.85 | 1.01 | 1.19 | 2.28 |
| 20                | 263.2            | 0.28                        | 0.38 | 0.50 | 0.63 | 0.78 | 0.95 | 1.13 | 1.32 | 2.54 |
| 25                | 236.1            | 0.31                        | 0.43 | 0.56 | 0.71 | 0.87 | 1.06 | 1.26 | 1.47 | 2.83 |
| 30                | 212.1            | 0.35                        | 0.48 | 0.62 | 0.79 | 0.97 | 1.18 | 1.40 | 1.64 | 3.15 |
| 35                | 191.4            | 0.39                        | 0.53 | 0.69 | 0.87 | 1.08 | 1.30 | 1.55 | 1.82 | 3.49 |
| 40                | 173.7            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.19 | 1.43 | 1.71 | 2.00 | 3.84 |
| 45                | 158.8            | 0.47                        | 0.64 | 0.83 | 1.05 | 1.30 | 1.57 | 1.87 | 2.19 | 4.20 |
| 50                | 146.3            | 0.51                        | 0.69 | 0.90 | 1.14 | 1.41 | 1.70 | 2.03 | 2.38 | 4.56 |
| 55                | 135.7            | 0.55                        | 0.74 | 0.97 | 1.23 | 1.52 | 1.84 | 2.19 | 2.57 | 4.92 |
| 60                | 126.7            | 0.59                        | 0.80 | 1.04 | 1.32 | 1.63 | 1.97 | 2.34 | 2.75 | 5.27 |
| 65                | 119.1            | 0.62                        | 0.85 | 1.11 | 1.40 | 1.73 | 2.09 | 2.49 | 2.92 | 5.61 |
| 70                | 112.5            | 0.66                        | 0.90 | 1.17 | 1.48 | 1.83 | 2.22 | 2.64 | 3.10 | 5.94 |
| 75                | 106.7            | 0.70                        | 0.95 | 1.24 | 1.56 | 1.93 | 2.34 | 2.78 | 3.26 | 6.26 |
| 80                | 101.7            | 0.73                        | 0.99 | 1.30 | 1.64 | 2.03 | 2.45 | 2.92 | 3.42 | 6.57 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 57 | <b>of</b> | 98 |

Appendix 27 - .05" or 50mm<sup>2</sup> AAAC - (All altitudes) - 120m Basic

## CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |         |
|------------------------------------------------------|---------|
| Conductor Code Name (if any)                         | HAZEL   |
| Greased Conductor Weight (kg/m)                      | 0.1648  |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 59.9    |
| Conductor Overall Diameter (mm)                      | 9.9     |
| Coefficient of Linear Expansion (/Degree C)          | 2.3E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 6016.33 |
| Rated Breaking Strength of Conductor (kgf)           | 1800.8  |
| Basic / Recommended Span (m)                         | 120     |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760     |
| Radial Ice Thickness (mm)                            | 0       |
| Ice Density (kg/m <sup>3</sup> )                     | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 720.3   |
| Temperature at MWT Limit (Degrees C)                 | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 360.2   |
| Temperature at EDT Limit (Degrees C)                 | 5       |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 685.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.165 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.767 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 394.1 |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |       |                               |   |
|------------|-------|-------------------------------|---|
| Conductor: | HAZEL | Temperature Shift (Deg. C)    | 0 |
|            |       | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.08                              | 0.21                                        | 0.22  | 0.24  | 0.25  | 0.26  | 0.27  |
| 80             | 0.30                              | 0.84                                        | 0.89  | 0.95  | 1.00  | 1.05  | 1.10  |
| 120            | 0.68                              | 1.89                                        | 2.01  | 2.13  | 2.25  | 2.36  | 2.47  |
| 160            | 1.22                              | 3.35                                        | 3.57  | 3.78  | 3.99  | 4.20  | 4.39  |
| 200            | 1.90                              | 5.24                                        | 5.58  | 5.91  | 6.24  | 6.55  | 6.86  |
| 240            | 2.73                              | 7.54                                        | 8.03  | 8.51  | 8.98  | 9.44  | 9.88  |
| 280            | 3.72                              | 10.27                                       | 10.94 | 11.59 | 12.23 | 12.85 | 13.45 |
| 320            | 4.86                              | 13.41                                       | 14.28 | 15.14 | 15.97 | 16.78 | 17.57 |
| 360            | 6.15                              | 16.98                                       | 18.08 | 19.16 | 20.21 | 21.24 | 22.24 |
| 400            | 7.60                              | 20.96                                       | 22.32 | 23.65 | 24.95 | 26.22 | 27.45 |
| 440            | 9.19                              | 25.36                                       | 27.01 | 28.62 | 30.19 | 31.73 | 33.22 |
| 480            | 10.94                             | 30.18                                       | 32.14 | 34.06 | 35.93 | 37.76 | 39.53 |
| 520            | 12.84                             | 35.42                                       | 37.72 | 39.97 | 42.17 | 44.31 | 46.40 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 58 | <b>of</b> | 98 |  |

## Appendix 27a - .05" or 50mm<sup>2</sup> AAAC - (All altitudes) - 120m Basic

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Conductor Code Name                           | HAZEL                                   |
| Basic / Recommended Span (m)                  | 120                                     |
| Temperature Shift for Creep (Deg. C)          | 0 <b>Insert minus sign as necessary</b> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                             |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                               |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 433.9            | 0.17                        | 0.23 | 0.30 | 0.38 | 0.47 | 0.57 | 0.68 | 0.80 | 1.54 |
| 0                 | 394.1            | 0.19                        | 0.26 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.88 | 1.69 |
| 5                 | 360.2            | 0.21                        | 0.28 | 0.37 | 0.46 | 0.57 | 0.69 | 0.82 | 0.97 | 1.85 |
| 10                | 328.0            | 0.23                        | 0.31 | 0.40 | 0.51 | 0.63 | 0.76 | 0.90 | 1.06 | 2.03 |
| 15                | 298.1            | 0.25                        | 0.34 | 0.44 | 0.56 | 0.69 | 0.84 | 1.00 | 1.17 | 2.24 |
| 20                | 270.7            | 0.27                        | 0.37 | 0.49 | 0.62 | 0.76 | 0.92 | 1.10 | 1.29 | 2.47 |
| 25                | 246.1            | 0.30                        | 0.41 | 0.54 | 0.68 | 0.84 | 1.01 | 1.21 | 1.41 | 2.71 |
| 30                | 224.3            | 0.33                        | 0.45 | 0.59 | 0.74 | 0.92 | 1.11 | 1.32 | 1.55 | 2.98 |
| 35                | 205.4            | 0.36                        | 0.49 | 0.64 | 0.81 | 1.00 | 1.21 | 1.44 | 1.69 | 3.25 |
| 40                | 189.1            | 0.39                        | 0.53 | 0.70 | 0.88 | 1.09 | 1.32 | 1.57 | 1.84 | 3.53 |
| 45                | 175.0            | 0.42                        | 0.58 | 0.75 | 0.95 | 1.18 | 1.42 | 1.69 | 1.99 | 3.81 |
| 50                | 163.0            | 0.46                        | 0.62 | 0.81 | 1.02 | 1.26 | 1.53 | 1.82 | 2.14 | 4.10 |
| 55                | 152.6            | 0.49                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.28 | 4.37 |
| 60                | 143.6            | 0.52                        | 0.70 | 0.92 | 1.16 | 1.43 | 1.74 | 2.07 | 2.42 | 4.65 |
| 65                | 135.8            | 0.55                        | 0.74 | 0.97 | 1.23 | 1.52 | 1.83 | 2.18 | 2.56 | 4.91 |
| 70                | 129.0            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.60 | 1.93 | 2.30 | 2.70 | 5.17 |
| 75                | 123.0            | 0.60                        | 0.82 | 1.07 | 1.36 | 1.68 | 2.03 | 2.41 | 2.83 | 5.43 |
| 80                | 117.6            | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 5.67 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 59 | <b>of</b> | 98 |

Appendix 28 - .025" or 16mm<sup>2</sup> HDBC - (All altitudes) - 90m Basic

## CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |            |
|------------------------------------------------------|------------|
| Conductor Code Name (if any)                         | .025" HDBC |
| Greased Conductor Weight (kg/m)                      | 0.1473     |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 16.4416    |
| Conductor Overall Diameter (mm)                      | 5.689      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05   |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3    |
| Rated Breaking Strength of Conductor (kgf)           | 689.4      |
| Basic / Recommended Span (m)                         | 90         |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760        |
| Radial Ice Thickness (mm)                            | 0          |
| Ice Density (kg/m <sup>3</sup> )                     | 913        |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 275.8      |
| Temperature at MWT Limit (Degrees C)                 | -5.6       |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 229.8      |
| Temperature at EDT Limit (Degrees C)                 | 5          |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 275.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.147 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.441 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 137.3 |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                       |                               |   |
|-----------------------|-------------------------------|---|
| Conductor: .025" HDBC | Temperature Shift (Deg. C)    | 0 |
|                       | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.20                              | 0.36                                        | 0.37  | 0.38  | 0.39  | 0.40  | 0.41  |
| 80             | 0.80                              | 1.42                                        | 1.47  | 1.51  | 1.56  | 1.60  | 1.65  |
| 120            | 1.81                              | 3.20                                        | 3.30  | 3.41  | 3.51  | 3.61  | 3.71  |
| 160            | 3.21                              | 5.68                                        | 5.87  | 6.06  | 6.24  | 6.42  | 6.60  |
| 200            | 5.01                              | 8.88                                        | 9.17  | 9.46  | 9.75  | 10.03 | 10.31 |
| 240            | 7.22                              | 12.78                                       | 13.21 | 13.63 | 14.04 | 14.44 | 14.84 |
| 280            | 9.83                              | 17.40                                       | 17.98 | 18.55 | 19.11 | 19.66 | 20.20 |
| 320            | 12.84                             | 22.73                                       | 23.49 | 24.23 | 24.96 | 25.68 | 26.38 |
| 360            | 16.25                             | 28.77                                       | 29.72 | 30.67 | 31.59 | 32.50 | 33.39 |
| 400            | 20.06                             | 35.51                                       | 36.70 | 37.86 | 39.00 | 40.12 | 41.22 |
| 440            | 24.27                             | 42.97                                       | 44.40 | 45.81 | 47.19 | 48.55 | 49.88 |
| 480            | 28.89                             | 51.14                                       | 52.84 | 54.52 | 56.16 | 57.78 | 59.36 |
| 520            | 33.90                             | 60.02                                       | 62.02 | 63.98 | 65.91 | 67.81 | 69.67 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 60 | <b>of</b> | 98 |  |

## Appendix 28a - .025" or 16mm HDBC - (All altitudes) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name .025" HDBC  
 Basic / Recommended Span (m) 90  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 146.9            | 0.45                        | 0.61 | 0.80 | 1.02 | 1.25 | 1.52 | 1.81 | 2.12 | 4.06 |
| 0                 | 137.3            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.34 | 1.62 | 1.93 | 2.27 | 4.35 |
| 5                 | 129.5            | 0.51                        | 0.70 | 0.91 | 1.15 | 1.42 | 1.72 | 2.05 | 2.40 | 4.61 |
| 10                | 122.5            | 0.54                        | 0.74 | 0.96 | 1.22 | 1.50 | 1.82 | 2.16 | 2.54 | 4.87 |
| 15                | 116.2            | 0.57                        | 0.78 | 1.01 | 1.28 | 1.58 | 1.92 | 2.28 | 2.68 | 5.13 |
| 20                | 110.5            | 0.60                        | 0.82 | 1.07 | 1.35 | 1.67 | 2.02 | 2.40 | 2.82 | 5.40 |
| 25                | 105.3            | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.95 | 5.66 |
| 30                | 100.7            | 0.66                        | 0.90 | 1.17 | 1.48 | 1.83 | 2.21 | 2.63 | 3.09 | 5.93 |
| 35                | 96.4             | 0.69                        | 0.94 | 1.22 | 1.55 | 1.91 | 2.31 | 2.75 | 3.23 | 6.19 |
| 40                | 92.6             | 0.72                        | 0.97 | 1.27 | 1.61 | 1.99 | 2.41 | 2.86 | 3.36 | 6.44 |
| 45                | 89.1             | 0.74                        | 1.01 | 1.32 | 1.67 | 2.07 | 2.50 | 2.98 | 3.49 | 6.69 |
| 50                | 85.9             | 0.77                        | 1.05 | 1.37 | 1.74 | 2.14 | 2.59 | 3.09 | 3.62 | 6.94 |
| 55                | 83.0             | 0.80                        | 1.09 | 1.42 | 1.80 | 2.22 | 2.68 | 3.19 | 3.75 | 7.19 |
| 60                | 80.3             | 0.83                        | 1.12 | 1.47 | 1.86 | 2.29 | 2.77 | 3.30 | 3.87 | 7.43 |
| 65                | 77.9             | 0.85                        | 1.16 | 1.51 | 1.92 | 2.36 | 2.86 | 3.41 | 4.00 | 7.66 |
| 70                | 75.6             | 0.88                        | 1.19 | 1.56 | 1.97 | 2.44 | 2.95 | 3.51 | 4.12 | 7.89 |
| 75                | 73.5             | 0.90                        | 1.23 | 1.60 | 2.03 | 2.51 | 3.03 | 3.61 | 4.24 | 8.12 |
| 80                | 71.5             | 0.93                        | 1.26 | 1.65 | 2.09 | 2.58 | 3.12 | 3.71 | 4.35 | 8.34 |

|                    |     |                 |         |      |    |    |    |
|--------------------|-----|-----------------|---------|------|----|----|----|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 61 | of | 98 |

## Appendix 29 - .025" or 16mm<sup>2</sup> HDBC - (All altitudes) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |            |
|------------------------------------------------------|------------|
| Conductor Code Name (if any)                         | .025" HDBC |
| Greased Conductor Weight (kg/m)                      | 0.1473     |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 16.4416    |
| Conductor Overall Diameter (mm)                      | 5.689      |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05   |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3    |
| Rated Breaking Strength of Conductor (kgf)           | 689.4      |
| Basic / Recommended Span (m)                         | 100        |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 760        |
| Radial Ice Thickness (mm)                            | 0          |
| Ice Density (kg/m <sup>3</sup> )                     | 913        |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 275.8      |
| Temperature at MWT Limit (Degrees C)                 | -5.6       |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 229.8      |
| Temperature at EDT Limit (Degrees C)                 | 5          |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 275.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.147 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.441 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 126.7 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                       |                               |   |
|-----------------------|-------------------------------|---|
| Conductor: .025" HDBC | Temperature Shift (Deg. C)    | 0 |
|                       | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.22                              | 0.35                                        | 0.36  | 0.37  | 0.38  | 0.39  | 0.40  |
| 80             | 0.88                              | 1.42                                        | 1.46  | 1.50  | 1.54  | 1.57  | 1.61  |
| 120            | 1.98                              | 3.19                                        | 3.28  | 3.37  | 3.45  | 3.54  | 3.62  |
| 160            | 3.52                              | 5.67                                        | 5.83  | 5.99  | 6.14  | 6.29  | 6.44  |
| 200            | 5.49                              | 8.86                                        | 9.11  | 9.36  | 9.59  | 9.83  | 10.06 |
| 240            | 7.91                              | 12.76                                       | 13.12 | 13.47 | 13.82 | 14.16 | 14.49 |
| 280            | 10.77                             | 17.37                                       | 17.86 | 18.34 | 18.81 | 19.27 | 19.72 |
| 320            | 14.06                             | 22.69                                       | 23.33 | 23.95 | 24.56 | 25.16 | 25.76 |
| 360            | 17.80                             | 28.72                                       | 29.52 | 30.31 | 31.09 | 31.85 | 32.60 |
| 400            | 21.97                             | 35.46                                       | 36.45 | 37.42 | 38.38 | 39.32 | 40.25 |
| 440            | 26.59                             | 42.90                                       | 44.10 | 45.28 | 46.44 | 47.58 | 48.70 |
| 480            | 31.64                             | 51.06                                       | 52.49 | 53.89 | 55.27 | 56.62 | 57.95 |
| 520            | 37.14                             | 59.92                                       | 61.60 | 63.24 | 64.86 | 66.45 | 68.01 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 62 | <b>of</b> | 98 |  |

## Appendix 29a - .025" or 16mm<sup>2</sup> HDBC - (All altitudes) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name .025" HDBC  
 Basic / Recommended Span (m) 100  
 Temperature Shift for Creep (Deg. C) 0 Insert minus sign as necessary  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 134.1            | 0.49                        | 0.67 | 0.88 | 1.11 | 1.37 | 1.66 | 1.98 | 2.32 | 4.45 |
| 0                 | 126.7            | 0.52                        | 0.71 | 0.93 | 1.18 | 1.45 | 1.76 | 2.09 | 2.46 | 4.71 |
| 5                 | 120.8            | 0.55                        | 0.75 | 0.98 | 1.23 | 1.52 | 1.84 | 2.19 | 2.58 | 4.94 |
| 10                | 115.4            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.60 | 1.93 | 2.30 | 2.70 | 5.17 |
| 15                | 110.5            | 0.60                        | 0.82 | 1.07 | 1.35 | 1.67 | 2.02 | 2.40 | 2.82 | 5.40 |
| 20                | 106.1            | 0.62                        | 0.85 | 1.11 | 1.41 | 1.74 | 2.10 | 2.50 | 2.93 | 5.62 |
| 25                | 102.0            | 0.65                        | 0.88 | 1.16 | 1.46 | 1.81 | 2.18 | 2.60 | 3.05 | 5.85 |
| 30                | 98.3             | 0.67                        | 0.92 | 1.20 | 1.52 | 1.87 | 2.27 | 2.70 | 3.17 | 6.07 |
| 35                | 94.9             | 0.70                        | 0.95 | 1.24 | 1.57 | 1.94 | 2.35 | 2.80 | 3.28 | 6.29 |
| 40                | 91.7             | 0.72                        | 0.98 | 1.28 | 1.63 | 2.01 | 2.43 | 2.89 | 3.39 | 6.50 |
| 45                | 88.8             | 0.75                        | 1.02 | 1.33 | 1.68 | 2.07 | 2.51 | 2.98 | 3.50 | 6.72 |
| 50                | 86.2             | 0.77                        | 1.05 | 1.37 | 1.73 | 2.14 | 2.59 | 3.08 | 3.61 | 6.92 |
| 55                | 83.7             | 0.79                        | 1.08 | 1.41 | 1.78 | 2.20 | 2.66 | 3.17 | 3.72 | 7.13 |
| 60                | 81.4             | 0.81                        | 1.11 | 1.45 | 1.83 | 2.26 | 2.74 | 3.26 | 3.82 | 7.33 |
| 65                | 79.2             | 0.84                        | 1.14 | 1.49 | 1.88 | 2.32 | 2.81 | 3.35 | 3.93 | 7.53 |
| 70                | 77.2             | 0.86                        | 1.17 | 1.53 | 1.93 | 2.38 | 2.88 | 3.43 | 4.03 | 7.72 |
| 75                | 75.4             | 0.88                        | 1.20 | 1.56 | 1.98 | 2.44 | 2.96 | 3.52 | 4.13 | 7.91 |
| 80                | 73.6             | 0.90                        | 1.23 | 1.60 | 2.03 | 2.50 | 3.03 | 3.60 | 4.23 | 8.10 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 63 | <b>of</b> | 98 |

## Appendix 30 - 3 SWG HDBC - (All altitudes) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |          |
|----------------------------------------------------|----------|
| Conductor Code Name (if any)                       | 3SWG     |
| Greased Conductor Weight (kg/m)                    | 0.2861   |
| Cross Sectional Area of Conductor (mm²)            | 32.18    |
| Conductor Overall Diameter (mm)                    | 6.4      |
| Coefficient of Linear Expansion (/Degree C)        | 1.70E-05 |
| Modulus of Elasticity (kg/mm²)                     | 12655.3  |
| Rated Breaking Strength of Conductor (kgf)         | 1301.9   |
| Basic / Recommended Span (m)                       | 90       |
| Wind Pressure on Conductor (N/m²)                  | 760      |
| Radial Ice Thickness (mm)                          | 0        |
| Ice Density (kg/m³)                                | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 520.8    |
| Temperature at MWT Limit (Degrees C)               | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 434.0    |
| Temperature at EDT Limit (Degrees C)               | 5        |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 520.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.286 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.496 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 389.8 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |   |
|------------|------|-------------------------------|---|
| Conductor: | 3SWG | Temperature Shift (Deg. C)    | 0 |
|            |      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.14                              | 0.28                                        | 0.29  | 0.30  | 0.32  | 0.33  | 0.34  |
| 80             | 0.55                              | 1.11                                        | 1.16  | 1.21  | 1.26  | 1.31  | 1.36  |
| 120            | 1.23                              | 2.50                                        | 2.61  | 2.73  | 2.84  | 2.96  | 3.07  |
| 160            | 2.19                              | 4.44                                        | 4.65  | 4.85  | 5.06  | 5.26  | 5.45  |
| 200            | 3.42                              | 6.94                                        | 7.26  | 7.58  | 7.90  | 8.21  | 8.52  |
| 240            | 4.92                              | 9.99                                        | 10.46 | 10.92 | 11.38 | 11.83 | 12.27 |
| 280            | 6.70                              | 13.59                                       | 14.23 | 14.86 | 15.48 | 16.10 | 16.70 |
| 320            | 8.75                              | 17.75                                       | 18.59 | 19.41 | 20.22 | 21.02 | 21.81 |
| 360            | 11.07                             | 22.47                                       | 23.52 | 24.57 | 25.60 | 26.61 | 27.61 |
| 400            | 13.66                             | 27.74                                       | 29.04 | 30.33 | 31.60 | 32.85 | 34.08 |
| 440            | 16.53                             | 33.57                                       | 35.14 | 36.70 | 38.24 | 39.75 | 41.24 |
| 480            | 19.68                             | 39.95                                       | 41.82 | 43.67 | 45.50 | 47.31 | 49.08 |
| 520            | 23.09                             | 46.88                                       | 49.08 | 51.26 | 53.40 | 55.52 | 57.60 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 64 | <b>of</b> | 98 |  |

## Appendix 30a - 3 SWG HDBC - (All altitudes ) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 3SWG  
 Basic / Recommended Span (m) 90  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 418.8            | 0.31                        | 0.42 | 0.55 | 0.69 | 0.85 | 1.03 | 1.23 | 1.44 | 2.77 |
| 0                 | 389.8            | 0.33                        | 0.45 | 0.59 | 0.74 | 0.92 | 1.11 | 1.32 | 1.55 | 2.97 |
| 5                 | 365.4            | 0.35                        | 0.48 | 0.63 | 0.79 | 0.98 | 1.18 | 1.41 | 1.65 | 3.17 |
| 10                | 342.5            | 0.38                        | 0.51 | 0.67 | 0.85 | 1.04 | 1.26 | 1.50 | 1.76 | 3.38 |
| 15                | 321.1            | 0.40                        | 0.55 | 0.71 | 0.90 | 1.11 | 1.35 | 1.60 | 1.88 | 3.61 |
| 20                | 301.3            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.19 | 1.44 | 1.71 | 2.01 | 3.85 |
| 25                | 283.1            | 0.45                        | 0.62 | 0.81 | 1.02 | 1.26 | 1.53 | 1.82 | 2.13 | 4.09 |
| 30                | 266.5            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.34 | 1.62 | 1.93 | 2.27 | 4.35 |
| 35                | 251.4            | 0.51                        | 0.70 | 0.91 | 1.15 | 1.42 | 1.72 | 2.05 | 2.40 | 4.61 |
| 40                | 237.8            | 0.54                        | 0.74 | 0.96 | 1.22 | 1.50 | 1.82 | 2.17 | 2.54 | 4.87 |
| 45                | 225.5            | 0.57                        | 0.78 | 1.01 | 1.28 | 1.59 | 1.92 | 2.28 | 2.68 | 5.14 |
| 50                | 214.4            | 0.60                        | 0.82 | 1.07 | 1.35 | 1.67 | 2.02 | 2.40 | 2.82 | 5.40 |
| 55                | 204.4            | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 5.67 |
| 60                | 195.3            | 0.66                        | 0.90 | 1.17 | 1.48 | 1.83 | 2.22 | 2.64 | 3.09 | 5.93 |
| 65                | 187.1            | 0.69                        | 0.94 | 1.22 | 1.55 | 1.91 | 2.31 | 2.75 | 3.23 | 6.19 |
| 70                | 179.7            | 0.72                        | 0.98 | 1.27 | 1.61 | 1.99 | 2.41 | 2.87 | 3.36 | 6.45 |
| 75                | 172.9            | 0.74                        | 1.01 | 1.32 | 1.68 | 2.07 | 2.50 | 2.98 | 3.50 | 6.70 |
| 80                | 166.7            | 0.77                        | 1.05 | 1.37 | 1.74 | 2.15 | 2.60 | 3.09 | 3.63 | 6.95 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 65 | <b>of</b> | 98 |

## Appendix 31 - 3 SWG HDBC - (All altitude ) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |          |
|----------------------------------------------------|----------|
| Conductor Code Name (if any)                       | 3SWG     |
| Greased Conductor Weight (kg/m)                    | 0.2861   |
| Cross Sectional Area of Conductor (mm²)            | 32.18    |
| Conductor Overall Diameter (mm)                    | 6.4      |
| Coefficient of Linear Expansion (/Degree C)        | 1.70E-05 |
| Modulus of Elasticity (kg/mm²)                     | 12655.3  |
| Rated Breaking Strength of Conductor (kgf)         | 1301.9   |
| Basic / Recommended Span (m)                       | 100      |
| Wind Pressure on Conductor (N/m²)                  | 760      |
| Radial Ice Thickness (mm)                          | 0        |
| Ice Density (kg/m³)                                | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 520.8    |
| Temperature at MWT Limit (Degrees C)               | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 434.0    |
| Temperature at EDT Limit (Degrees C)               | 5        |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 520.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.286 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.496 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 375.4 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |   |
|------------|------|-------------------------------|---|
| Conductor: | 3SWG | Temperature Shift (Deg. C)    | 0 |
|            |      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.14                              | 0.27                                        | 0.28  | 0.29  | 0.30  | 0.31  | 0.33  |
| 80             | 0.57                              | 1.08                                        | 1.13  | 1.17  | 1.22  | 1.26  | 1.30  |
| 120            | 1.28                              | 2.44                                        | 2.54  | 2.64  | 2.74  | 2.83  | 2.93  |
| 160            | 2.28                              | 4.34                                        | 4.51  | 4.69  | 4.87  | 5.04  | 5.21  |
| 200            | 3.56                              | 6.77                                        | 7.05  | 7.33  | 7.60  | 7.87  | 8.13  |
| 240            | 5.13                              | 9.75                                        | 10.16 | 10.55 | 10.95 | 11.33 | 11.71 |
| 280            | 6.98                              | 13.28                                       | 13.82 | 14.37 | 14.90 | 15.43 | 15.94 |
| 320            | 9.12                              | 17.34                                       | 18.06 | 18.76 | 19.46 | 20.15 | 20.83 |
| 360            | 11.54                             | 21.95                                       | 22.85 | 23.75 | 24.63 | 25.50 | 26.36 |
| 400            | 14.24                             | 27.09                                       | 28.21 | 29.32 | 30.41 | 31.48 | 32.54 |
| 440            | 17.24                             | 32.78                                       | 34.14 | 35.48 | 36.79 | 38.09 | 39.37 |
| 480            | 20.51                             | 39.02                                       | 40.63 | 42.22 | 43.79 | 45.33 | 46.86 |
| 520            | 24.07                             | 45.79                                       | 47.68 | 49.55 | 51.39 | 53.20 | 54.99 |

|                    |     |                 |  |         |      |    |    |    |
|--------------------|-----|-----------------|--|---------|------|----|----|----|
| Document reference |     | NSP/004/042/001 |  |         |      |    |    |    |
| Version:-          | 2.0 | Date of Issue:- |  | Sept 10 | Page | 66 | of | 98 |

## Appendix 31 - a3 SWG HDBC - (All altitudes) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |      |                                |
|-----------------------------------------------|------|--------------------------------|
| Conductor Code Name                           | 3SWG |                                |
| Basic / Recommended Span (m)                  | 100  |                                |
| Temperature Shift for Creep (Deg. C)          | 0    | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 0.0  | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 0    | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 401.7            | 0.32                        | 0.44 | 0.57 | 0.72 | 0.89 | 1.08 | 1.28 | 1.50 | 2.88 |
| 0                 | 375.4            | 0.34                        | 0.47 | 0.61 | 0.77 | 0.95 | 1.15 | 1.37 | 1.61 | 3.09 |
| 5                 | 353.4            | 0.36                        | 0.50 | 0.65 | 0.82 | 1.01 | 1.22 | 1.46 | 1.71 | 3.28 |
| 10                | 332.9            | 0.39                        | 0.53 | 0.69 | 0.87 | 1.07 | 1.30 | 1.55 | 1.82 | 3.48 |
| 15                | 313.9            | 0.41                        | 0.56 | 0.73 | 0.92 | 1.14 | 1.38 | 1.64 | 1.93 | 3.69 |
| 20                | 296.4            | 0.43                        | 0.59 | 0.77 | 0.98 | 1.21 | 1.46 | 1.74 | 2.04 | 3.91 |
| 25                | 280.4            | 0.46                        | 0.62 | 0.82 | 1.03 | 1.28 | 1.54 | 1.84 | 2.16 | 4.13 |
| 30                | 265.8            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.27 | 4.36 |
| 35                | 252.5            | 0.51                        | 0.69 | 0.91 | 1.15 | 1.42 | 1.71 | 2.04 | 2.39 | 4.59 |
| 40                | 240.4            | 0.54                        | 0.73 | 0.95 | 1.21 | 1.49 | 1.80 | 2.14 | 2.51 | 4.82 |
| 45                | 229.4            | 0.56                        | 0.76 | 1.00 | 1.26 | 1.56 | 1.89 | 2.25 | 2.64 | 5.05 |
| 50                | 219.4            | 0.59                        | 0.80 | 1.04 | 1.32 | 1.63 | 1.97 | 2.35 | 2.76 | 5.28 |
| 55                | 210.3            | 0.61                        | 0.83 | 1.09 | 1.38 | 1.70 | 2.06 | 2.45 | 2.87 | 5.51 |
| 60                | 202.0            | 0.64                        | 0.87 | 1.13 | 1.43 | 1.77 | 2.14 | 2.55 | 2.99 | 5.74 |
| 65                | 194.4            | 0.66                        | 0.90 | 1.18 | 1.49 | 1.84 | 2.23 | 2.65 | 3.11 | 5.96 |
| 70                | 187.5            | 0.69                        | 0.93 | 1.22 | 1.55 | 1.91 | 2.31 | 2.75 | 3.22 | 6.18 |
| 75                | 181.1            | 0.71                        | 0.97 | 1.26 | 1.60 | 1.97 | 2.39 | 2.84 | 3.34 | 6.40 |
| 80                | 175.2            | 0.73                        | 1.00 | 1.31 | 1.65 | 2.04 | 2.47 | 2.94 | 3.45 | 6.61 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 67 | <b>of</b> | 98 |

## Appendix 32 - 5SWG HDBC - (All altitudes ) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |          |
|----------------------------------------------------|----------|
| Conductor Code Name (if any)                       | 5SWG     |
| Greased Conductor Weight (kg/m)                    | 0.2024   |
| Cross Sectional Area of Conductor (mm²)            | 22.77    |
| Conductor Overall Diameter (mm)                    | 5.38     |
| Coefficient of Linear Expansion (/Degree C)        | 1.70E-05 |
| Modulus of Elasticity (kg/mm²)                     | 12655.3  |
| Rated Breaking Strength of Conductor (kgf)         | 945      |
| Basic / Recommended Span (m)                       | 90       |
| Wind Pressure on Conductor (N/m²)                  | 760      |
| Radial Ice Thickness (mm)                          | 0        |
| Ice Density (kg/m³)                                | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 378.0    |
| Temperature at MWT Limit (Degrees C)               | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 315.0    |
| Temperature at EDT Limit (Degrees C)               | 5        |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 378.0 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.202 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.417 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 262.3 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |   |
|------------|------|-------------------------------|---|
| Conductor: | 5SWG | Temperature Shift (Deg. C)    | 0 |
|            |      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.14                              | 0.29                                        | 0.30  | 0.31  | 0.33  | 0.34  | 0.35  |
| 80             | 0.57                              | 1.15                                        | 1.20  | 1.25  | 1.30  | 1.35  | 1.40  |
| 120            | 1.29                              | 2.59                                        | 2.71  | 2.82  | 2.93  | 3.04  | 3.15  |
| 160            | 2.30                              | 4.60                                        | 4.81  | 5.01  | 5.21  | 5.41  | 5.61  |
| 200            | 3.59                              | 7.19                                        | 7.52  | 7.83  | 8.15  | 8.46  | 8.76  |
| 240            | 5.17                              | 10.36                                       | 10.82 | 11.28 | 11.73 | 12.18 | 12.62 |
| 280            | 7.04                              | 14.10                                       | 14.73 | 15.36 | 15.97 | 16.58 | 17.17 |
| 320            | 9.19                              | 18.41                                       | 19.24 | 20.06 | 20.86 | 21.65 | 22.43 |
| 360            | 11.63                             | 23.31                                       | 24.35 | 25.38 | 26.40 | 27.40 | 28.39 |
| 400            | 14.36                             | 28.77                                       | 30.06 | 31.34 | 32.59 | 33.83 | 35.04 |
| 440            | 17.37                             | 34.82                                       | 36.38 | 37.92 | 39.44 | 40.93 | 42.40 |
| 480            | 20.68                             | 41.43                                       | 43.29 | 45.13 | 46.93 | 48.71 | 50.46 |
| 520            | 24.27                             | 48.63                                       | 50.81 | 52.96 | 55.08 | 57.17 | 59.22 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 68 | <b>of</b> | 98 |  |

## Appendix 32a - 5SWG HDBC - (All altitudes) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 5SWG  
 Basic / Recommended Span (m) 90  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 281.9            | 0.32                        | 0.44 | 0.57 | 0.73 | 0.90 | 1.09 | 1.29 | 1.52 | 2.91 |
| 0                 | 262.3            | 0.35                        | 0.47 | 0.62 | 0.78 | 0.96 | 1.17 | 1.39 | 1.63 | 3.13 |
| 5                 | 245.8            | 0.37                        | 0.50 | 0.66 | 0.83 | 1.03 | 1.25 | 1.48 | 1.74 | 3.33 |
| 10                | 230.4            | 0.40                        | 0.54 | 0.70 | 0.89 | 1.10 | 1.33 | 1.58 | 1.86 | 3.56 |
| 15                | 216.2            | 0.42                        | 0.57 | 0.75 | 0.95 | 1.17 | 1.42 | 1.69 | 1.98 | 3.79 |
| 20                | 203.0            | 0.45                        | 0.61 | 0.80 | 1.01 | 1.25 | 1.51 | 1.79 | 2.11 | 4.04 |
| 25                | 191.0            | 0.48                        | 0.65 | 0.85 | 1.07 | 1.32 | 1.60 | 1.91 | 2.24 | 4.29 |
| 30                | 180.2            | 0.51                        | 0.69 | 0.90 | 1.14 | 1.40 | 1.70 | 2.02 | 2.37 | 4.55 |
| 35                | 170.3            | 0.53                        | 0.73 | 0.95 | 1.20 | 1.49 | 1.80 | 2.14 | 2.51 | 4.81 |
| 40                | 161.4            | 0.56                        | 0.77 | 1.00 | 1.27 | 1.57 | 1.90 | 2.26 | 2.65 | 5.08 |
| 45                | 153.3            | 0.59                        | 0.81 | 1.06 | 1.34 | 1.65 | 2.00 | 2.38 | 2.79 | 5.35 |
| 50                | 146.1            | 0.62                        | 0.85 | 1.11 | 1.40 | 1.73 | 2.10 | 2.49 | 2.93 | 5.61 |
| 55                | 139.5            | 0.65                        | 0.89 | 1.16 | 1.47 | 1.81 | 2.19 | 2.61 | 3.06 | 5.87 |
| 60                | 133.6            | 0.68                        | 0.93 | 1.21 | 1.53 | 1.89 | 2.29 | 2.73 | 3.20 | 6.14 |
| 65                | 128.2            | 0.71                        | 0.97 | 1.26 | 1.60 | 1.97 | 2.39 | 2.84 | 3.33 | 6.39 |
| 70                | 123.3            | 0.74                        | 1.01 | 1.31 | 1.66 | 2.05 | 2.48 | 2.95 | 3.47 | 6.65 |
| 75                | 118.9            | 0.77                        | 1.04 | 1.36 | 1.72 | 2.13 | 2.58 | 3.06 | 3.60 | 6.90 |
| 80                | 114.8            | 0.79                        | 1.08 | 1.41 | 1.79 | 2.20 | 2.67 | 3.17 | 3.72 | 7.14 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 69 | <b>of</b> | 98 |

## Appendix 33 - 5SWG HDBC - (All altitude) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |          |
|----------------------------------------------------|----------|
| Conductor Code Name (if any)                       | 5SWG     |
| Greased Conductor Weight (kg/m)                    | 0.2024   |
| Cross Sectional Area of Conductor (mm²)            | 22.77    |
| Conductor Overall Diameter (mm)                    | 5.38     |
| Coefficient of Linear Expansion (/Degree C)        | 1.70E-05 |
| Modulus of Elasticity (kg/mm²)                     | 12655.3  |
| Rated Breaking Strength of Conductor (kgf)         | 945      |
| Basic / Recommended Span (m)                       | 100      |
| Wind Pressure on Conductor (N/m²)                  | 760      |
| Radial Ice Thickness (mm)                          | 0        |
| Ice Density (kg/m³)                                | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 378.0    |
| Temperature at MWT Limit (Degrees C)               | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 315.0    |
| Temperature at EDT Limit (Degrees C)               | 5        |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 378.0 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.202 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.417 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 249.2 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |   |
|------------|------|-------------------------------|---|
| Conductor: | 5SWG | Temperature Shift (Deg. C)    | 0 |
|            |      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.15                              | 0.28                                        | 0.29  | 0.30  | 0.32  | 0.33  | 0.34  |
| 80             | 0.61                              | 1.13                                        | 1.18  | 1.22  | 1.26  | 1.31  | 1.35  |
| 120            | 1.37                              | 2.55                                        | 2.65  | 2.74  | 2.84  | 2.94  | 3.03  |
| 160            | 2.43                              | 4.53                                        | 4.71  | 4.88  | 5.05  | 5.22  | 5.39  |
| 200            | 3.80                              | 7.08                                        | 7.35  | 7.62  | 7.89  | 8.16  | 8.42  |
| 240            | 5.46                              | 10.19                                       | 10.59 | 10.98 | 11.37 | 11.75 | 12.12 |
| 280            | 7.44                              | 13.87                                       | 14.41 | 14.94 | 15.47 | 15.99 | 16.50 |
| 320            | 9.72                              | 18.12                                       | 18.82 | 19.52 | 20.21 | 20.88 | 21.55 |
| 360            | 12.30                             | 22.93                                       | 23.82 | 24.70 | 25.57 | 26.43 | 27.27 |
| 400            | 15.18                             | 28.31                                       | 29.41 | 30.50 | 31.57 | 32.63 | 33.67 |
| 440            | 18.37                             | 34.25                                       | 35.59 | 36.90 | 38.20 | 39.48 | 40.73 |
| 480            | 21.86                             | 40.76                                       | 42.35 | 43.92 | 45.46 | 46.98 | 48.48 |
| 520            | 25.65                             | 47.84                                       | 49.70 | 51.54 | 53.35 | 55.14 | 56.89 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 70 | <b>of</b> | 98 |  |

## Appendix 33a - 5SWG HDBC - (All altitudes) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 5SWG  
 Basic / Recommended Span (m) 100  
 Temperature Shift for Creep (Deg. C) 0 Insert minus sign as necessary  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 266.7            | 0.34                        | 0.46 | 0.61 | 0.77 | 0.95 | 1.15 | 1.37 | 1.60 | 3.07 |
| 0                 | 249.2            | 0.37                        | 0.50 | 0.65 | 0.82 | 1.02 | 1.23 | 1.46 | 1.72 | 3.29 |
| 5                 | 234.8            | 0.39                        | 0.53 | 0.69 | 0.87 | 1.08 | 1.30 | 1.55 | 1.82 | 3.49 |
| 10                | 221.4            | 0.41                        | 0.56 | 0.73 | 0.93 | 1.14 | 1.38 | 1.65 | 1.93 | 3.70 |
| 15                | 209.1            | 0.44                        | 0.59 | 0.77 | 0.98 | 1.21 | 1.46 | 1.74 | 2.04 | 3.92 |
| 20                | 197.8            | 0.46                        | 0.63 | 0.82 | 1.04 | 1.28 | 1.55 | 1.84 | 2.16 | 4.14 |
| 25                | 187.5            | 0.49                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.28 | 4.37 |
| 30                | 178.1            | 0.51                        | 0.70 | 0.91 | 1.15 | 1.42 | 1.72 | 2.05 | 2.40 | 4.60 |
| 35                | 169.6            | 0.54                        | 0.73 | 0.95 | 1.21 | 1.49 | 1.80 | 2.15 | 2.52 | 4.83 |
| 40                | 161.9            | 0.56                        | 0.77 | 1.00 | 1.27 | 1.56 | 1.89 | 2.25 | 2.64 | 5.06 |
| 45                | 154.8            | 0.59                        | 0.80 | 1.05 | 1.32 | 1.63 | 1.98 | 2.35 | 2.76 | 5.29 |
| 50                | 148.4            | 0.61                        | 0.84 | 1.09 | 1.38 | 1.70 | 2.06 | 2.45 | 2.88 | 5.52 |
| 55                | 142.6            | 0.64                        | 0.87 | 1.14 | 1.44 | 1.77 | 2.15 | 2.56 | 3.00 | 5.75 |
| 60                | 137.2            | 0.66                        | 0.90 | 1.18 | 1.49 | 1.84 | 2.23 | 2.65 | 3.12 | 5.97 |
| 65                | 132.4            | 0.69                        | 0.94 | 1.22 | 1.55 | 1.91 | 2.31 | 2.75 | 3.23 | 6.19 |
| 70                | 127.9            | 0.71                        | 0.97 | 1.27 | 1.60 | 1.98 | 2.39 | 2.85 | 3.34 | 6.41 |
| 75                | 123.8            | 0.74                        | 1.00 | 1.31 | 1.66 | 2.04 | 2.47 | 2.94 | 3.45 | 6.62 |
| 80                | 120.0            | 0.76                        | 1.03 | 1.35 | 1.71 | 2.11 | 2.55 | 3.04 | 3.56 | 6.83 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 71 | <b>of</b> | 98 |

## Appendix 34 - 6SWG HDBC - (All altitudes) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |          |
|----------------------------------------------------|----------|
| Conductor Code Name (if any)                       | 6SWG     |
| Greased Conductor Weight (kg/m)                    | 0.16606  |
| Cross Sectional Area of Conductor (mm²)            | 18.68    |
| Conductor Overall Diameter (mm)                    | 4.88     |
| Coefficient of Linear Expansion (/Degree C)        | 1.70E-05 |
| Modulus of Elasticity (kg/mm²)                     | 12655.29 |
| Rated Breaking Strength of Conductor (kgf)         | 787      |
| Basic / Recommended Span (m)                       | 90       |
| Wind Pressure on Conductor (N/m²)                  | 760      |
| Radial Ice Thickness (mm)                          | 0        |
| Ice Density (kg/m³)                                | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 314.8    |
| Temperature at MWT Limit (Degrees C)               | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 262.3    |
| Temperature at EDT Limit (Degrees C)               | 5        |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 314.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.166 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.378 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 206.5 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |   |
|------------|------|-------------------------------|---|
| Conductor: | 6SWG | Temperature Shift (Deg. C)    | 0 |
|            |      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.15                              | 0.30                                        | 0.31  | 0.32  | 0.33  | 0.35  | 0.36  |
| 80             | 0.60                              | 1.18                                        | 1.24  | 1.29  | 1.34  | 1.38  | 1.43  |
| 120            | 1.35                              | 2.66                                        | 2.78  | 2.89  | 3.01  | 3.12  | 3.22  |
| 160            | 2.39                              | 4.74                                        | 4.94  | 5.14  | 5.34  | 5.54  | 5.73  |
| 200            | 3.74                              | 7.40                                        | 7.72  | 8.04  | 8.35  | 8.65  | 8.95  |
| 240            | 5.38                              | 10.66                                       | 11.12 | 11.57 | 12.02 | 12.46 | 12.89 |
| 280            | 7.33                              | 14.51                                       | 15.14 | 15.75 | 16.36 | 16.96 | 17.55 |
| 320            | 9.57                              | 18.95                                       | 19.77 | 20.58 | 21.37 | 22.15 | 22.92 |
| 360            | 12.11                             | 23.98                                       | 25.02 | 26.04 | 27.05 | 28.04 | 29.01 |
| 400            | 14.96                             | 29.61                                       | 30.89 | 32.15 | 33.39 | 34.62 | 35.82 |
| 440            | 18.10                             | 35.83                                       | 37.38 | 38.90 | 40.41 | 41.89 | 43.34 |
| 480            | 21.54                             | 42.64                                       | 44.48 | 46.30 | 48.09 | 49.85 | 51.58 |
| 520            | 25.28                             | 50.04                                       | 52.20 | 54.34 | 56.44 | 58.50 | 60.53 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 72 | <b>of</b> | 98 |  |

## Appendix 34a - 6SWG HDBC - (All altitudes) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |      |                                |
|-----------------------------------------------|------|--------------------------------|
| Conductor Code Name                           | 6SWG |                                |
| Basic / Recommended Span (m)                  | 90   |                                |
| Temperature Shift for Creep (Deg. C)          | 0    | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 0.0  | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 0    | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 222.1            | 0.34                        | 0.46 | 0.60 | 0.76 | 0.93 | 1.13 | 1.35 | 1.58 | 3.03 |
| 0                 | 206.5            | 0.36                        | 0.49 | 0.64 | 0.81 | 1.01 | 1.22 | 1.45 | 1.70 | 3.26 |
| 5                 | 193.6            | 0.39                        | 0.53 | 0.69 | 0.87 | 1.07 | 1.30 | 1.54 | 1.81 | 3.47 |
| 10                | 181.5            | 0.41                        | 0.56 | 0.73 | 0.93 | 1.14 | 1.38 | 1.65 | 1.93 | 3.71 |
| 15                | 170.4            | 0.44                        | 0.60 | 0.78 | 0.99 | 1.22 | 1.47 | 1.75 | 2.06 | 3.95 |
| 20                | 160.2            | 0.47                        | 0.63 | 0.83 | 1.05 | 1.30 | 1.57 | 1.87 | 2.19 | 4.20 |
| 25                | 151.0            | 0.49                        | 0.67 | 0.88 | 1.11 | 1.37 | 1.66 | 1.98 | 2.32 | 4.45 |
| 30                | 142.6            | 0.52                        | 0.71 | 0.93 | 1.18 | 1.46 | 1.76 | 2.10 | 2.46 | 4.72 |
| 35                | 135.0            | 0.55                        | 0.75 | 0.98 | 1.25 | 1.54 | 1.86 | 2.21 | 2.60 | 4.98 |
| 40                | 128.2            | 0.58                        | 0.79 | 1.04 | 1.31 | 1.62 | 1.96 | 2.33 | 2.74 | 5.25 |
| 45                | 122.0            | 0.61                        | 0.83 | 1.09 | 1.38 | 1.70 | 2.06 | 2.45 | 2.88 | 5.51 |
| 50                | 116.4            | 0.64                        | 0.87 | 1.14 | 1.44 | 1.78 | 2.16 | 2.57 | 3.01 | 5.78 |
| 55                | 111.4            | 0.67                        | 0.91 | 1.19 | 1.51 | 1.86 | 2.26 | 2.68 | 3.15 | 6.04 |
| 60                | 106.8            | 0.70                        | 0.95 | 1.24 | 1.57 | 1.94 | 2.35 | 2.80 | 3.29 | 6.30 |
| 65                | 102.6            | 0.73                        | 0.99 | 1.29 | 1.64 | 2.02 | 2.45 | 2.91 | 3.42 | 6.55 |
| 70                | 98.8             | 0.76                        | 1.03 | 1.34 | 1.70 | 2.10 | 2.54 | 3.02 | 3.55 | 6.80 |
| 75                | 95.4             | 0.78                        | 1.07 | 1.39 | 1.76 | 2.18 | 2.63 | 3.13 | 3.68 | 7.05 |
| 80                | 92.2             | 0.81                        | 1.10 | 1.44 | 1.82 | 2.25 | 2.72 | 3.24 | 3.80 | 7.29 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 73 | <b>of</b> | 98 |

## Appendix 35 - 6SWG HDBC - (All altitudes) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |          |
|----------------------------------------------------|----------|
| Conductor Code Name (if any)                       | 6SWG     |
| Greased Conductor Weight (kg/m)                    | 0.16606  |
| Cross Sectional Area of Conductor (mm²)            | 18.68    |
| Conductor Overall Diameter (mm)                    | 4.88     |
| Coefficient of Linear Expansion (/Degree C)        | 1.70E-05 |
| Modulus of Elasticity (kg/mm²)                     | 12655.29 |
| Rated Breaking Strength of Conductor (kgf)         | 787      |
| Basic / Recommended Span (m)                       | 100      |
| Wind Pressure on Conductor (N/m²)                  | 760      |
| Radial Ice Thickness (mm)                          | 0        |
| Ice Density (kg/m³)                                | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 314.8    |
| Temperature at MWT Limit (Degrees C)               | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 262.3    |
| Temperature at EDT Limit (Degrees C)               | 5        |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 314.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.166 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.378 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 194.5 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |   |
|------------|------|-------------------------------|---|
| Conductor: | 6SWG | Temperature Shift (Deg. C)    | 0 |
|            |      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.16                              | 0.29                                        | 0.30  | 0.31  | 0.32  | 0.34  | 0.35  |
| 80             | 0.64                              | 1.17                                        | 1.21  | 1.26  | 1.30  | 1.34  | 1.38  |
| 120            | 1.44                              | 2.63                                        | 2.73  | 2.83  | 2.92  | 3.02  | 3.11  |
| 160            | 2.55                              | 4.68                                        | 4.86  | 5.03  | 5.20  | 5.36  | 5.53  |
| 200            | 3.99                              | 7.32                                        | 7.59  | 7.86  | 8.12  | 8.38  | 8.64  |
| 240            | 5.75                              | 10.53                                       | 10.93 | 11.31 | 11.69 | 12.07 | 12.44 |
| 280            | 7.82                              | 14.34                                       | 14.87 | 15.40 | 15.92 | 16.43 | 16.93 |
| 320            | 10.22                             | 18.73                                       | 19.43 | 20.11 | 20.79 | 21.46 | 22.11 |
| 360            | 12.93                             | 23.70                                       | 24.59 | 25.46 | 26.31 | 27.16 | 27.99 |
| 400            | 15.97                             | 29.26                                       | 30.35 | 31.43 | 32.49 | 33.53 | 34.55 |
| 440            | 19.32                             | 35.41                                       | 36.73 | 38.03 | 39.31 | 40.57 | 41.81 |
| 480            | 22.99                             | 42.14                                       | 43.71 | 45.26 | 46.78 | 48.28 | 49.75 |
| 520            | 26.98                             | 49.45                                       | 51.30 | 53.11 | 54.90 | 56.66 | 58.39 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 74 | <b>of</b> | 98 |  |

## Appendix 35a - 6SWG HDBC - (All altitudes) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 6SWG  
 Basic / Recommended Span (m) 100  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 208.0            | 0.36                        | 0.49 | 0.64 | 0.81 | 1.00 | 1.21 | 1.44 | 1.69 | 3.23 |
| 0                 | 194.5            | 0.38                        | 0.52 | 0.68 | 0.86 | 1.07 | 1.29 | 1.54 | 1.80 | 3.46 |
| 5                 | 183.4            | 0.41                        | 0.55 | 0.72 | 0.92 | 1.13 | 1.37 | 1.63 | 1.91 | 3.67 |
| 10                | 173.1            | 0.43                        | 0.59 | 0.77 | 0.97 | 1.20 | 1.45 | 1.73 | 2.03 | 3.88 |
| 15                | 163.7            | 0.46                        | 0.62 | 0.81 | 1.03 | 1.27 | 1.53 | 1.83 | 2.14 | 4.11 |
| 20                | 155.2            | 0.48                        | 0.66 | 0.86 | 1.08 | 1.34 | 1.62 | 1.93 | 2.26 | 4.33 |
| 25                | 147.4            | 0.51                        | 0.69 | 0.90 | 1.14 | 1.41 | 1.70 | 2.03 | 2.38 | 4.56 |
| 30                | 140.2            | 0.53                        | 0.73 | 0.95 | 1.20 | 1.48 | 1.79 | 2.13 | 2.50 | 4.80 |
| 35                | 133.8            | 0.56                        | 0.76 | 0.99 | 1.26 | 1.55 | 1.88 | 2.23 | 2.62 | 5.03 |
| 40                | 127.9            | 0.58                        | 0.80 | 1.04 | 1.31 | 1.62 | 1.96 | 2.34 | 2.74 | 5.26 |
| 45                | 122.6            | 0.61                        | 0.83 | 1.08 | 1.37 | 1.69 | 2.05 | 2.44 | 2.86 | 5.49 |
| 50                | 117.7            | 0.63                        | 0.86 | 1.13 | 1.43 | 1.76 | 2.13 | 2.54 | 2.98 | 5.71 |
| 55                | 113.3            | 0.66                        | 0.90 | 1.17 | 1.48 | 1.83 | 2.22 | 2.64 | 3.10 | 5.94 |
| 60                | 109.2            | 0.68                        | 0.93 | 1.22 | 1.54 | 1.90 | 2.30 | 2.74 | 3.21 | 6.16 |
| 65                | 105.5            | 0.71                        | 0.96 | 1.26 | 1.59 | 1.97 | 2.38 | 2.83 | 3.32 | 6.37 |
| 70                | 102.1            | 0.73                        | 1.00 | 1.30 | 1.65 | 2.03 | 2.46 | 2.93 | 3.44 | 6.59 |
| 75                | 98.9             | 0.76                        | 1.03 | 1.34 | 1.70 | 2.10 | 2.54 | 3.02 | 3.55 | 6.80 |
| 80                | 96.0             | 0.78                        | 1.06 | 1.38 | 1.75 | 2.16 | 2.62 | 3.11 | 3.65 | 7.01 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 75 | <b>of</b> | 98 |

## Appendix 36 - 7SWG HDBC - (All altitudes) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |          |
|----------------------------------------------------|----------|
| Conductor Code Name (if any)                       | 7SWG     |
| Greased Conductor Weight (kg/m)                    | 0.13953  |
| Cross Sectional Area of Conductor (mm²)            | 15.7     |
| Conductor Overall Diameter (mm)                    | 4.47     |
| Coefficient of Linear Expansion (/Degree C)        | 1.70E-05 |
| Modulus of Elasticity (kg/mm²)                     | 12655.3  |
| Rated Breaking Strength of Conductor (kgf)         | 671      |
| Basic / Recommended Span (m)                       | 90       |
| Wind Pressure on Conductor (N/m²)                  | 760      |
| Radial Ice Thickness (mm)                          | 0        |
| Ice Density (kg/m³)                                | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 268.4    |
| Temperature at MWT Limit (Degrees C)               | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 223.7    |
| Temperature at EDT Limit (Degrees C)               | 5        |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 268.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.140 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.346 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 166.7 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |   |
|------------|------|-------------------------------|---|
| Conductor: | 7SWG | Temperature Shift (Deg. C)    | 0 |
|            |      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.16                              | 0.30                                        | 0.32  | 0.33  | 0.34  | 0.35  | 0.37  |
| 80             | 0.62                              | 1.22                                        | 1.27  | 1.32  | 1.37  | 1.42  | 1.46  |
| 120            | 1.40                              | 2.74                                        | 2.85  | 2.97  | 3.08  | 3.19  | 3.29  |
| 160            | 2.49                              | 4.87                                        | 5.07  | 5.27  | 5.47  | 5.66  | 5.85  |
| 200            | 3.89                              | 7.61                                        | 7.93  | 8.24  | 8.55  | 8.85  | 9.14  |
| 240            | 5.61                              | 10.96                                       | 11.41 | 11.86 | 12.30 | 12.74 | 13.17 |
| 280            | 7.63                              | 14.91                                       | 15.53 | 16.15 | 16.75 | 17.34 | 17.92 |
| 320            | 9.97                              | 19.48                                       | 20.29 | 21.09 | 21.88 | 22.65 | 23.41 |
| 360            | 12.61                             | 24.65                                       | 25.68 | 26.69 | 27.69 | 28.67 | 29.63 |
| 400            | 15.57                             | 30.43                                       | 31.70 | 32.95 | 34.18 | 35.39 | 36.58 |
| 440            | 18.84                             | 36.82                                       | 38.36 | 39.87 | 41.36 | 42.82 | 44.26 |
| 480            | 22.42                             | 43.82                                       | 45.65 | 47.45 | 49.22 | 50.96 | 52.67 |
| 520            | 26.32                             | 51.43                                       | 53.57 | 55.69 | 57.76 | 59.81 | 61.82 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 76 | <b>of</b> | 98 |  |

## Appendix 36a - 7SWG HDBC - (All altitudes) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 7SWG  
 Basic / Recommended Span (m) 90  
 Temperature Shift for Creep (Deg. C) 0 Insert minus sign as necessary  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 179.2            | 0.35                        | 0.48 | 0.62 | 0.79 | 0.97 | 1.18 | 1.40 | 1.64 | 3.15 |
| 0                 | 166.7            | 0.38                        | 0.51 | 0.67 | 0.85 | 1.05 | 1.27 | 1.51 | 1.77 | 3.39 |
| 5                 | 156.2            | 0.40                        | 0.55 | 0.71 | 0.90 | 1.12 | 1.35 | 1.61 | 1.89 | 3.62 |
| 10                | 146.6            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.19 | 1.44 | 1.71 | 2.01 | 3.85 |
| 15                | 137.8            | 0.46                        | 0.62 | 0.81 | 1.03 | 1.27 | 1.53 | 1.82 | 2.14 | 4.10 |
| 20                | 129.7            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.34 | 1.63 | 1.94 | 2.27 | 4.36 |
| 25                | 122.4            | 0.51                        | 0.70 | 0.91 | 1.15 | 1.43 | 1.72 | 2.05 | 2.41 | 4.62 |
| 30                | 115.8            | 0.54                        | 0.74 | 0.96 | 1.22 | 1.51 | 1.82 | 2.17 | 2.55 | 4.88 |
| 35                | 109.8            | 0.57                        | 0.78 | 1.02 | 1.29 | 1.59 | 1.92 | 2.29 | 2.69 | 5.15 |
| 40                | 104.4            | 0.60                        | 0.82 | 1.07 | 1.35 | 1.67 | 2.02 | 2.41 | 2.82 | 5.41 |
| 45                | 99.5             | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 5.68 |
| 50                | 95.1             | 0.66                        | 0.90 | 1.17 | 1.49 | 1.83 | 2.22 | 2.64 | 3.10 | 5.94 |
| 55                | 91.1             | 0.69                        | 0.94 | 1.23 | 1.55 | 1.91 | 2.32 | 2.76 | 3.23 | 6.20 |
| 60                | 87.5             | 0.72                        | 0.98 | 1.28 | 1.61 | 1.99 | 2.41 | 2.87 | 3.37 | 6.46 |
| 65                | 84.2             | 0.75                        | 1.01 | 1.33 | 1.68 | 2.07 | 2.51 | 2.98 | 3.50 | 6.71 |
| 70                | 81.2             | 0.77                        | 1.05 | 1.37 | 1.74 | 2.15 | 2.60 | 3.09 | 3.63 | 6.96 |
| 75                | 78.5             | 0.80                        | 1.09 | 1.42 | 1.80 | 2.22 | 2.69 | 3.20 | 3.76 | 7.20 |
| 80                | 75.9             | 0.83                        | 1.13 | 1.47 | 1.86 | 2.30 | 2.78 | 3.31 | 3.88 | 7.44 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 77 | <b>of</b> | 98 |

## Appendix 37 - 7SWG HDBC - (All altitude ) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |          |
|----------------------------------------------------|----------|
| Conductor Code Name (if any)                       | 7SWG     |
| Greased Conductor Weight (kg/m)                    | 0.13953  |
| Cross Sectional Area of Conductor (mm²)            | 15.7     |
| Conductor Overall Diameter (mm)                    | 4.47     |
| Coefficient of Linear Expansion (/Degree C)        | 1.70E-05 |
| Modulus of Elasticity (kg/mm²)                     | 12655.3  |
| Rated Breaking Strength of Conductor (kgf)         | 671      |
| Basic / Recommended Span (m)                       | 100      |
| Wind Pressure on Conductor (N/m²)                  | 760      |
| Radial Ice Thickness (mm)                          | 0        |
| Ice Density (kg/m³)                                | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 268.4    |
| Temperature at MWT Limit (Degrees C)               | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 223.7    |
| Temperature at EDT Limit (Degrees C)               | 5        |

|                                         |    |        |       |
|-----------------------------------------|----|--------|-------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 268.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 0.140 |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 0.346 |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 155.7 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |   |
|------------|------|-------------------------------|---|
| Conductor: | 7SWG | Temperature Shift (Deg. C)    | 0 |
|            |      | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.17                              | 0.30                                        | 0.31  | 0.32  | 0.33  | 0.34  | 0.35  |
| 80             | 0.67                              | 1.21                                        | 1.25  | 1.29  | 1.34  | 1.38  | 1.42  |
| 120            | 1.51                              | 2.72                                        | 2.81  | 2.91  | 3.00  | 3.10  | 3.19  |
| 160            | 2.68                              | 4.83                                        | 5.00  | 5.17  | 5.34  | 5.50  | 5.67  |
| 200            | 4.19                              | 7.55                                        | 7.82  | 8.08  | 8.34  | 8.60  | 8.85  |
| 240            | 6.04                              | 10.87                                       | 11.26 | 11.64 | 12.02 | 12.39 | 12.75 |
| 280            | 8.22                              | 14.80                                       | 15.32 | 15.84 | 16.36 | 16.86 | 17.35 |
| 320            | 10.74                             | 19.33                                       | 20.02 | 20.69 | 21.36 | 22.02 | 22.66 |
| 360            | 13.59                             | 24.46                                       | 25.33 | 26.19 | 27.04 | 27.87 | 28.68 |
| 400            | 16.77                             | 30.20                                       | 31.27 | 32.34 | 33.38 | 34.40 | 35.41 |
| 440            | 20.30                             | 36.54                                       | 37.84 | 39.13 | 40.39 | 41.63 | 42.85 |
| 480            | 24.16                             | 43.48                                       | 45.04 | 46.56 | 48.07 | 49.54 | 51.00 |
| 520            | 28.35                             | 51.03                                       | 52.85 | 54.65 | 56.41 | 58.14 | 59.85 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 78 | <b>of</b> | 98 |  |

## Appendix 37a - 7SWG HDBC - (All altitude) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 7SWG  
 Basic / Recommended Span (m) 100  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 166.4            | 0.38                        | 0.51 | 0.67 | 0.85 | 1.05 | 1.27 | 1.51 | 1.77 | 3.40 |
| 0                 | 155.7            | 0.40                        | 0.55 | 0.72 | 0.91 | 1.12 | 1.36 | 1.61 | 1.89 | 3.63 |
| 5                 | 147.0            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.19 | 1.44 | 1.71 | 2.01 | 3.85 |
| 10                | 138.9            | 0.45                        | 0.62 | 0.80 | 1.02 | 1.26 | 1.52 | 1.81 | 2.12 | 4.07 |
| 15                | 131.6            | 0.48                        | 0.65 | 0.85 | 1.07 | 1.33 | 1.60 | 1.91 | 2.24 | 4.29 |
| 20                | 124.9            | 0.50                        | 0.68 | 0.89 | 1.13 | 1.40 | 1.69 | 2.01 | 2.36 | 4.52 |
| 25                | 118.9            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.78 | 2.11 | 2.48 | 4.75 |
| 30                | 113.3            | 0.55                        | 0.75 | 0.98 | 1.25 | 1.54 | 1.86 | 2.22 | 2.60 | 4.99 |
| 35                | 108.3            | 0.58                        | 0.79 | 1.03 | 1.30 | 1.61 | 1.95 | 2.32 | 2.72 | 5.22 |
| 40                | 103.8            | 0.61                        | 0.82 | 1.08 | 1.36 | 1.68 | 2.03 | 2.42 | 2.84 | 5.45 |
| 45                | 99.6             | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 5.67 |
| 50                | 95.8             | 0.66                        | 0.89 | 1.16 | 1.47 | 1.82 | 2.20 | 2.62 | 3.08 | 5.90 |
| 55                | 92.4             | 0.68                        | 0.93 | 1.21 | 1.53 | 1.89 | 2.28 | 2.72 | 3.19 | 6.12 |
| 60                | 89.2             | 0.70                        | 0.96 | 1.25 | 1.58 | 1.96 | 2.37 | 2.82 | 3.31 | 6.34 |
| 65                | 86.3             | 0.73                        | 0.99 | 1.29 | 1.64 | 2.02 | 2.45 | 2.91 | 3.42 | 6.55 |
| 70                | 83.6             | 0.75                        | 1.02 | 1.34 | 1.69 | 2.09 | 2.53 | 3.01 | 3.53 | 6.76 |
| 75                | 81.1             | 0.77                        | 1.05 | 1.38 | 1.74 | 2.15 | 2.60 | 3.10 | 3.64 | 6.97 |
| 80                | 78.8             | 0.80                        | 1.09 | 1.42 | 1.79 | 2.21 | 2.68 | 3.19 | 3.74 | 7.17 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 79 | <b>of</b> | 98 |

Appendix 38 - .175" ACSR / 175mm ACSR (Lynx) - (All altitudes ) - 76.2mm Basic  
Matched to Table 5 & 6 of CE/C/36 Specification.

## CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                    |         |
|----------------------------------------------------|---------|
| Conductor Code Name (if any)                       | LYNX    |
| Greased Conductor Weight (kg/m)                    | 0.8626  |
| Cross Sectional Area of Conductor (mm²)            | 226.2   |
| Conductor Overall Diameter (mm)                    | 19.5    |
| Coefficient of Linear Expansion (/Degree C)        | 1.8E-05 |
| Modulus of Elasticity (kg/mm²)                     | 8157.73 |
| Rated Breaking Strength of Conductor (kgf)         | 8137.3  |
| Basic / Recommended Span (m)                       | 76.2    |
| Wind Pressure on Conductor (N/m²)                  | 380     |
| Radial Ice Thickness (mm)                          | 9.5     |
| Ice Density (kg/m³)                                | 913     |
| Absolute Maximum Working Tension (MWT) Limit (kgf) | 2312.8  |
| Temperature at MWT Limit (Degrees C)               | -5.6    |
| Maximum "Everyday" Tension (EDT) Limit (kgf)       | 1627.5  |
| Temperature at EDT Limit (Degrees C)               | 5       |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 2277.1 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.653  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.492  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 1772.1 |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |      |                               |    |
|------------|------|-------------------------------|----|
| Conductor: | LYNX | Temperature Shift (Deg. C)    | 0  |
|            |      | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.09                              | 0.27                                        | 0.29  | 0.31  | 0.32  | 0.34  | 0.36  |
| 80             | 0.36                              | 1.07                                        | 1.15  | 1.22  | 1.30  | 1.37  | 1.44  |
| 120            | 0.80                              | 2.41                                        | 2.58  | 2.76  | 2.92  | 3.09  | 3.25  |
| 160            | 1.42                              | 4.28                                        | 4.59  | 4.90  | 5.20  | 5.49  | 5.77  |
| 200            | 2.22                              | 6.69                                        | 7.17  | 7.65  | 8.12  | 8.58  | 9.02  |
| 240            | 3.20                              | 9.63                                        | 10.33 | 11.02 | 11.69 | 12.35 | 12.99 |
| 280            | 4.36                              | 13.11                                       | 14.06 | 15.00 | 15.92 | 16.81 | 17.68 |
| 320            | 5.70                              | 17.12                                       | 18.37 | 19.59 | 20.79 | 21.96 | 23.09 |
| 360            | 7.21                              | 21.67                                       | 23.25 | 24.80 | 26.31 | 27.79 | 29.23 |
| 400            | 8.90                              | 26.75                                       | 28.70 | 30.61 | 32.48 | 34.31 | 36.08 |
| 440            | 10.77                             | 32.37                                       | 34.73 | 37.04 | 39.31 | 41.51 | 43.66 |
| 480            | 12.81                             | 38.52                                       | 41.33 | 44.08 | 46.78 | 49.40 | 51.96 |
| 520            | 15.04                             | 45.21                                       | 48.50 | 51.74 | 54.90 | 57.98 | 60.98 |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 80 | of | 98 |  |

**Appendix 38a - .175" ACSR / 175mm ACSR (Lynx) - (All altitudes) - 76.2mm Basic  
Matched to Table 5 & 6 of CE/C/36 Specification.**

**Table 6 - Existing conductors re-sagged**

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Conductor Code Name                           | LYNX                                    |
| Basic / Recommended Span (m)                  | 76.2                                    |
| Temperature Shift for Creep (Deg. C)          | 0 <b>Insert minus sign as necessary</b> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                             |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                               |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 75   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 1938.6           | 0.20                        | 0.31 | 0.36 | 0.45 | 0.56 | 0.67 | 0.80 | 0.94 | 1.80 |
| 0                 | 1772.1           | 0.22                        | 0.34 | 0.39 | 0.49 | 0.61 | 0.74 | 0.88 | 1.03 | 1.97 |
| 5                 | 1627.5           | 0.24                        | 0.37 | 0.42 | 0.54 | 0.66 | 0.80 | 0.95 | 1.12 | 2.15 |
| 10                | 1487.9           | 0.26                        | 0.41 | 0.46 | 0.59 | 0.72 | 0.88 | 1.04 | 1.22 | 2.35 |
| 15                | 1354.6           | 0.29                        | 0.45 | 0.51 | 0.64 | 0.80 | 0.96 | 1.15 | 1.35 | 2.58 |
| 20                | 1229.2           | 0.32                        | 0.49 | 0.56 | 0.71 | 0.88 | 1.06 | 1.26 | 1.48 | 2.84 |
| 25                | 1113.2           | 0.35                        | 0.54 | 0.62 | 0.78 | 0.97 | 1.17 | 1.39 | 1.64 | 3.14 |
| 30                | 1007.9           | 0.39                        | 0.60 | 0.68 | 0.87 | 1.07 | 1.29 | 1.54 | 1.81 | 3.47 |
| 35                | 914.2            | 0.42                        | 0.66 | 0.75 | 0.96 | 1.18 | 1.43 | 1.70 | 1.99 | 3.82 |
| 40                | 832.1            | 0.47                        | 0.73 | 0.83 | 1.05 | 1.30 | 1.57 | 1.87 | 2.19 | 4.20 |
| 45                | 761.3            | 0.51                        | 0.80 | 0.91 | 1.15 | 1.42 | 1.71 | 2.04 | 2.39 | 4.59 |
| 50                | 700.7            | 0.55                        | 0.87 | 0.98 | 1.25 | 1.54 | 1.86 | 2.22 | 2.60 | 4.99 |
| 55                | 648.8            | 0.60                        | 0.93 | 1.06 | 1.35 | 1.66 | 2.01 | 2.39 | 2.81 | 5.38 |
| 60                | 604.5            | 0.64                        | 1.00 | 1.14 | 1.44 | 1.78 | 2.16 | 2.57 | 3.01 | 5.78 |
| 65                | 566.5            | 0.69                        | 1.07 | 1.22 | 1.54 | 1.90 | 2.30 | 2.74 | 3.22 | 6.17 |
| 70                | 533.6            | 0.73                        | 1.14 | 1.29 | 1.64 | 2.02 | 2.44 | 2.91 | 3.41 | 6.55 |
| 75                | 505.1            | 0.77                        | 1.20 | 1.37 | 1.73 | 2.13 | 2.58 | 3.07 | 3.61 | 6.92 |
| 80                | 480.1            | 0.81                        | 1.26 | 1.44 | 1.82 | 2.25 | 2.72 | 3.23 | 3.80 | 7.28 |

**Table 5 - New conductors with 8% overtension**

|                                               |                                         |
|-----------------------------------------------|-----------------------------------------|
| Conductor Code Name                           | LYNX                                    |
| Basic / Recommended Span (m)                  | 76.2                                    |
| Temperature Shift for Creep (Deg. C)          | 0 <b>Insert minus sign as necessary</b> |
| Equivalent Percentage Increase in Tension (%) | 8.0 at 15°C                             |
| Required Percentage Increase in Tension (%)   | 8 at 15°C                               |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 75   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 2062.5           | 0.19                        | 0.29 | 0.33 | 0.42 | 0.52 | 0.63 | 0.75 | 0.88 | 1.69 |
| 0                 | 1893.2           | 0.21                        | 0.32 | 0.36 | 0.46 | 0.57 | 0.69 | 0.82 | 0.96 | 1.85 |
| 5                 | 1745.3           | 0.22                        | 0.35 | 0.40 | 0.50 | 0.62 | 0.75 | 0.89 | 1.04 | 2.00 |
| 10                | 1601.5           | 0.24                        | 0.38 | 0.43 | 0.55 | 0.67 | 0.81 | 0.97 | 1.14 | 2.18 |
| 15                | 1463.0           | 0.27                        | 0.41 | 0.47 | 0.60 | 0.74 | 0.89 | 1.06 | 1.25 | 2.39 |
| 20                | 1331.1           | 0.29                        | 0.46 | 0.52 | 0.66 | 0.81 | 0.98 | 1.17 | 1.37 | 2.62 |
| 25                | 1207.2           | 0.32                        | 0.50 | 0.57 | 0.72 | 0.89 | 1.08 | 1.29 | 1.51 | 2.89 |
| 30                | 1093.1           | 0.36                        | 0.55 | 0.63 | 0.80 | 0.99 | 1.19 | 1.42 | 1.67 | 3.20 |
| 35                | 989.9            | 0.39                        | 0.61 | 0.70 | 0.88 | 1.09 | 1.32 | 1.57 | 1.84 | 3.53 |
| 40                | 898.3            | 0.43                        | 0.68 | 0.77 | 0.97 | 1.20 | 1.45 | 1.73 | 2.03 | 3.89 |
| 45                | 818.4            | 0.47                        | 0.74 | 0.84 | 1.07 | 1.32 | 1.59 | 1.90 | 2.23 | 4.27 |
| 50                | 749.5            | 0.52                        | 0.81 | 0.92 | 1.17 | 1.44 | 1.74 | 2.07 | 2.43 | 4.66 |
| 55                | 690.6            | 0.56                        | 0.88 | 1.00 | 1.26 | 1.56 | 1.89 | 2.25 | 2.64 | 5.06 |
| 60                | 640.2            | 0.61                        | 0.95 | 1.08 | 1.36 | 1.68 | 2.04 | 2.43 | 2.85 | 5.46 |
| 65                | 597.1            | 0.65                        | 1.02 | 1.16 | 1.46 | 1.81 | 2.18 | 2.60 | 3.05 | 5.85 |
| 70                | 560.1            | 0.69                        | 1.08 | 1.23 | 1.56 | 1.93 | 2.33 | 2.77 | 3.25 | 6.24 |
| 75                | 528.1            | 0.74                        | 1.15 | 1.31 | 1.65 | 2.04 | 2.47 | 2.94 | 3.45 | 6.62 |
| 80                | 500.2            | 0.78                        | 1.21 | 1.38 | 1.75 | 2.16 | 2.61 | 3.10 | 3.64 | 6.98 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 81 | <b>of</b> | 98 |

## Appendix 39 - .1" / 100mm<sup>2</sup> ACSR (Normal altitude) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |        |
|------------------------------------------------------|--------|
| Conductor Code Name (if any)                         | DOG    |
| Greased Conductor Weight (kg/m)                      | 0.3962 |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 118.5  |
| Conductor Overall Diameter (mm)                      | 14.2   |
| Coefficient of Linear Expansion (/Degree C)          | 2E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 7699   |
| Rated Breaking Strength of Conductor (kgf)           | 3329.4 |
| Basic / Recommended Span (m)                         | 90     |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 380    |
| Radial Ice Thickness (mm)                            | 9.5    |
| Ice Density (kg/m <sup>3</sup> )                     | 913    |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 1664.7 |
| Temperature at MWT Limit (Degrees C)                 | -5.6   |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 665.9  |
| Temperature at EDT Limit (Degrees C)                 | 5      |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1270.9 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.042  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.286  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 736.3  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | DOG | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.10                              | 0.29                                        | 0.30  | 0.32  | 0.33  | 0.35  | 0.36  |
| 80             | 0.39                              | 1.14                                        | 1.21  | 1.27  | 1.33  | 1.40  | 1.45  |
| 120            | 0.87                              | 2.57                                        | 2.72  | 2.86  | 3.00  | 3.14  | 3.27  |
| 160            | 1.55                              | 4.56                                        | 4.83  | 5.09  | 5.34  | 5.58  | 5.82  |
| 200            | 2.42                              | 7.13                                        | 7.55  | 7.95  | 8.34  | 8.72  | 9.09  |
| 240            | 3.48                              | 10.27                                       | 10.87 | 11.45 | 12.01 | 12.56 | 13.09 |
| 280            | 4.73                              | 13.98                                       | 14.79 | 15.58 | 16.35 | 17.10 | 17.82 |
| 320            | 6.18                              | 18.26                                       | 19.32 | 20.35 | 21.36 | 22.33 | 23.28 |
| 360            | 7.83                              | 23.11                                       | 24.45 | 25.76 | 27.03 | 28.26 | 29.46 |
| 400            | 9.66                              | 28.53                                       | 30.19 | 31.80 | 33.37 | 34.89 | 36.37 |
| 440            | 11.69                             | 34.52                                       | 36.53 | 38.48 | 40.38 | 42.22 | 44.01 |
| 480            | 13.91                             | 41.08                                       | 43.47 | 45.80 | 48.05 | 50.25 | 52.37 |
| 520            | 16.33                             | 48.22                                       | 51.02 | 53.75 | 56.40 | 58.97 | 61.47 |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 82 | of | 98 |  |

## Appendix 39a - .1" / 100mm2 ACSR (Normal altitude) - 90m Basic

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | DOG                                                               |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 820.2            | 0.22                        | 0.30 | 0.39 | 0.49 | 0.60 | 0.73 | 0.87 | 1.02 | 1.96 |
| 0                 | 736.3            | 0.24                        | 0.33 | 0.43 | 0.54 | 0.67 | 0.81 | 0.97 | 1.14 | 2.18 |
| 5                 | 665.9            | 0.27                        | 0.36 | 0.48 | 0.60 | 0.74 | 0.90 | 1.07 | 1.26 | 2.41 |
| 10                | 600.6            | 0.30                        | 0.40 | 0.53 | 0.67 | 0.82 | 1.00 | 1.19 | 1.39 | 2.67 |
| 15                | 541.2            | 0.33                        | 0.45 | 0.59 | 0.74 | 0.92 | 1.11 | 1.32 | 1.55 | 2.96 |
| 20                | 488.5            | 0.36                        | 0.50 | 0.65 | 0.82 | 1.01 | 1.23 | 1.46 | 1.71 | 3.28 |
| 25                | 442.5            | 0.40                        | 0.55 | 0.72 | 0.91 | 1.12 | 1.35 | 1.61 | 1.89 | 3.63 |
| 30                | 402.9            | 0.44                        | 0.60 | 0.79 | 1.00 | 1.23 | 1.49 | 1.77 | 2.08 | 3.98 |
| 35                | 369.3            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.34 | 1.62 | 1.93 | 2.27 | 4.34 |
| 40                | 340.8            | 0.52                        | 0.71 | 0.93 | 1.18 | 1.45 | 1.76 | 2.09 | 2.46 | 4.71 |
| 45                | 316.6            | 0.56                        | 0.77 | 1.00 | 1.27 | 1.56 | 1.89 | 2.25 | 2.64 | 5.07 |
| 50                | 295.9            | 0.60                        | 0.82 | 1.07 | 1.36 | 1.67 | 2.03 | 2.41 | 2.83 | 5.42 |
| 55                | 278.2            | 0.64                        | 0.87 | 1.14 | 1.44 | 1.78 | 2.15 | 2.56 | 3.01 | 5.77 |
| 60                | 262.9            | 0.68                        | 0.92 | 1.21 | 1.53 | 1.88 | 2.28 | 2.71 | 3.18 | 6.10 |
| 65                | 249.5            | 0.71                        | 0.97 | 1.27 | 1.61 | 1.99 | 2.40 | 2.86 | 3.35 | 6.43 |
| 70                | 237.8            | 0.75                        | 1.02 | 1.33 | 1.69 | 2.08 | 2.52 | 3.00 | 3.52 | 6.75 |
| 75                | 227.4            | 0.78                        | 1.07 | 1.39 | 1.76 | 2.18 | 2.64 | 3.14 | 3.68 | 7.06 |
| 80                | 218.1            | 0.82                        | 1.11 | 1.45 | 1.84 | 2.27 | 2.75 | 3.27 | 3.84 | 7.36 |

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | DOG                                                               |
| Basic / Recommended Span (m)                  | 90                                                                |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 10.0 at 15°C                                                      |
| Required Percentage Increase in Tension (%)   | 10 at 15°C                                                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 891.9            | 0.20                        | 0.27 | 0.36 | 0.45 | 0.56 | 0.67 | 0.80 | 0.94 | 1.80 |
| 0                 | 804.6            | 0.22                        | 0.30 | 0.39 | 0.50 | 0.62 | 0.74 | 0.89 | 1.04 | 1.99 |
| 5                 | 730.3            | 0.24                        | 0.33 | 0.43 | 0.55 | 0.68 | 0.82 | 0.98 | 1.15 | 2.20 |
| 10                | 660.2            | 0.27                        | 0.37 | 0.48 | 0.61 | 0.75 | 0.91 | 1.08 | 1.27 | 2.43 |
| 15                | 595.3            | 0.30                        | 0.41 | 0.53 | 0.67 | 0.83 | 1.01 | 1.20 | 1.41 | 2.70 |
| 20                | 536.6            | 0.33                        | 0.45 | 0.59 | 0.75 | 0.92 | 1.12 | 1.33 | 1.56 | 2.99 |
| 25                | 484.4            | 0.37                        | 0.50 | 0.65 | 0.83 | 1.02 | 1.24 | 1.47 | 1.73 | 3.31 |
| 30                | 438.9            | 0.41                        | 0.55 | 0.72 | 0.91 | 1.13 | 1.37 | 1.62 | 1.91 | 3.66 |
| 35                | 399.9            | 0.45                        | 0.61 | 0.79 | 1.00 | 1.24 | 1.50 | 1.78 | 2.09 | 4.01 |
| 40                | 366.7            | 0.49                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.28 | 4.38 |
| 45                | 338.6            | 0.53                        | 0.72 | 0.94 | 1.18 | 1.46 | 1.77 | 2.11 | 2.47 | 4.74 |
| 50                | 314.7            | 0.57                        | 0.77 | 1.01 | 1.27 | 1.57 | 1.90 | 2.27 | 2.66 | 5.10 |
| 55                | 294.3            | 0.61                        | 0.82 | 1.08 | 1.36 | 1.68 | 2.04 | 2.42 | 2.84 | 5.45 |
| 60                | 276.8            | 0.64                        | 0.88 | 1.14 | 1.45 | 1.79 | 2.16 | 2.58 | 3.02 | 5.80 |
| 65                | 261.7            | 0.68                        | 0.93 | 1.21 | 1.53 | 1.89 | 2.29 | 2.73 | 3.20 | 6.13 |
| 70                | 248.5            | 0.72                        | 0.98 | 1.28 | 1.61 | 1.99 | 2.41 | 2.87 | 3.37 | 6.46 |
| 75                | 236.8            | 0.75                        | 1.02 | 1.34 | 1.69 | 2.09 | 2.53 | 3.01 | 3.53 | 6.78 |
| 80                | 226.5            | 0.79                        | 1.07 | 1.40 | 1.77 | 2.19 | 2.65 | 3.15 | 3.69 | 7.08 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 83 | <b>of</b> | 98 |

Appendix 40 - .1" / 100mm<sup>2</sup> ACSR (Normal altitude) - 100m Basic

## CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |        |
|------------------------------------------------------|--------|
| Conductor Code Name (if any)                         | DOG    |
| Greased Conductor Weight (kg/m)                      | 0.3962 |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 118.5  |
| Conductor Overall Diameter (mm)                      | 14.2   |
| Coefficient of Linear Expansion (/Degree C)          | 2E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 7699   |
| Rated Breaking Strength of Conductor (kgf)           | 3329.4 |
| Basic / Recommended Span (m)                         | 100    |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 380    |
| Radial Ice Thickness (mm)                            | 9.5    |
| Ice Density (kg/m <sup>3</sup> )                     | 913    |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 1664.7 |
| Temperature at MWT Limit (Degrees C)                 | -5.6   |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 665.9  |
| Temperature at EDT Limit (Degrees C)                 | 5      |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1320.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.042  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.286  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 732.8  |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | DOG | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.10                              | 0.27                                        | 0.28  | 0.30  | 0.31  | 0.32  | 0.34  |
| 80             | 0.39                              | 1.07                                        | 1.13  | 1.18  | 1.24  | 1.29  | 1.34  |
| 120            | 0.88                              | 2.41                                        | 2.54  | 2.66  | 2.78  | 2.90  | 3.02  |
| 160            | 1.56                              | 4.28                                        | 4.51  | 4.73  | 4.95  | 5.16  | 5.37  |
| 200            | 2.44                              | 6.68                                        | 7.04  | 7.39  | 7.74  | 8.07  | 8.39  |
| 240            | 3.51                              | 9.63                                        | 10.14 | 10.65 | 11.14 | 11.62 | 12.08 |
| 280            | 4.78                              | 13.10                                       | 13.81 | 14.49 | 15.16 | 15.81 | 16.45 |
| 320            | 6.24                              | 17.11                                       | 18.03 | 18.93 | 19.80 | 20.65 | 21.48 |
| 360            | 7.89                              | 21.66                                       | 22.82 | 23.96 | 25.06 | 26.14 | 27.19 |
| 400            | 9.75                              | 26.74                                       | 28.18 | 29.58 | 30.94 | 32.27 | 33.56 |
| 440            | 11.79                             | 32.35                                       | 34.09 | 35.79 | 37.44 | 39.05 | 40.61 |
| 480            | 14.03                             | 38.50                                       | 40.57 | 42.59 | 44.56 | 46.47 | 48.33 |
| 520            | 16.47                             | 45.18                                       | 47.62 | 49.99 | 52.30 | 54.54 | 56.72 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 84 | <b>of</b> | 98 |  |

## Appendix 40a - .1" / 100mm<sup>2</sup> ACSR (Normal altitude) - 100m Basic

Conductor Code Name DOG  
 Basic / Recommended Span (m) 100  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 813.0            | 0.22                        | 0.30 | 0.39 | 0.49 | 0.61 | 0.74 | 0.88 | 1.03 | 1.97 |
| 0                 | 732.8            | 0.24                        | 0.33 | 0.43 | 0.55 | 0.68 | 0.82 | 0.97 | 1.14 | 2.19 |
| 5                 | 665.9            | 0.27                        | 0.36 | 0.48 | 0.60 | 0.74 | 0.90 | 1.07 | 1.26 | 2.41 |
| 10                | 604.4            | 0.30                        | 0.40 | 0.52 | 0.66 | 0.82 | 0.99 | 1.18 | 1.38 | 2.66 |
| 15                | 548.8            | 0.32                        | 0.44 | 0.58 | 0.73 | 0.90 | 1.09 | 1.30 | 1.53 | 2.92 |
| 20                | 499.5            | 0.36                        | 0.49 | 0.63 | 0.80 | 0.99 | 1.20 | 1.43 | 1.68 | 3.21 |
| 25                | 456.4            | 0.39                        | 0.53 | 0.69 | 0.88 | 1.09 | 1.31 | 1.56 | 1.83 | 3.52 |
| 30                | 419.2            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.18 | 1.43 | 1.70 | 2.00 | 3.83 |
| 35                | 387.3            | 0.46                        | 0.63 | 0.82 | 1.04 | 1.28 | 1.55 | 1.84 | 2.16 | 4.14 |
| 40                | 359.9            | 0.50                        | 0.67 | 0.88 | 1.11 | 1.38 | 1.67 | 1.98 | 2.33 | 4.46 |
| 45                | 336.3            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.78 | 2.12 | 2.49 | 4.77 |
| 50                | 316.0            | 0.56                        | 0.77 | 1.00 | 1.27 | 1.57 | 1.90 | 2.26 | 2.65 | 5.08 |
| 55                | 298.4            | 0.60                        | 0.81 | 1.06 | 1.34 | 1.66 | 2.01 | 2.39 | 2.81 | 5.38 |
| 60                | 283.0            | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 5.67 |
| 65                | 269.4            | 0.66                        | 0.90 | 1.18 | 1.49 | 1.84 | 2.22 | 2.65 | 3.11 | 5.96 |
| 70                | 257.4            | 0.69                        | 0.94 | 1.23 | 1.56 | 1.92 | 2.33 | 2.77 | 3.25 | 6.23 |
| 75                | 246.8            | 0.72                        | 0.98 | 1.28 | 1.63 | 2.01 | 2.43 | 2.89 | 3.39 | 6.50 |
| 80                | 237.2            | 0.75                        | 1.02 | 1.34 | 1.69 | 2.09 | 2.53 | 3.01 | 3.53 | 6.77 |

Conductor Code Name DOG  
 Basic / Recommended Span (m) 100  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 10.0 at 15°C  
 Required Percentage Increase in Tension (%) 10 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 887.8            | 0.20                        | 0.27 | 0.36 | 0.45 | 0.56 | 0.68 | 0.80 | 0.94 | 1.81 |
| 0                 | 803.3            | 0.22                        | 0.30 | 0.39 | 0.50 | 0.62 | 0.75 | 0.89 | 1.04 | 2.00 |
| 5                 | 731.9            | 0.24                        | 0.33 | 0.43 | 0.55 | 0.68 | 0.82 | 0.97 | 1.14 | 2.19 |
| 10                | 665.1            | 0.27                        | 0.36 | 0.48 | 0.60 | 0.74 | 0.90 | 1.07 | 1.26 | 2.41 |
| 15                | 603.7            | 0.30                        | 0.40 | 0.53 | 0.66 | 0.82 | 0.99 | 1.18 | 1.39 | 2.66 |
| 20                | 548.2            | 0.33                        | 0.44 | 0.58 | 0.73 | 0.90 | 1.09 | 1.30 | 1.53 | 2.93 |
| 25                | 499.0            | 0.36                        | 0.49 | 0.64 | 0.80 | 0.99 | 1.20 | 1.43 | 1.68 | 3.22 |
| 30                | 456.0            | 0.39                        | 0.53 | 0.70 | 0.88 | 1.09 | 1.31 | 1.56 | 1.84 | 3.52 |
| 35                | 418.8            | 0.43                        | 0.58 | 0.76 | 0.96 | 1.18 | 1.43 | 1.70 | 2.00 | 3.83 |
| 40                | 386.9            | 0.46                        | 0.63 | 0.82 | 1.04 | 1.28 | 1.55 | 1.84 | 2.16 | 4.15 |
| 45                | 359.6            | 0.50                        | 0.67 | 0.88 | 1.12 | 1.38 | 1.67 | 1.98 | 2.33 | 4.46 |
| 50                | 336.1            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.78 | 2.12 | 2.49 | 4.77 |
| 55                | 315.8            | 0.56                        | 0.77 | 1.00 | 1.27 | 1.57 | 1.90 | 2.26 | 2.65 | 5.08 |
| 60                | 298.2            | 0.60                        | 0.81 | 1.06 | 1.35 | 1.66 | 2.01 | 2.39 | 2.81 | 5.38 |
| 65                | 282.8            | 0.63                        | 0.86 | 1.12 | 1.42 | 1.75 | 2.12 | 2.52 | 2.96 | 5.67 |
| 70                | 269.3            | 0.66                        | 0.90 | 1.18 | 1.49 | 1.84 | 2.23 | 2.65 | 3.11 | 5.96 |
| 75                | 257.3            | 0.69                        | 0.94 | 1.23 | 1.56 | 1.92 | 2.33 | 2.77 | 3.25 | 6.24 |
| 80                | 246.6            | 0.72                        | 0.98 | 1.29 | 1.63 | 2.01 | 2.43 | 2.89 | 3.39 | 6.51 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 85 | <b>of</b> | 98 |

## Appendix 41 - .1" / 70mm<sup>2</sup> HDBC (Normal altitude) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |                  |
|------------------------------------------------------|------------------|
| Conductor Code Name (if any)                         | <b>COPPER 70</b> |
| Greased Conductor Weight (kg/m)                      | <b>0.621</b>     |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | <b>69.28</b>     |
| Conductor Overall Diameter (mm)                      | <b>10.65</b>     |
| Coefficient of Linear Expansion (/Degree C)          | <b>1.70E-05</b>  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | <b>12655.3</b>   |
| Rated Breaking Strength of Conductor (kgf)           | <b>2740.9</b>    |
| Basic / Recommended Span (m)                         | <b>90</b>        |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | <b>380</b>       |
| Radial Ice Thickness (mm)                            | <b>9.5</b>       |
| Ice Density (kg/m <sup>3</sup> )                     | <b>913</b>       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | <b>1370.5</b>    |
| Temperature at MWT Limit (Degrees C)                 | <b>-5.6</b>      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | <b>913.6</b>     |
| Temperature at EDT Limit (Degrees C)                 | <b>5</b>         |

|                                         |    |        |               |
|-----------------------------------------|----|--------|---------------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | <b>1363.1</b> |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | <b>1.170</b>  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | <b>1.149</b>  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | <b>972.2</b>  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                             |                               |          |
|-----------------------------|-------------------------------|----------|
| Conductor: <b>COPPER 70</b> | Temperature Shift (Deg. C)    | <b>0</b> |
|                             | Tension Reduction at 15°C (%) | <b>5</b> |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |           |           |           |           |           |
|----------------|-----------------------------------|---------------------------------------------|-----------|-----------|-----------|-----------|-----------|
|                |                                   | <b>50</b>                                   | <b>55</b> | <b>60</b> | <b>65</b> | <b>70</b> | <b>75</b> |
| <b>40</b>      | 0.12                              | 0.25                                        | 0.26      | 0.27      | 0.29      | 0.30      | 0.31      |
| <b>80</b>      | 0.48                              | 0.99                                        | 1.04      | 1.10      | 1.15      | 1.20      | 1.25      |
| <b>120</b>     | 1.07                              | 2.23                                        | 2.35      | 2.47      | 2.58      | 2.70      | 2.81      |
| <b>160</b>     | 1.91                              | 3.96                                        | 4.17      | 4.38      | 4.59      | 4.80      | 5.00      |
| <b>200</b>     | 2.98                              | 6.20                                        | 6.52      | 6.85      | 7.18      | 7.50      | 7.82      |
| <b>240</b>     | 4.30                              | 8.92                                        | 9.39      | 9.86      | 10.33     | 10.80     | 11.25     |
| <b>280</b>     | 5.85                              | 12.14                                       | 12.78     | 13.43     | 14.06     | 14.69     | 15.32     |
| <b>320</b>     | 7.64                              | 15.86                                       | 16.70     | 17.54     | 18.37     | 19.19     | 20.01     |
| <b>360</b>     | 9.67                              | 20.07                                       | 21.13     | 22.19     | 23.25     | 24.29     | 25.32     |
| <b>400</b>     | 11.94                             | 24.78                                       | 26.09     | 27.40     | 28.70     | 29.99     | 31.26     |
| <b>440</b>     | 14.45                             | 29.98                                       | 31.57     | 33.15     | 34.73     | 36.29     | 37.83     |
| <b>480</b>     | 17.19                             | 35.68                                       | 37.57     | 39.46     | 41.33     | 43.18     | 45.02     |
| <b>520</b>     | 20.18                             | 41.88                                       | 44.09     | 46.31     | 48.50     | 50.68     | 52.83     |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 86 | of | 98 |  |

## Appendix 41a - .1" / 70mm<sup>2</sup> HDBC (Normal altitude) - 90m Basic

### Common Design & Erection Chart

#### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |           |                                |
|-----------------------------------------------|-----------|--------------------------------|
| Conductor Code Name                           | COPPER 70 |                                |
| Basic / Recommended Span (m)                  | 90        |                                |
| Temperature Shift for Creep (Deg. C)          | 0         | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 0.0       | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 0         | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 1040.4           | 0.27                        | 0.37 | 0.48 | 0.60 | 0.75 | 0.90 | 1.07 | 1.26 | 2.42 |
| 0                 | 972.2            | 0.29                        | 0.39 | 0.51 | 0.65 | 0.80 | 0.97 | 1.15 | 1.35 | 2.59 |
| 5                 | 913.6            | 0.31                        | 0.42 | 0.54 | 0.69 | 0.85 | 1.03 | 1.22 | 1.44 | 2.75 |
| 10                | 857.6            | 0.33                        | 0.44 | 0.58 | 0.73 | 0.91 | 1.10 | 1.30 | 1.53 | 2.93 |
| 15                | 804.3            | 0.35                        | 0.47 | 0.62 | 0.78 | 0.97 | 1.17 | 1.39 | 1.63 | 3.13 |
| 20                | 754.0            | 0.37                        | 0.50 | 0.66 | 0.83 | 1.03 | 1.25 | 1.48 | 1.74 | 3.34 |
| 25                | 707.1            | 0.40                        | 0.54 | 0.70 | 0.89 | 1.10 | 1.33 | 1.58 | 1.86 | 3.56 |
| 30                | 663.5            | 0.42                        | 0.57 | 0.75 | 0.95 | 1.17 | 1.42 | 1.68 | 1.98 | 3.79 |
| 35                | 623.4            | 0.45                        | 0.61 | 0.80 | 1.01 | 1.25 | 1.51 | 1.79 | 2.10 | 4.03 |
| 40                | 586.7            | 0.48                        | 0.65 | 0.85 | 1.07 | 1.32 | 1.60 | 1.91 | 2.24 | 4.29 |
| 45                | 553.4            | 0.50                        | 0.69 | 0.90 | 1.14 | 1.40 | 1.70 | 2.02 | 2.37 | 4.54 |
| 50                | 523.2            | 0.53                        | 0.73 | 0.95 | 1.20 | 1.48 | 1.80 | 2.14 | 2.51 | 4.81 |
| 55                | 495.8            | 0.56                        | 0.77 | 1.00 | 1.27 | 1.57 | 1.89 | 2.25 | 2.65 | 5.07 |
| 60                | 471.2            | 0.59                        | 0.81 | 1.05 | 1.33 | 1.65 | 1.99 | 2.37 | 2.78 | 5.34 |
| 65                | 448.9            | 0.62                        | 0.85 | 1.11 | 1.40 | 1.73 | 2.09 | 2.49 | 2.92 | 5.60 |
| 70                | 428.8            | 0.65                        | 0.89 | 1.16 | 1.47 | 1.81 | 2.19 | 2.61 | 3.06 | 5.87 |
| 75                | 410.6            | 0.68                        | 0.93 | 1.21 | 1.53 | 1.89 | 2.29 | 2.72 | 3.20 | 6.13 |
| 80                | 394.0            | 0.71                        | 0.97 | 1.26 | 1.60 | 1.97 | 2.38 | 2.84 | 3.33 | 6.38 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 87 | <b>of</b> | 98 |

## Appendix 42 - .15" / 100mm<sup>2</sup> HDBC (Normal altitude) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |          |
|------------------------------------------------------|----------|
| Conductor Code Name (if any)                         | HDBC 100 |
| Greased Conductor Weight (kg/m)                      | 0.911    |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 101.654  |
| Conductor Overall Diameter (mm)                      | 12.9     |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3  |
| Rated Breaking Strength of Conductor (kgf)           | 3838.2   |
| Basic / Recommended Span (m)                         | 90       |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 380      |
| Radial Ice Thickness (mm)                            | 9.5      |
| Ice Density (kg/m <sup>3</sup> )                     | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 1814.4   |
| Temperature at MWT Limit (Degrees C)                 | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 1279.4   |
| Temperature at EDT Limit (Degrees C)                 | 5        |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1803.8 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.522  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.236  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 1362.7 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |          |                               |   |
|------------|----------|-------------------------------|---|
| Conductor: | HDBC 100 | Temperature Shift (Deg. C)    | 0 |
|            |          | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.12                              | 0.26                                        | 0.27  | 0.28  | 0.30  | 0.31  | 0.32  |
| 80             | 0.50                              | 1.03                                        | 1.08  | 1.13  | 1.19  | 1.24  | 1.29  |
| 120            | 1.12                              | 2.32                                        | 2.44  | 2.55  | 2.67  | 2.78  | 2.90  |
| 160            | 2.00                              | 4.12                                        | 4.33  | 4.54  | 4.74  | 4.95  | 5.15  |
| 200            | 3.12                              | 6.44                                        | 6.76  | 7.09  | 7.41  | 7.73  | 8.05  |
| 240            | 4.49                              | 9.27                                        | 9.74  | 10.21 | 10.68 | 11.14 | 11.59 |
| 280            | 6.12                              | 12.62                                       | 13.26 | 13.90 | 14.53 | 15.16 | 15.77 |
| 320            | 7.99                              | 16.48                                       | 17.32 | 18.15 | 18.98 | 19.80 | 20.60 |
| 360            | 10.11                             | 20.86                                       | 21.92 | 22.97 | 24.02 | 25.05 | 26.08 |
| 400            | 12.48                             | 25.75                                       | 27.06 | 28.36 | 29.66 | 30.93 | 32.19 |
| 440            | 15.10                             | 31.16                                       | 32.74 | 34.32 | 35.88 | 37.43 | 38.95 |
| 480            | 17.97                             | 37.08                                       | 38.97 | 40.84 | 42.70 | 44.54 | 46.36 |
| 520            | 21.09                             | 43.52                                       | 45.73 | 47.93 | 50.12 | 52.28 | 54.40 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 88 | <b>of</b> | 98 |

## Appendix 42a - .15" / 100mm<sup>2</sup> HDBC (Normal altitude) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |          |                                |
|-----------------------------------------------|----------|--------------------------------|
| Conductor Code Name                           | HDBC 100 |                                |
| Basic / Recommended Span (m)                  | 90       |                                |
| Temperature Shift for Creep (Deg. C)          | 0        | Insert minus sign as necessary |
| Equivalent Percentage Increase in Tension (%) | 0.0      | at 15°C                        |
| Required Percentage Increase in Tension (%)   | 0        | at 15°C                        |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 1460.1           | 0.28                        | 0.38 | 0.50 | 0.63 | 0.78 | 0.94 | 1.12 | 1.32 | 2.53 |
| 0                 | 1362.7           | 0.30                        | 0.41 | 0.53 | 0.68 | 0.84 | 1.01 | 1.20 | 1.41 | 2.71 |
| 5                 | 1279.4           | 0.32                        | 0.44 | 0.57 | 0.72 | 0.89 | 1.08 | 1.28 | 1.50 | 2.88 |
| 10                | 1200.1           | 0.34                        | 0.47 | 0.61 | 0.77 | 0.95 | 1.15 | 1.37 | 1.60 | 3.08 |
| 15                | 1125.2           | 0.36                        | 0.50 | 0.65 | 0.82 | 1.01 | 1.22 | 1.46 | 1.71 | 3.28 |
| 20                | 1055.0           | 0.39                        | 0.53 | 0.69 | 0.87 | 1.08 | 1.31 | 1.55 | 1.82 | 3.50 |
| 25                | 989.8            | 0.41                        | 0.56 | 0.74 | 0.93 | 1.15 | 1.39 | 1.66 | 1.94 | 3.73 |
| 30                | 929.6            | 0.44                        | 0.60 | 0.78 | 0.99 | 1.23 | 1.48 | 1.76 | 2.07 | 3.97 |
| 35                | 874.5            | 0.47                        | 0.64 | 0.83 | 1.06 | 1.30 | 1.58 | 1.88 | 2.20 | 4.22 |
| 40                | 824.3            | 0.50                        | 0.68 | 0.88 | 1.12 | 1.38 | 1.67 | 1.99 | 2.34 | 4.48 |
| 45                | 778.8            | 0.53                        | 0.72 | 0.94 | 1.18 | 1.46 | 1.77 | 2.11 | 2.47 | 4.74 |
| 50                | 737.6            | 0.56                        | 0.76 | 0.99 | 1.25 | 1.54 | 1.87 | 2.22 | 2.61 | 5.00 |
| 55                | 700.4            | 0.59                        | 0.80 | 1.04 | 1.32 | 1.63 | 1.97 | 2.34 | 2.75 | 5.27 |
| 60                | 666.9            | 0.61                        | 0.84 | 1.09 | 1.38 | 1.71 | 2.07 | 2.46 | 2.89 | 5.53 |
| 65                | 636.6            | 0.64                        | 0.88 | 1.15 | 1.45 | 1.79 | 2.16 | 2.58 | 3.02 | 5.80 |
| 70                | 609.1            | 0.67                        | 0.92 | 1.20 | 1.51 | 1.87 | 2.26 | 2.69 | 3.16 | 6.06 |
| 75                | 584.3            | 0.70                        | 0.96 | 1.25 | 1.58 | 1.95 | 2.36 | 2.81 | 3.29 | 6.32 |
| 80                | 561.7            | 0.73                        | 0.99 | 1.30 | 1.64 | 2.03 | 2.45 | 2.92 | 3.43 | 6.57 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 89 | <b>of</b> | 98 |

## Appendix 43 - .2" / 125mm<sup>2</sup> HDBC (Normal altitude) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |          |
|------------------------------------------------------|----------|
| Conductor Code Name (if any)                         | HDBC 125 |
| Greased Conductor Weight (kg/m)                      | 1.130    |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 125.499  |
| Conductor Overall Diameter (mm)                      | 14.5     |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.29 |
| Rated Breaking Strength of Conductor (kgf)           | 4929.3   |
| Basic / Recommended Span (m)                         | 100      |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 380      |
| Radial Ice Thickness (mm)                            | 9.5      |
| Ice Density (kg/m <sup>3</sup> )                     | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 1814.4   |
| Temperature at MWT Limit (Degrees C)                 | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 1643.1   |
| Temperature at EDT Limit (Degrees C)                 | 5        |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1814.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.784  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 1.298  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 1237.0 |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |          |                               |   |
|------------|----------|-------------------------------|---|
| Conductor: | HDBC 125 | Temperature Shift (Deg. C)    | 0 |
|            |          | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.17                              | 0.31                                        | 0.32  | 0.33  | 0.34  | 0.35  | 0.36  |
| 80             | 0.68                              | 1.22                                        | 1.26  | 1.31  | 1.35  | 1.39  | 1.43  |
| 120            | 1.54                              | 2.75                                        | 2.84  | 2.94  | 3.03  | 3.12  | 3.21  |
| 160            | 2.74                              | 4.88                                        | 5.05  | 5.22  | 5.39  | 5.55  | 5.71  |
| 200            | 4.28                              | 7.63                                        | 7.90  | 8.16  | 8.42  | 8.68  | 8.93  |
| 240            | 6.16                              | 10.99                                       | 11.37 | 11.75 | 12.13 | 12.49 | 12.85 |
| 280            | 8.39                              | 14.96                                       | 15.48 | 16.00 | 16.50 | 17.00 | 17.49 |
| 320            | 10.95                             | 19.53                                       | 20.22 | 20.89 | 21.56 | 22.21 | 22.85 |
| 360            | 13.86                             | 24.72                                       | 25.59 | 26.44 | 27.28 | 28.11 | 28.92 |
| 400            | 17.11                             | 30.52                                       | 31.59 | 32.64 | 33.68 | 34.70 | 35.70 |
| 440            | 20.71                             | 36.93                                       | 38.23 | 39.50 | 40.75 | 41.99 | 43.20 |
| 480            | 24.65                             | 43.95                                       | 45.49 | 47.01 | 48.50 | 49.97 | 51.41 |
| 520            | 28.92                             | 51.58                                       | 53.39 | 55.17 | 56.92 | 58.64 | 60.34 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 90 | <b>of</b> | 98 |  |

## Appendix 43a - .2" / 125mm<sup>2</sup> HDBC (Normal altitude) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

|                                               |                                                                   |
|-----------------------------------------------|-------------------------------------------------------------------|
| Conductor Code Name                           | HDBC 125                                                          |
| Basic / Recommended Span (m)                  | 100                                                               |
| Temperature Shift for Creep (Deg. C)          | 0 <span style="color: red;">Insert minus sign as necessary</span> |
| Equivalent Percentage Increase in Tension (%) | 0.0 at 15°C                                                       |
| Required Percentage Increase in Tension (%)   | 0 at 15°C                                                         |

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 1320.5           | 0.39                        | 0.52 | 0.68 | 0.87 | 1.07 | 1.29 | 1.54 | 1.81 | 3.47 |
| 0                 | 1237.0           | 0.41                        | 0.56 | 0.73 | 0.92 | 1.14 | 1.38 | 1.64 | 1.93 | 3.70 |
| 5                 | 1168.5           | 0.44                        | 0.59 | 0.77 | 0.98 | 1.21 | 1.46 | 1.74 | 2.04 | 3.92 |
| 10                | 1105.8           | 0.46                        | 0.63 | 0.82 | 1.03 | 1.28 | 1.55 | 1.84 | 2.16 | 4.14 |
| 15                | 1048.4           | 0.49                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.28 | 4.37 |
| 20                | 996.2            | 0.51                        | 0.69 | 0.91 | 1.15 | 1.42 | 1.72 | 2.04 | 2.40 | 4.59 |
| 25                | 948.6            | 0.54                        | 0.73 | 0.95 | 1.21 | 1.49 | 1.80 | 2.14 | 2.52 | 4.82 |
| 30                | 905.4            | 0.56                        | 0.76 | 1.00 | 1.26 | 1.56 | 1.89 | 2.25 | 2.64 | 5.05 |
| 35                | 866.1            | 0.59                        | 0.80 | 1.04 | 1.32 | 1.63 | 1.97 | 2.35 | 2.76 | 5.28 |
| 40                | 830.3            | 0.61                        | 0.83 | 1.09 | 1.38 | 1.70 | 2.06 | 2.45 | 2.87 | 5.51 |
| 45                | 797.7            | 0.64                        | 0.87 | 1.13 | 1.43 | 1.77 | 2.14 | 2.55 | 2.99 | 5.74 |
| 50                | 767.9            | 0.66                        | 0.90 | 1.18 | 1.49 | 1.84 | 2.23 | 2.65 | 3.11 | 5.96 |
| 55                | 740.5            | 0.69                        | 0.93 | 1.22 | 1.54 | 1.91 | 2.31 | 2.75 | 3.22 | 6.18 |
| 60                | 715.5            | 0.71                        | 0.97 | 1.26 | 1.60 | 1.97 | 2.39 | 2.84 | 3.34 | 6.40 |
| 65                | 692.4            | 0.73                        | 1.00 | 1.31 | 1.65 | 2.04 | 2.47 | 2.94 | 3.45 | 6.61 |
| 70                | 671.1            | 0.76                        | 1.03 | 1.35 | 1.70 | 2.10 | 2.55 | 3.03 | 3.56 | 6.82 |
| 75                | 651.4            | 0.78                        | 1.06 | 1.39 | 1.76 | 2.17 | 2.62 | 3.12 | 3.66 | 7.03 |
| 80                | 633.1            | 0.80                        | 1.09 | 1.43 | 1.81 | 2.23 | 2.70 | 3.21 | 3.77 | 7.23 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 91 | <b>of</b> | 98 |

## Appendix 44 - .1" / 100mm<sup>2</sup> ACSR (High altitude) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |        |
|------------------------------------------------------|--------|
| Conductor Code Name (if any)                         | DOG    |
| Greased Conductor Weight (kg/m)                      | 0.3962 |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 118.5  |
| Conductor Overall Diameter (mm)                      | 14.2   |
| Coefficient of Linear Expansion (/Degree C)          | 2E-05  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 7699   |
| Rated Breaking Strength of Conductor (kgf)           | 3329.4 |
| Basic / Recommended Span (m)                         | 90     |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 570    |
| Radial Ice Thickness (mm)                            | 12.5   |
| Ice Density (kg/m <sup>3</sup> )                     | 913    |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 1664.7 |
| Temperature at MWT Limit (Degrees C)                 | -5.6   |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 665.9  |
| Temperature at EDT Limit (Degrees C)                 | 5      |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1596.6 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.353  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 2.278  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 736.3  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |     |                               |    |
|------------|-----|-------------------------------|----|
| Conductor: | DOG | Temperature Shift (Deg. C)    | 0  |
|            |     | Tension Reduction at 15°C (%) | 10 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.10                              | 0.29                                        | 0.30  | 0.32  | 0.33  | 0.35  | 0.36  |
| 80             | 0.39                              | 1.14                                        | 1.21  | 1.27  | 1.33  | 1.40  | 1.45  |
| 120            | 0.87                              | 2.57                                        | 2.72  | 2.86  | 3.00  | 3.14  | 3.27  |
| 160            | 1.55                              | 4.56                                        | 4.83  | 5.09  | 5.34  | 5.58  | 5.82  |
| 200            | 2.42                              | 7.13                                        | 7.55  | 7.95  | 8.34  | 8.72  | 9.09  |
| 240            | 3.48                              | 10.27                                       | 10.87 | 11.45 | 12.01 | 12.56 | 13.09 |
| 280            | 4.73                              | 13.98                                       | 14.79 | 15.58 | 16.35 | 17.10 | 17.82 |
| 320            | 6.18                              | 18.26                                       | 19.32 | 20.35 | 21.36 | 22.33 | 23.28 |
| 360            | 7.83                              | 23.11                                       | 24.45 | 25.76 | 27.03 | 28.26 | 29.46 |
| 400            | 9.66                              | 28.53                                       | 30.19 | 31.80 | 33.37 | 34.89 | 36.37 |
| 440            | 11.69                             | 34.52                                       | 36.53 | 38.48 | 40.38 | 42.22 | 44.01 |
| 480            | 13.91                             | 41.08                                       | 43.47 | 45.80 | 48.05 | 50.25 | 52.37 |
| 520            | 16.33                             | 48.22                                       | 51.02 | 53.75 | 56.40 | 58.97 | 61.47 |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 92 | of | 98 |  |

## Appendix 44a - .1" / 100mm2 ACSR (High altitude) - 90m Basic

Conductor Code Name DOG  
 Basic / Recommended Span (m) 90  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 820.2            | 0.22                        | 0.30 | 0.39 | 0.49 | 0.60 | 0.73 | 0.87 | 1.02 | 1.96 |
| 0                 | 736.3            | 0.24                        | 0.33 | 0.43 | 0.54 | 0.67 | 0.81 | 0.97 | 1.14 | 2.18 |
| 5                 | 665.9            | 0.27                        | 0.36 | 0.48 | 0.60 | 0.74 | 0.90 | 1.07 | 1.26 | 2.41 |
| 10                | 600.6            | 0.30                        | 0.40 | 0.53 | 0.67 | 0.82 | 1.00 | 1.19 | 1.39 | 2.67 |
| 15                | 541.2            | 0.33                        | 0.45 | 0.59 | 0.74 | 0.92 | 1.11 | 1.32 | 1.55 | 2.96 |
| 20                | 488.5            | 0.36                        | 0.50 | 0.65 | 0.82 | 1.01 | 1.23 | 1.46 | 1.71 | 3.28 |
| 25                | 442.5            | 0.40                        | 0.55 | 0.72 | 0.91 | 1.12 | 1.35 | 1.61 | 1.89 | 3.63 |
| 30                | 402.9            | 0.44                        | 0.60 | 0.79 | 1.00 | 1.23 | 1.49 | 1.77 | 2.08 | 3.98 |
| 35                | 369.3            | 0.48                        | 0.66 | 0.86 | 1.09 | 1.34 | 1.62 | 1.93 | 2.27 | 4.34 |
| 40                | 340.8            | 0.52                        | 0.71 | 0.93 | 1.18 | 1.45 | 1.76 | 2.09 | 2.46 | 4.71 |
| 45                | 316.6            | 0.56                        | 0.77 | 1.00 | 1.27 | 1.56 | 1.89 | 2.25 | 2.64 | 5.07 |
| 50                | 295.9            | 0.60                        | 0.82 | 1.07 | 1.36 | 1.67 | 2.03 | 2.41 | 2.83 | 5.42 |
| 55                | 278.2            | 0.64                        | 0.87 | 1.14 | 1.44 | 1.78 | 2.15 | 2.56 | 3.01 | 5.77 |
| 60                | 262.9            | 0.68                        | 0.92 | 1.21 | 1.53 | 1.88 | 2.28 | 2.71 | 3.18 | 6.10 |
| 65                | 249.5            | 0.71                        | 0.97 | 1.27 | 1.61 | 1.99 | 2.40 | 2.86 | 3.35 | 6.43 |
| 70                | 237.8            | 0.75                        | 1.02 | 1.33 | 1.69 | 2.08 | 2.52 | 3.00 | 3.52 | 6.75 |
| 75                | 227.4            | 0.78                        | 1.07 | 1.39 | 1.76 | 2.18 | 2.64 | 3.14 | 3.68 | 7.06 |
| 80                | 218.1            | 0.82                        | 1.11 | 1.45 | 1.84 | 2.27 | 2.75 | 3.27 | 3.84 | 7.36 |

Conductor Code Name DOG  
 Basic / Recommended Span (m) 90  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 10.0 at 15°C  
 Required Percentage Increase in Tension (%) 10 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | ERECTION TABLE              |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 891.9            | 0.20                        | 0.27 | 0.36 | 0.45 | 0.56 | 0.67 | 0.80 | 0.94 | 1.80 |
| 0                 | 804.6            | 0.22                        | 0.30 | 0.39 | 0.50 | 0.62 | 0.74 | 0.89 | 1.04 | 1.99 |
| 5                 | 730.3            | 0.24                        | 0.33 | 0.43 | 0.55 | 0.68 | 0.82 | 0.98 | 1.15 | 2.20 |
| 10                | 660.2            | 0.27                        | 0.37 | 0.48 | 0.61 | 0.75 | 0.91 | 1.08 | 1.27 | 2.43 |
| 15                | 595.3            | 0.30                        | 0.41 | 0.53 | 0.67 | 0.83 | 1.01 | 1.20 | 1.41 | 2.70 |
| 20                | 536.6            | 0.33                        | 0.45 | 0.59 | 0.75 | 0.92 | 1.12 | 1.33 | 1.56 | 2.99 |
| 25                | 484.4            | 0.37                        | 0.50 | 0.65 | 0.83 | 1.02 | 1.24 | 1.47 | 1.73 | 3.31 |
| 30                | 438.9            | 0.41                        | 0.55 | 0.72 | 0.91 | 1.13 | 1.37 | 1.62 | 1.91 | 3.66 |
| 35                | 399.9            | 0.45                        | 0.61 | 0.79 | 1.00 | 1.24 | 1.50 | 1.78 | 2.09 | 4.01 |
| 40                | 366.7            | 0.49                        | 0.66 | 0.86 | 1.09 | 1.35 | 1.63 | 1.94 | 2.28 | 4.38 |
| 45                | 338.6            | 0.53                        | 0.72 | 0.94 | 1.18 | 1.46 | 1.77 | 2.11 | 2.47 | 4.74 |
| 50                | 314.7            | 0.57                        | 0.77 | 1.01 | 1.27 | 1.57 | 1.90 | 2.27 | 2.66 | 5.10 |
| 55                | 294.3            | 0.61                        | 0.82 | 1.08 | 1.36 | 1.68 | 2.04 | 2.42 | 2.84 | 5.45 |
| 60                | 276.8            | 0.64                        | 0.88 | 1.14 | 1.45 | 1.79 | 2.16 | 2.58 | 3.02 | 5.80 |
| 65                | 261.7            | 0.68                        | 0.93 | 1.21 | 1.53 | 1.89 | 2.29 | 2.73 | 3.20 | 6.13 |
| 70                | 248.5            | 0.72                        | 0.98 | 1.28 | 1.61 | 1.99 | 2.41 | 2.87 | 3.37 | 6.46 |
| 75                | 236.8            | 0.75                        | 1.02 | 1.34 | 1.69 | 2.09 | 2.53 | 3.01 | 3.53 | 6.78 |
| 80                | 226.5            | 0.79                        | 1.07 | 1.40 | 1.77 | 2.19 | 2.65 | 3.15 | 3.69 | 7.08 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 93 | <b>of</b> | 98 |

## Appendix 45 - .1" / 70mm<sup>2</sup> HDBC (High altitude) - 90m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |                  |
|------------------------------------------------------|------------------|
| Conductor Code Name (if any)                         | <b>COPPER 70</b> |
| Greased Conductor Weight (kg/m)                      | <b>0.621</b>     |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | <b>69.28</b>     |
| Conductor Overall Diameter (mm)                      | <b>10.65</b>     |
| Coefficient of Linear Expansion (/Degree C)          | <b>1.70E-05</b>  |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | <b>12655.3</b>   |
| Rated Breaking Strength of Conductor (kgf)           | <b>2740.9</b>    |
| Basic / Recommended Span (m)                         | <b>90</b>        |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | <b>570</b>       |
| Radial Ice Thickness (mm)                            | <b>12.5</b>      |
| Ice Density (kg/m <sup>3</sup> )                     | <b>913</b>       |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | <b>1370.5</b>    |
| Temperature at MWT Limit (Degrees C)                 | <b>-5.6</b>      |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | <b>913.6</b>     |
| Temperature at EDT Limit (Degrees C)                 | <b>5</b>         |

|                                         |    |        |               |
|-----------------------------------------|----|--------|---------------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | <b>1370.5</b> |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | <b>1.451</b>  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | <b>2.072</b>  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | <b>597.9</b>  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|                             |                               |          |
|-----------------------------|-------------------------------|----------|
| Conductor: <b>COPPER 70</b> | Temperature Shift (Deg. C)    | <b>0</b> |
|                             | Tension Reduction at 15°C (%) | <b>5</b> |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |           |           |           |           |           |
|----------------|-----------------------------------|---------------------------------------------|-----------|-----------|-----------|-----------|-----------|
|                |                                   | <b>50</b>                                   | <b>55</b> | <b>60</b> | <b>65</b> | <b>70</b> | <b>75</b> |
| <b>40</b>      | 0.19                              | 0.35                                        | 0.36      | 0.37      | 0.38      | 0.39      | 0.41      |
| <b>80</b>      | 0.78                              | 1.39                                        | 1.44      | 1.49      | 1.53      | 1.58      | 1.62      |
| <b>120</b>     | 1.75                              | 3.13                                        | 3.24      | 3.35      | 3.45      | 3.55      | 3.65      |
| <b>160</b>     | 3.10                              | 5.57                                        | 5.76      | 5.95      | 6.14      | 6.32      | 6.49      |
| <b>200</b>     | 4.85                              | 8.70                                        | 9.00      | 9.30      | 9.59      | 9.87      | 10.15     |
| <b>240</b>     | 6.98                              | 12.53                                       | 12.97     | 13.39     | 13.80     | 14.21     | 14.61     |
| <b>280</b>     | 9.50                              | 17.06                                       | 17.65     | 18.22     | 18.79     | 19.34     | 19.89     |
| <b>320</b>     | 12.41                             | 22.28                                       | 23.05     | 23.80     | 24.54     | 25.27     | 25.98     |
| <b>360</b>     | 15.71                             | 28.20                                       | 29.17     | 30.12     | 31.06     | 31.98     | 32.88     |
| <b>400</b>     | 19.39                             | 34.82                                       | 36.02     | 37.19     | 38.34     | 39.48     | 40.59     |
| <b>440</b>     | 23.46                             | 42.13                                       | 43.58     | 45.00     | 46.40     | 47.77     | 49.11     |
| <b>480</b>     | 27.93                             | 50.14                                       | 51.86     | 53.55     | 55.22     | 56.85     | 58.45     |
| <b>520</b>     | 32.77                             | 58.84                                       | 60.87     | 62.85     | 64.80     | 66.72     | 68.60     |

|                    |     |                 |         |      |    |    |    |  |
|--------------------|-----|-----------------|---------|------|----|----|----|--|
| Document reference |     | NSP/004/042/001 |         |      |    |    |    |  |
| Version:-          | 2.0 | Date of Issue:- | Sept 10 | Page | 94 | of | 98 |  |

## Appendix 45a - .1" / 70mm<sup>2</sup> HDBC (High altitude) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name **COPPER 70**  
 Basic / Recommended Span (m) **90**  
 Temperature Shift for Creep (Deg. C) **0** **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) **0.0** **at 15°C**  
 Required Percentage Increase in Tension (%) **0** **at 15°C**

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 640.5            | 0.44                        | 0.59 | 0.78 | 0.98 | 1.21 | 1.47 | 1.75 | 2.05 | 3.93 |
| 0                 | 597.9            | 0.47                        | 0.64 | 0.83 | 1.05 | 1.30 | 1.57 | 1.87 | 2.19 | 4.21 |
| 5                 | 563.6            | 0.50                        | 0.67 | 0.88 | 1.12 | 1.38 | 1.67 | 1.98 | 2.33 | 4.46 |
| 10                | 532.4            | 0.52                        | 0.71 | 0.93 | 1.18 | 1.46 | 1.76 | 2.10 | 2.46 | 4.72 |
| 15                | 504.2            | 0.55                        | 0.75 | 0.99 | 1.25 | 1.54 | 1.86 | 2.22 | 2.60 | 4.99 |
| 20                | 478.7            | 0.58                        | 0.79 | 1.04 | 1.31 | 1.62 | 1.96 | 2.34 | 2.74 | 5.25 |
| 25                | 455.7            | 0.61                        | 0.83 | 1.09 | 1.38 | 1.70 | 2.06 | 2.45 | 2.88 | 5.52 |
| 30                | 434.9            | 0.64                        | 0.87 | 1.14 | 1.45 | 1.78 | 2.16 | 2.57 | 3.02 | 5.78 |
| 35                | 416.1            | 0.67                        | 0.91 | 1.19 | 1.51 | 1.87 | 2.26 | 2.69 | 3.15 | 6.04 |
| 40                | 399.1            | 0.70                        | 0.95 | 1.24 | 1.58 | 1.95 | 2.35 | 2.80 | 3.29 | 6.30 |
| 45                | 383.6            | 0.73                        | 0.99 | 1.30 | 1.64 | 2.02 | 2.45 | 2.91 | 3.42 | 6.56 |
| 50                | 369.5            | 0.76                        | 1.03 | 1.34 | 1.70 | 2.10 | 2.54 | 3.03 | 3.55 | 6.81 |
| 55                | 356.6            | 0.78                        | 1.07 | 1.39 | 1.76 | 2.18 | 2.63 | 3.13 | 3.68 | 7.05 |
| 60                | 344.7            | 0.81                        | 1.10 | 1.44 | 1.82 | 2.25 | 2.72 | 3.24 | 3.81 | 7.30 |
| 65                | 333.8            | 0.84                        | 1.14 | 1.49 | 1.88 | 2.33 | 2.81 | 3.35 | 3.93 | 7.53 |
| 70                | 323.8            | 0.86                        | 1.17 | 1.53 | 1.94 | 2.40 | 2.90 | 3.45 | 4.05 | 7.77 |
| 75                | 314.5            | 0.89                        | 1.21 | 1.58 | 2.00 | 2.47 | 2.99 | 3.55 | 4.17 | 8.00 |
| 80                | 305.9            | 0.91                        | 1.24 | 1.62 | 2.06 | 2.54 | 3.07 | 3.65 | 4.29 | 8.22 |

|                           |     |                        |         |             |    |           |    |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 95 | <b>of</b> | 98 |

Appendix 46 - .15" / 100mm<sup>2</sup> HDBC (High altitude) - 90m Basic

## CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |          |
|------------------------------------------------------|----------|
| Conductor Code Name (if any)                         | HDBC 100 |
| Greased Conductor Weight (kg/m)                      | 0.911    |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 101.654  |
| Conductor Overall Diameter (mm)                      | 12.9     |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.3  |
| Rated Breaking Strength of Conductor (kgf)           | 3838.2   |
| Basic / Recommended Span (m)                         | 90       |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 570      |
| Radial Ice Thickness (mm)                            | 12.5     |
| Ice Density (kg/m <sup>3</sup> )                     | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 1814.4   |
| Temperature at MWT Limit (Degrees C)                 | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 1279.4   |
| Temperature at EDT Limit (Degrees C)                 | 5        |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1814.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 1.822  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 2.203  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 985.4  |

## ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |          |                               |   |
|------------|----------|-------------------------------|---|
| Conductor: | HDBC 100 | Temperature Shift (Deg. C)    | 0 |
|            |          | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.17                              | 0.32                                        | 0.34  | 0.35  | 0.36  | 0.37  | 0.38  |
| 80             | 0.69                              | 1.30                                        | 1.35  | 1.39  | 1.44  | 1.49  | 1.54  |
| 120            | 1.55                              | 2.92                                        | 3.03  | 3.14  | 3.25  | 3.35  | 3.46  |
| 160            | 2.75                              | 5.19                                        | 5.38  | 5.58  | 5.77  | 5.96  | 6.14  |
| 200            | 4.30                              | 8.10                                        | 8.41  | 8.72  | 9.02  | 9.31  | 9.60  |
| 240            | 6.20                              | 11.67                                       | 12.11 | 12.55 | 12.98 | 13.41 | 13.82 |
| 280            | 8.44                              | 15.88                                       | 16.49 | 17.08 | 17.67 | 18.25 | 18.81 |
| 320            | 11.02                             | 20.74                                       | 21.54 | 22.31 | 23.08 | 23.83 | 24.57 |
| 360            | 13.95                             | 26.26                                       | 27.26 | 28.24 | 29.21 | 30.16 | 31.10 |
| 400            | 17.22                             | 32.41                                       | 33.65 | 34.87 | 36.06 | 37.24 | 38.39 |
| 440            | 20.83                             | 39.22                                       | 40.72 | 42.19 | 43.64 | 45.06 | 46.45 |
| 480            | 24.79                             | 46.68                                       | 48.46 | 50.21 | 51.93 | 53.62 | 55.28 |
| 520            | 29.10                             | 54.78                                       | 56.87 | 58.93 | 60.95 | 62.93 | 64.88 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 96 | <b>of</b> | 98 |  |

## Appendix 46a - .15" / 100mm<sup>2</sup> HDBC (High altitude) - 90m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name HDBC 100  
 Basic / Recommended Span (m) 90  
 Temperature Shift for Creep (Deg. C) 0 Insert minus sign as necessary  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 1058.4           | 0.39                        | 0.53 | 0.69 | 0.87 | 1.08 | 1.30 | 1.55 | 1.82 | 3.49 |
| 0                 | 985.4            | 0.42                        | 0.57 | 0.74 | 0.94 | 1.16 | 1.40 | 1.66 | 1.95 | 3.75 |
| 5                 | 925.6            | 0.44                        | 0.60 | 0.79 | 1.00 | 1.23 | 1.49 | 1.77 | 2.08 | 3.99 |
| 10                | 870.8            | 0.47                        | 0.64 | 0.84 | 1.06 | 1.31 | 1.58 | 1.88 | 2.21 | 4.24 |
| 15                | 820.9            | 0.50                        | 0.68 | 0.89 | 1.12 | 1.39 | 1.68 | 2.00 | 2.34 | 4.50 |
| 20                | 775.7            | 0.53                        | 0.72 | 0.94 | 1.19 | 1.47 | 1.78 | 2.11 | 2.48 | 4.76 |
| 25                | 734.9            | 0.56                        | 0.76 | 0.99 | 1.26 | 1.55 | 1.88 | 2.23 | 2.62 | 5.02 |
| 30                | 698.0            | 0.59                        | 0.80 | 1.04 | 1.32 | 1.63 | 1.97 | 2.35 | 2.76 | 5.29 |
| 35                | 664.7            | 0.62                        | 0.84 | 1.10 | 1.39 | 1.71 | 2.07 | 2.47 | 2.90 | 5.55 |
| 40                | 634.6            | 0.65                        | 0.88 | 1.15 | 1.45 | 1.79 | 2.17 | 2.58 | 3.03 | 5.82 |
| 45                | 607.3            | 0.68                        | 0.92 | 1.20 | 1.52 | 1.88 | 2.27 | 2.70 | 3.17 | 6.08 |
| 50                | 582.6            | 0.70                        | 0.96 | 1.25 | 1.58 | 1.95 | 2.37 | 2.82 | 3.30 | 6.33 |
| 55                | 560.2            | 0.73                        | 1.00 | 1.30 | 1.65 | 2.03 | 2.46 | 2.93 | 3.44 | 6.59 |
| 60                | 539.7            | 0.76                        | 1.03 | 1.35 | 1.71 | 2.11 | 2.55 | 3.04 | 3.57 | 6.84 |
| 65                | 520.9            | 0.79                        | 1.07 | 1.40 | 1.77 | 2.19 | 2.65 | 3.15 | 3.70 | 7.08 |
| 70                | 503.8            | 0.81                        | 1.11 | 1.45 | 1.83 | 2.26 | 2.74 | 3.26 | 3.82 | 7.33 |
| 75                | 487.9            | 0.84                        | 1.14 | 1.49 | 1.89 | 2.33 | 2.82 | 3.36 | 3.94 | 7.56 |
| 80                | 473.4            | 0.87                        | 1.18 | 1.54 | 1.95 | 2.41 | 2.91 | 3.46 | 4.07 | 7.80 |

|                           |                 |                        |         |             |    |           |    |
|---------------------------|-----------------|------------------------|---------|-------------|----|-----------|----|
| <b>Document reference</b> | NSP/004/042/001 |                        |         |             |    |           |    |
| <b>Version:-</b>          | 2.0             | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 97 | <b>of</b> | 98 |

## Appendix 47 - .2" / 125mm<sup>2</sup> HDBC (High altitude) - 100m Basic

### CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

|                                                      |          |
|------------------------------------------------------|----------|
| Conductor Code Name (if any)                         | HDBC 125 |
| Greased Conductor Weight (kg/m)                      | 1.130    |
| Cross Sectional Area of Conductor (mm <sup>2</sup> ) | 125.499  |
| Conductor Overall Diameter (mm)                      | 14.5     |
| Coefficient of Linear Expansion (/Degree C)          | 1.70E-05 |
| Modulus of Elasticity (kg/mm <sup>2</sup> )          | 12655.29 |
| Rated Breaking Strength of Conductor (kgf)           | 4929.3   |
| Basic / Recommended Span (m)                         | 100      |
| Wind Pressure on Conductor (N/m <sup>2</sup> )       | 570      |
| Radial Ice Thickness (mm)                            | 12.5     |
| Ice Density (kg/m <sup>3</sup> )                     | 913      |
| Absolute Maximum Working Tension (MWT) Limit (kgf)   | 1814.4   |
| Temperature at MWT Limit (Degrees C)                 | -5.6     |
| Maximum "Everyday" Tension (EDT) Limit (kgf)         | 1643.1   |
| Temperature at EDT Limit (Degrees C)                 | 5        |

|                                         |    |        |        |
|-----------------------------------------|----|--------|--------|
| Maximum Conductor Tension (MCT) (kgf)   | at | -5.6°C | 1814.4 |
| Maximum Conductor Weight (MCW) (kg/m)   |    |        | 2.098  |
| Maximum Conductor Pressure (MCP) (kg/m) |    |        | 2.296  |
| Freezing Point Tension (FPT) (kgf)      | at | 0°C    | 860.3  |

### ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

|            |          |                               |   |
|------------|----------|-------------------------------|---|
| Conductor: | HDBC 125 | Temperature Shift (Deg. C)    | 0 |
|            |          | Tension Reduction at 15°C (%) | 5 |

| 2X Span<br>(m) | Sag (m) for<br>Cold Curve<br>-5.6 | Sag (m) for Range of Hot Curve Temperatures |       |       |       |       |       |
|----------------|-----------------------------------|---------------------------------------------|-------|-------|-------|-------|-------|
|                |                                   | 50                                          | 55    | 60    | 65    | 70    | 75    |
| 40             | 0.25                              | 0.38                                        | 0.39  | 0.40  | 0.41  | 0.42  | 0.43  |
| 80             | 1.00                              | 1.53                                        | 1.57  | 1.60  | 1.64  | 1.68  | 1.71  |
| 120            | 2.25                              | 3.44                                        | 3.53  | 3.61  | 3.69  | 3.77  | 3.85  |
| 160            | 4.00                              | 6.12                                        | 6.27  | 6.42  | 6.56  | 6.71  | 6.85  |
| 200            | 6.25                              | 9.56                                        | 9.79  | 10.02 | 10.25 | 10.48 | 10.70 |
| 240            | 9.00                              | 13.76                                       | 14.10 | 14.43 | 14.76 | 15.09 | 15.40 |
| 280            | 12.25                             | 18.73                                       | 19.19 | 19.65 | 20.09 | 20.53 | 20.97 |
| 320            | 16.00                             | 24.46                                       | 25.07 | 25.66 | 26.25 | 26.82 | 27.39 |
| 360            | 20.25                             | 30.96                                       | 31.73 | 32.48 | 33.22 | 33.94 | 34.66 |
| 400            | 25.00                             | 38.22                                       | 39.17 | 40.10 | 41.01 | 41.91 | 42.79 |
| 440            | 30.25                             | 46.25                                       | 47.39 | 48.52 | 49.62 | 50.71 | 51.78 |
| 480            | 36.00                             | 55.04                                       | 56.40 | 57.74 | 59.05 | 60.35 | 61.62 |
| 520            | 42.25                             | 64.60                                       | 66.19 | 67.76 | 69.30 | 70.82 | 72.32 |

|                           |     |                        |         |             |    |           |    |  |
|---------------------------|-----|------------------------|---------|-------------|----|-----------|----|--|
| <b>Document reference</b> |     | NSP/004/042/001        |         |             |    |           |    |  |
| <b>Version:-</b>          | 2.0 | <b>Date of Issue:-</b> | Sept 10 | <b>Page</b> | 98 | <b>of</b> | 98 |  |

## Appendix 47a - .2" / 125mm<sup>2</sup> HDBC (High altitude) - 100m Basic

### DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name HDBC 125  
 Basic / Recommended Span (m) 100  
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**  
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C  
 Required Percentage Increase in Tension (%) 0 at 15°C

| Temp.<br>(Deg. C) | Tension<br>(kgf) | DESIGN TABLE                |      |      |      |      |      |      |      |      |
|-------------------|------------------|-----------------------------|------|------|------|------|------|------|------|------|
|                   |                  | Sag (m) for Span Length (m) |      |      |      |      |      |      |      |      |
|                   |                  | 60                          | 70   | 80   | 90   | 100  | 110  | 120  | 130  | 180  |
| -5.6              | 904.0            | 0.56                        | 0.77 | 1.00 | 1.27 | 1.56 | 1.89 | 2.25 | 2.64 | 5.06 |
| 0                 | 860.3            | 0.59                        | 0.80 | 1.05 | 1.33 | 1.64 | 1.99 | 2.36 | 2.77 | 5.32 |
| 5                 | 825.1            | 0.62                        | 0.84 | 1.10 | 1.39 | 1.71 | 2.07 | 2.47 | 2.89 | 5.55 |
| 10                | 792.9            | 0.64                        | 0.87 | 1.14 | 1.44 | 1.78 | 2.16 | 2.57 | 3.01 | 5.77 |
| 15                | 763.5            | 0.67                        | 0.91 | 1.18 | 1.50 | 1.85 | 2.24 | 2.66 | 3.13 | 5.99 |
| 20                | 736.5            | 0.69                        | 0.94 | 1.23 | 1.55 | 1.92 | 2.32 | 2.76 | 3.24 | 6.21 |
| 25                | 711.7            | 0.71                        | 0.97 | 1.27 | 1.61 | 1.98 | 2.40 | 2.86 | 3.35 | 6.43 |
| 30                | 689.0            | 0.74                        | 1.00 | 1.31 | 1.66 | 2.05 | 2.48 | 2.95 | 3.46 | 6.64 |
| 35                | 667.9            | 0.76                        | 1.04 | 1.35 | 1.71 | 2.11 | 2.56 | 3.05 | 3.57 | 6.85 |
| 40                | 648.4            | 0.78                        | 1.07 | 1.39 | 1.76 | 2.18 | 2.64 | 3.14 | 3.68 | 7.06 |
| 45                | 630.4            | 0.81                        | 1.10 | 1.43 | 1.82 | 2.24 | 2.71 | 3.23 | 3.79 | 7.26 |
| 50                | 613.5            | 0.83                        | 1.13 | 1.47 | 1.86 | 2.30 | 2.79 | 3.32 | 3.89 | 7.46 |
| 55                | 597.9            | 0.85                        | 1.16 | 1.51 | 1.91 | 2.36 | 2.86 | 3.40 | 3.99 | 7.65 |
| 60                | 583.2            | 0.87                        | 1.19 | 1.55 | 1.96 | 2.42 | 2.93 | 3.49 | 4.09 | 7.85 |
| 65                | 569.5            | 0.89                        | 1.22 | 1.59 | 2.01 | 2.48 | 3.00 | 3.57 | 4.19 | 8.04 |
| 70                | 556.6            | 0.91                        | 1.24 | 1.62 | 2.06 | 2.54 | 3.07 | 3.65 | 4.29 | 8.22 |
| 75                | 544.4            | 0.93                        | 1.27 | 1.66 | 2.10 | 2.59 | 3.14 | 3.74 | 4.38 | 8.41 |
| 80                | 533.0            | 0.95                        | 1.30 | 1.70 | 2.15 | 2.65 | 3.21 | 3.82 | 4.48 | 8.59 |